UG0449
User Guide
SmartFusion2 and IGLOO2 Clocking Resources

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1 Revision History

The revision history describes the changes that were implemented in the document. The changes are listed by revision, starting with the most current publication.

1.1 Revision 10.0

The following is a summary of the changes in revision

- Added CCC Simulation Model Limitations, page 85.
- Updated GPD Operating Modes, page 56 regarding the GLx output phase alignments when General Purpose Dividers (GPDs) operate in Mode 3.

1.2 Revision 9.0

The following is a summary of the changes in revision 9.0 of this document.

- Information about Design Recommendations, page 22 was updated.
- Information about Fabric CCC Dynamic Configuration, page 60 was updated.
- Information about reference clock cycles was updated. See Lock Generation Circuit, page 52, PLL Lock Control, page 84, and MPLL, page 93.

1.3 Revision 8.0

The following is a summary of the changes in revision 8.0 of this document.

- A note saying simulation models does not support PLL LOCK Delay was added. See Simulation Support, page 85.
- Information about main crystal oscillator for Flash*Freeze was corrected. See Clocking Scheme Overview, page 5.

1.4 Revision 7.0

The following is a summary of the changes in revision 7.0 of this document.

- Information about MSS clock switching was added, see MSS Clock Switching from User Clock to Standby Clock, page 97.
- Corrected Fabric CCC locations (X, Y coordinates), see Table 10, page 44.

1.5 Revision 6.0

The following is a summary of the changes in revision 6.0 of this document.

- Added M2GL150 device to the Figure 7, page 14 title.
- Information about Managing Global Signals, Global Promotion and Demotion Using PDC, and Place and Route was added. For more information, see Design Recommendations, page 22.
- Information about fabric CCC locations was added. For more information, see Table 10, page 44.
- Updated Figure 38, page 42, Figure 39, page 43, and Figure 40, page 44.
- Information about PLL core was updated, see PLL Core, page 50. Libero does not support using both 2.5 V and 3.3 V power supplies for the PLLs used in a design.
- Information about reference clock and feedback clock was added as a note to PLL Core, page 50 section.
- Information about lock settings was added as a note to Lock Generation Circuit, page 52 section.
- Information about NGMUX reset was added. For more information, see NGMUX, page 58.
- Updated FCCC_PLL_CR10, page 69.
- Information about the PLL bypass mode was added to CCC Input Clock Sources, page 82 section.
- Information about clock delay adjustment was updated, see Clock Delay Adjustment, page 82.
- Information about speed grade was added under M3_CLK or HPMS_CLK and HCLK clocks. For more information, see Table 55, page 94.

1.6 Revision 5.0

Information about Lock Generation Circuit, page 52 was updated. For more information, see Fabric Clock Conditioning Circuitry, page 40 chapter.

1.7 Revision 4.0

Merged and updated all the chapters of SmartFusion 2 and IGLOO 2 Clocking resources user guide.

1.8 Revision 3.0

Modified Fabric Clock Conditioning Circuitry, page 40 Chapter

1.9 Revision 2.0

The following is a summary of the changes in revision 2.0 of this document.

- Updated MSS/HPMS Clock Conditioning Circuitry, page 90 Chapter.
- Updated Fabric Clock Conditioning Circuitry, page 40 Chapter.
- The Fabric CCC Dynamic Configuration, page 60 section was added.
- Updated On-Chip Oscillators, page 28 Chapter.

1.10 Revision 1.0

Revision 1.0 was the first publication of this document.

2 Clocking Resources Overview

This chapter provides an overview of SmartFusion® 2 and IGLOO® 2 clocking resources and device clocking scheme. The following table lists the maximum number of clocking resources available on each SmartFusion 2 and IGLOO 2 family device.

Table 1 • **Maximum Clocking Resources for SmartFusion 2 and IGLOO 2 Family Devices**

		SmartFusion® 2 and IGLOO® 2 Device Part Number						
Resource		**M2S005 and M2GL005**	**M2S010 and M2GL010**	**M2S025 and M2GL025**	**M2S050 and M2GL050**	**M2S060 and M2GL060**	**M2S090 and M2GL090**	**M2S150 and M2GL150**
On-chip oscillators	1 MHz RC oscillator	1	1	1	1	1	1	1
	50 MHz RC oscillator	1	1	1	1	1	1	1
	Main crystal oscillator	1	1	1	1	1	1	1
	Auxiliary crystal oscillator (SmartFusion 2 only)	1	1	1	0	1	1	1
Fabric CCCs[1]		2	2	6	6	6	6	8
MSS/HPMS CCC[1]		1	1	1	1	1	1	1
Global buffers		8	8	16	16	16	16	16
Dedicated global I/Os		16	16	32	32	32	32	32

1. Each CCC has a dedicated PLL for clock synchronization and clock synthesis.

As listed in the table, SmartFusion 2 and IGLOO 2 devices have the following on-chip oscillators to use as the primary source for generating free-running clocks:

- 1 MHz RC Oscillator
- 50 MHz RC Oscillator
- Main Crystal Oscillator
- Auxiliary Crystal Oscillator

These on-chip oscillators (except the auxiliary crystal oscillator) can be used in conjunction with the on-chip CCCs to generate clocks of varying frequencies and phases. The auxiliary crystal oscillator is dedicated to MSS RTC. See the On-Chip Oscillators, page 28 for more information.

The Fabric CCCs present in SmartFusion 2 and IGLOO 2 devices can generate four different clock outputs with a maximum frequency of 400 MHz. Each fabric CCC has a dedicated PLL for flexible clocking in the FPGA fabric. Fabric CCCs can also provide a base reference clock to the on-chip hard IP blocks: MSS/HPMS, FDDR, and high-speed serial interfaces. See the Fabric Clock Conditioning Circuitry, page 40 for more information.

SmartFusion 2 and IGLOO 2 devices contain a dedicated CCC (MSS/HPMS CCC) for the MSS/HPMS clocking. The MSS or HPMS CCC is responsible for generating various aligned clocks required by the MSS/HPMS for correct operation of the MSS/HPMS sub-blocks and synchronous communication with the user logic in the FPGA fabric. The MSS/HPMS CCC has a PLL (MPLL) which is specifically used for generation of the base clock and de-skewing the internal MSS/HPMS clock from the base clock. See the MSS/HPMS Clock Conditioning Circuitry, page 90 for more information on MSS/HPMS clocking.

SmartFusion 2/IGLOO 2 FPGA fabric offers a low-skew global network, which provides an effective clock distribution throughout the FPGA fabric and has extensive support for multiple clock domains. The global network is composed of global buffers (GBs) to distribute low-skew clock signals or high-fanout nets. As listed in Table 1, page 4, SmartFusion 2 and IGLOO 2 devices have either 8 or 16 GBs depending on the size of the device. Each GB produces a global signal.

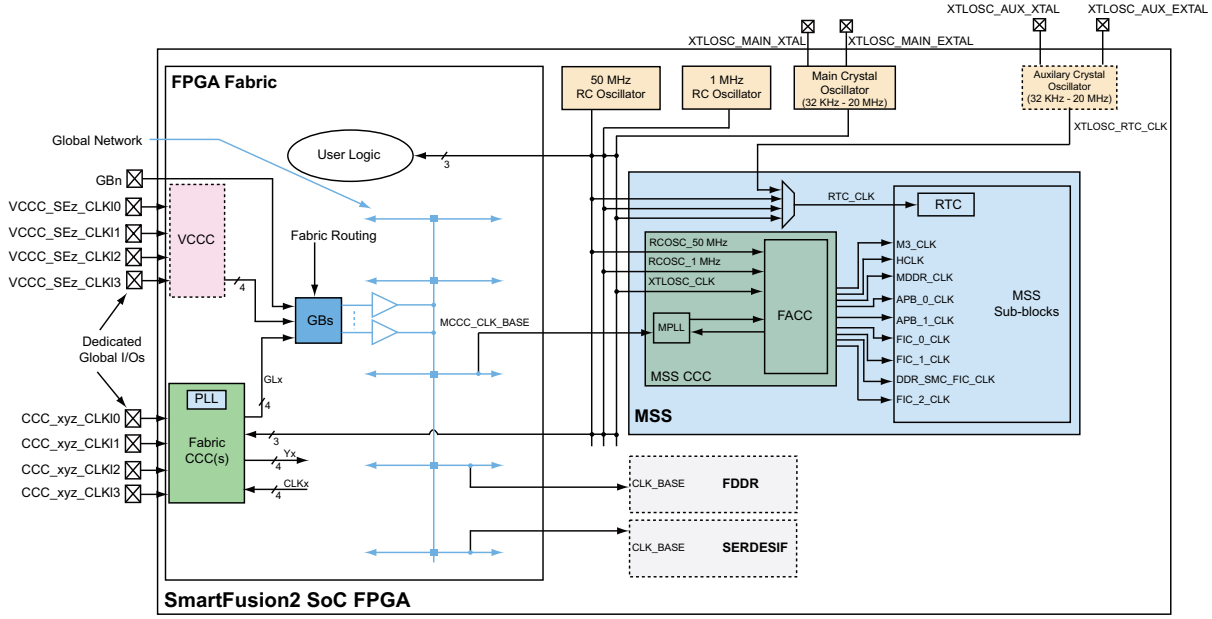
Dedicated global I/Os drive the GBs directly and are the primary source for connecting external clock inputs to the internal global clock network. The total number of dedicated global I/Os varies from 16 to 32, depending on the device selected. See the FPGA Fabric Global Network Architecture, page 8 for more information on the global network and dedicated global I/Os.

The Libero SoC design software provides clock management macros for static configuration of the on-chip oscillators and CCCs.

2.1 Clocking Scheme Overview

The following figure depicts the top-level SmartFusion 2 device clocking scheme. The figure shows the inputs and outputs for one fabric CCC; each fabric CCC has a similar set of inputs and outputs.

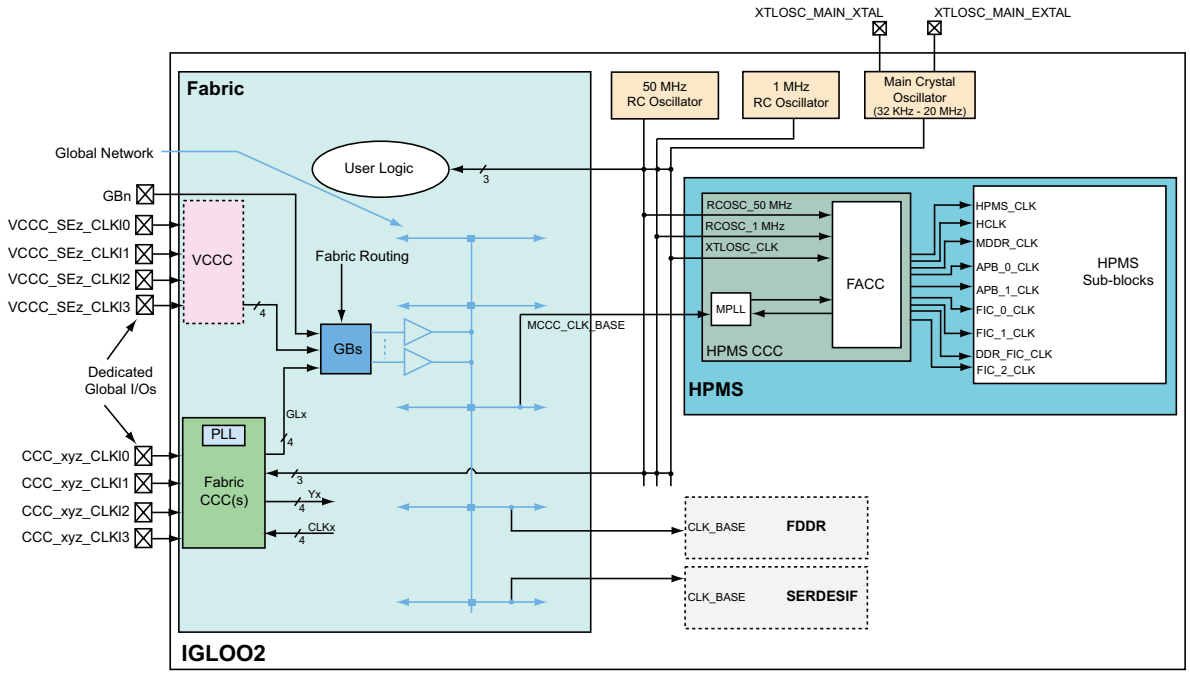
Figure 1 • **SmartFusion 2 SoC FPGA Device Clocking Scheme**



Note: The blocks marked with dotted lines are available only in selected devices of SmartFusion2 SoC FPGA family.

The following figure depicts the top-level IGLOO 2 device clocking scheme. The figure shows the inputs and outputs for one fabric CCC; each fabric CCC has a similar set of inputs and outputs.

Figure 2 • **IGLOO 2 FPGA Device Clocking Scheme**



Note: The blocks marked with dotted lines are available only in selected devices of IGLOO2 FPGA family.

The GBs in the FPGA fabric distribute global signals to the entire FPGA fabric with low skew. As shown in Figure 1, page 5 and Figure 2, page 6, GBs can be driven from multiple sources:

- Dedicated global I/Os
- Virtual CCCs (VCCCs)
- Fabric CCCs
- FPGA fabric routing

Some of the dedicated global I/Os have direct access to the GBs, whereas others must go through either VCCCs or fabric CCCs to reach the GBs. It is also possible to access the GBs from a regular I/O or FPGA fabric internal signal through FPGA fabric routing. The four global clock outputs (GLx, x = 0 to 3) of each fabric CCC have a hardwired connection to the GBs. In addition to global clock outputs (GLx), each fabric CCC can also generate four core clocks (Yx, x = 0 to 3) to drive the fabric routing resources in the FPGA fabric. Note that the core clocks are not routed through the global clock buffers. The core clocks may add skew in the signals due to routing. For more information on VCCCs, see the VCCC, page 16.

The on-chip oscillators—1 MHz RC oscillator and 50 MHz RC oscillator—have hardwired connections to the MSS/HPMS and all the fabric CCCs. On-chip oscillators can be configured as a clock source to any of the following:

- Fabric CCCs
- User logic clock nets through FPGA fabric routing
- MSS/HPMS during Flash*Freeze mode
- MSS RTC

Each fabric CCC can have an independent reference clock from one of the following clock sources:

- Four dedicated global I/Os (CCC_xyz_CLKIw, w = 0 to 3)
- FPGA fabric inputs (CLKx, x = 0 to 3)
- On-chip oscillators

Each fabric CCC has its own dedicated global I/Os. For instance, a dedicated global I/O belonging to northeast side CCCs is referred to as CCC_NEz_CLKIw where z represents CCC number and w represents one of the four associated dedicated global I/Os. The dedicated global I/Os have a hardwired connection to the fabric CCCs whereas the FPGA fabric inputs are routed through FPGA fabric routing

nets. Providing the reference clock through FPGA fabric inputs introduces delay because of fabric routing. See the Dedicated Global I/Os, page 12 for more information.

The base clocks (MSCC_CLK_BASE and CLK_BASE) to the on-chip hard IP blocks (MSS/HPMS, FDDR, and SERDESIF) must come through the FPGA fabric. Each base clock can be generated from any one of the fabric CCCs or a clock source (internal or external) through the global network. The MSS or HPMS CCC takes the base clock as a reference clock input to the MPLL and generates various aligned clocks. See the MSS/HPMS Clock Conditioning Circuitry, page 90 for more information on clocking requirements for synchronous communication between MSS/HPMS and FPGA fabric subsystems. The FDDR and SERDESIF subsystems also have their own clock controllers with a dedicated PLL for generating the required clocks. See the *UG0446: SmartFusion2 and IGLOO2 FPGA High-Speed DDR Interfaces User Guide* and *UG0447: SmartFusion2 and IGLOO2 FPGA High-Speed Serial Interfaces User Guide* for more information on FDDR and SERDESIF clocking.

The following chapters of this user guide describe the SmartFusion 2 and IGLOO 2 FPGA clocking resources in detail.

3 FPGA Fabric Global Network Architecture

SmartFusion 2 and IGLOO 2 FPGA fabric offer a low-skew global network for effective distribution of high-fanout nets including clock signals. The global network has an extensive support for multiple clock domains. This chapter describes the global network architecture and global resources.

3.1 Global Network Architecture

The SmartFusion 2 or IGLOO 2 global network is a tightly coupled, hardwired, and dedicated routing network between the following global resources:

- Dedicated Global I/Os
- Fabric CCC
- VCCC
- Global Buffer
- Row Global Buffers

The global network architecture depends on the number of global resources available on the device. The following table lists the maximum global resources available for SmartFusion 2 and IGLOO 2 devices.

Table 2 • **Maximum Global Resources for SmartFusion 2 and IGLOO 2 Devices**

	SmartFusion® 2/IGLOO 2 Device						
Resource	**M2S005 and M2GL005**	**M2S010 and M2GL010**	**M2S025 and M2GL025**	**M2S050 and M2GL050**	**M2S060 and M2GL060**	**M2S090 and M2GL090**	**M2S150 and M2GL150**
Fabric CCCs	2	2	6	6	6	6	8
VCCCs	0	0	2	2	2	2	0
Global buffers	8	8	16	16	16	16	16
Dedicated global I/Os	16	16	32	32	32	32	32

Each device has 8 or 16 global buffers (GBs), depending on the global network architecture. There are two types of global network architectures:

- 2VS16 (2 vertical stripes, 16 GBs)
- 1VS8 (1 vertical stripe, 8 GBs)

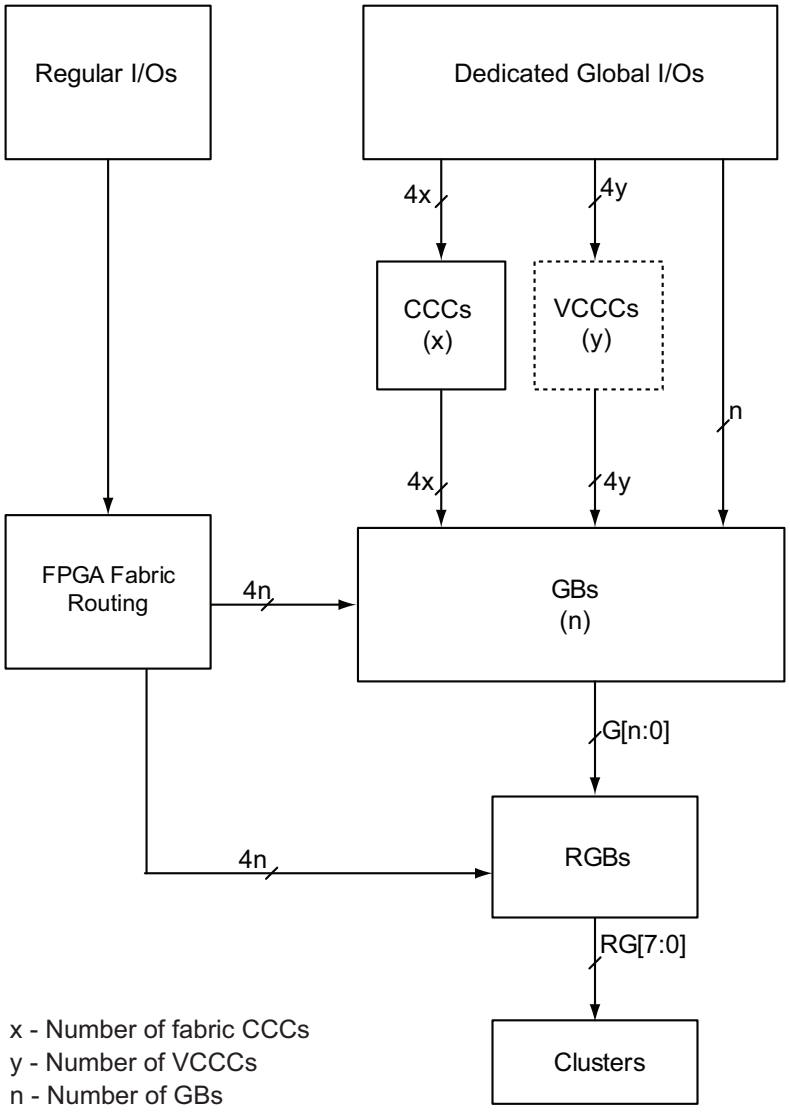
The 2VS16 global network architecture is used for M2S025/M2GL025, M2S050/M2GL050, M2S060/M2GL060, M2S090/M2GL090, and M2S150/M2GL150 devices. The fabric CCCs present on south-east side of the M2S150/M2GL150 devices are replaced with VCCCs on the M2S025/M2GL025, M2S050/M2GL050, M2S060/M2GL060, and M2S090/M2GL090 devices. It supports two vertical stripes, 16 GBs, 32 dedicated global I/Os (8 in each side), and up to eight fabric CCCs (2 in each quadrant).

The 1VS8 global network architecture is used for M2S005/M2GL005 and M2S010/M2GL010 devices, as shown in Figure 6, page 12. It supports 1 vertical stripe, 8 GBs, 16 dedicated global I/Os (4 in each side), and 2 fabric CCCs located in the northeast corner.

The global network is composed of global buffers (GBs) to distribute low-skew clock signals or high-fanout nets. The following figure shows the global signal routing hierarchy to logic clusters. Global signals (G[n:0]) reach the logic clusters through row global signals (RG[7:0]) generated by an associated row global buffer (RGB). RGBs are located on the vertical stripes. RGBs can be accessed from GBs and FPGA fabric routing. GBs can be accessed from dedicated global I/Os, CCC/VCCC global outputs, and the FPGA fabric routing. Clocks coming from regular I/Os can reach GBs or RGBs through FPGA fabric routing.

For external clocks, which do not require clock conditioning (frequency division, frequency multiplication, phase shifting, and delay operations), it is recommended to use dedicated global I/Os having direct access to GBs for clock distribution with less delay. Use CCC to perform clock conditioning on the external clocks or internally generated clocks and CCC generated clocks can be connected to GBs for clock distribution.

Figure 3 • **Global Signal Routing**



The following figures show the global network architecture for SmartFusion 2 and IGLOO 2 family devices.

Figure 4 • **Global Network Architecture for SmartFusion 2 and IGLOO 2 M2S150/M2GL150 Devices**

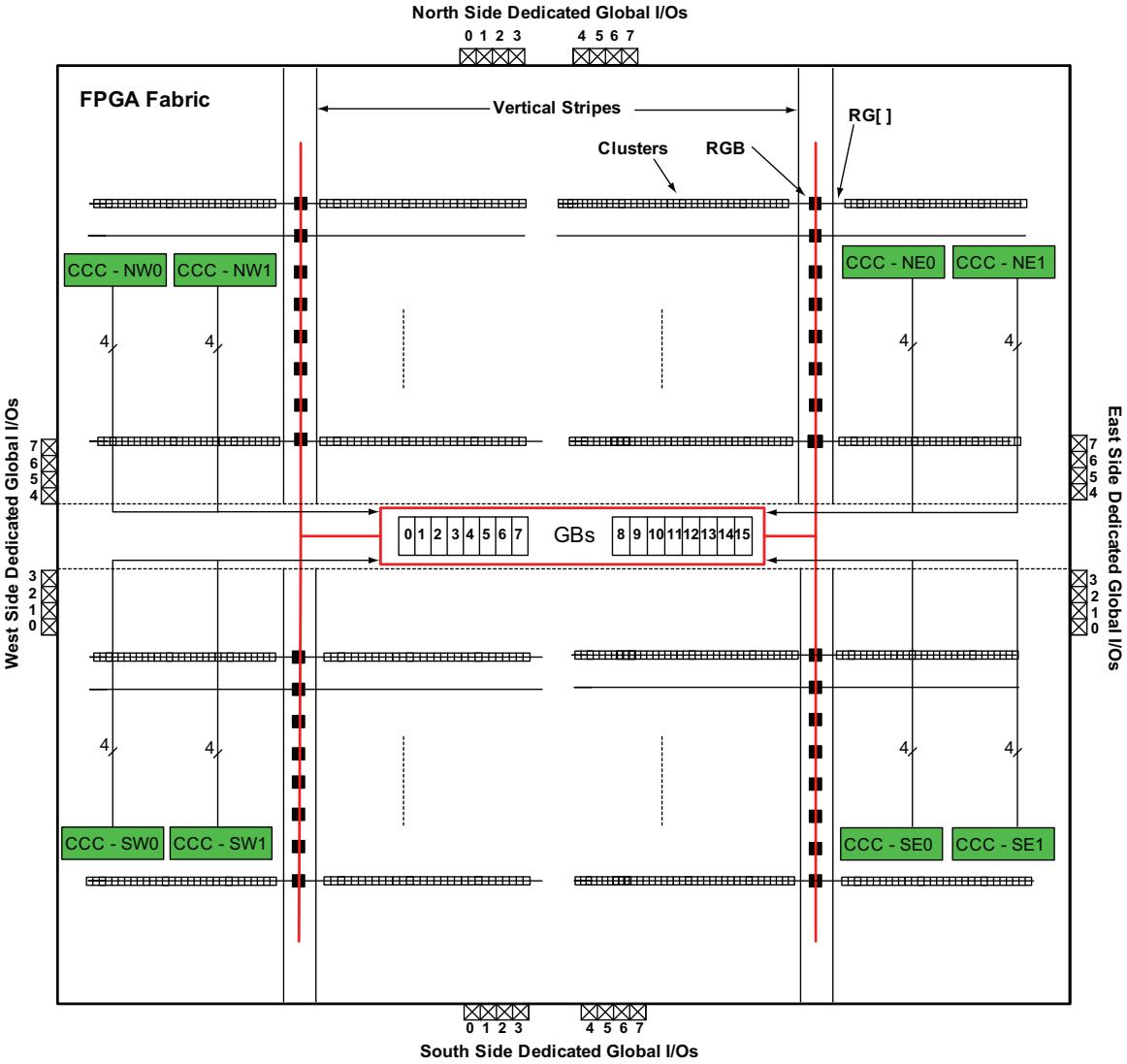


Figure 5 • **Global Network Architecture for SmartFusion 2 and IGLOO 2 M2S025/M2GL025, M2S050/M2GL050, M2S060/M2GL060, and M2S090/M2GL090 Devices**

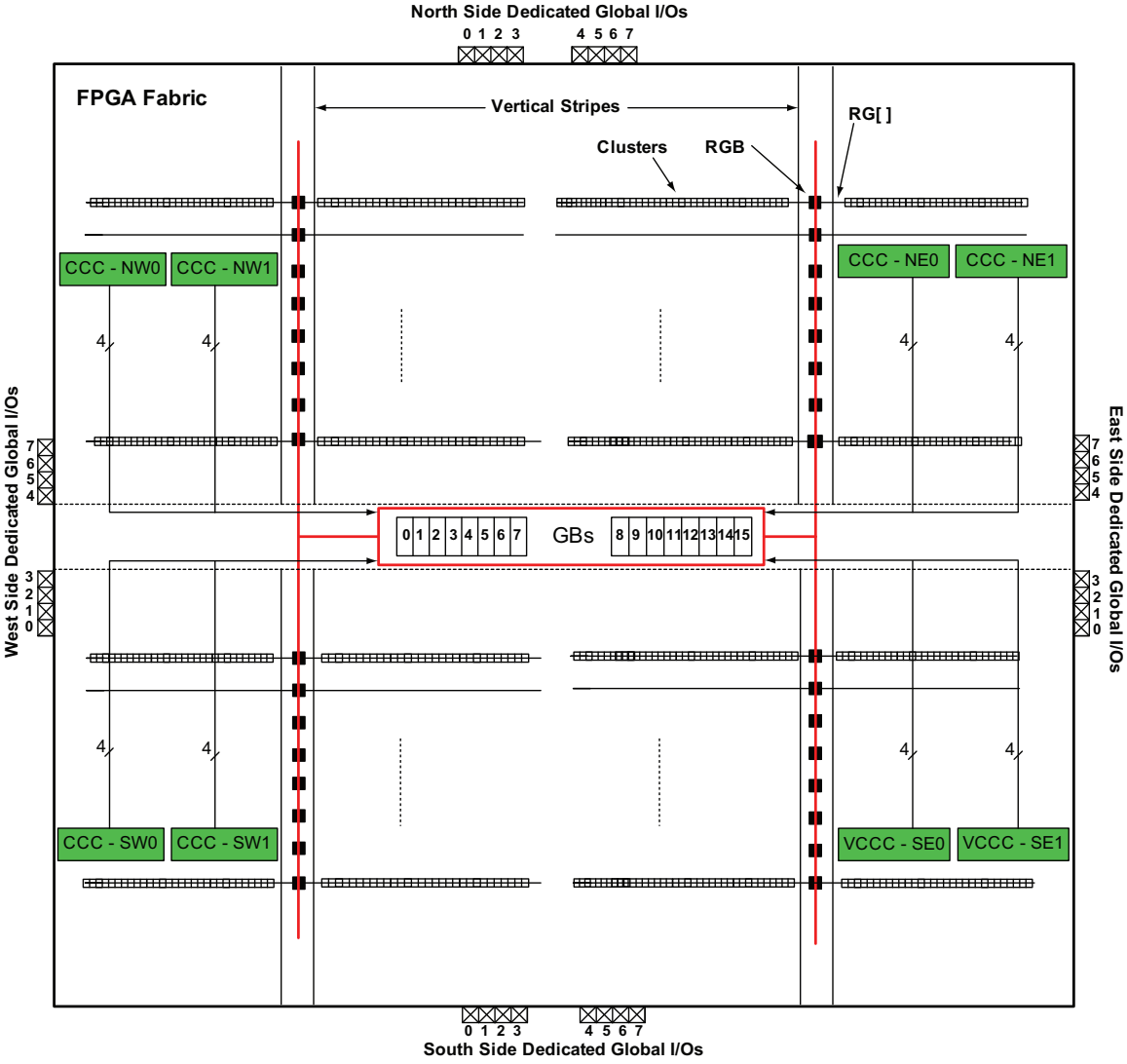
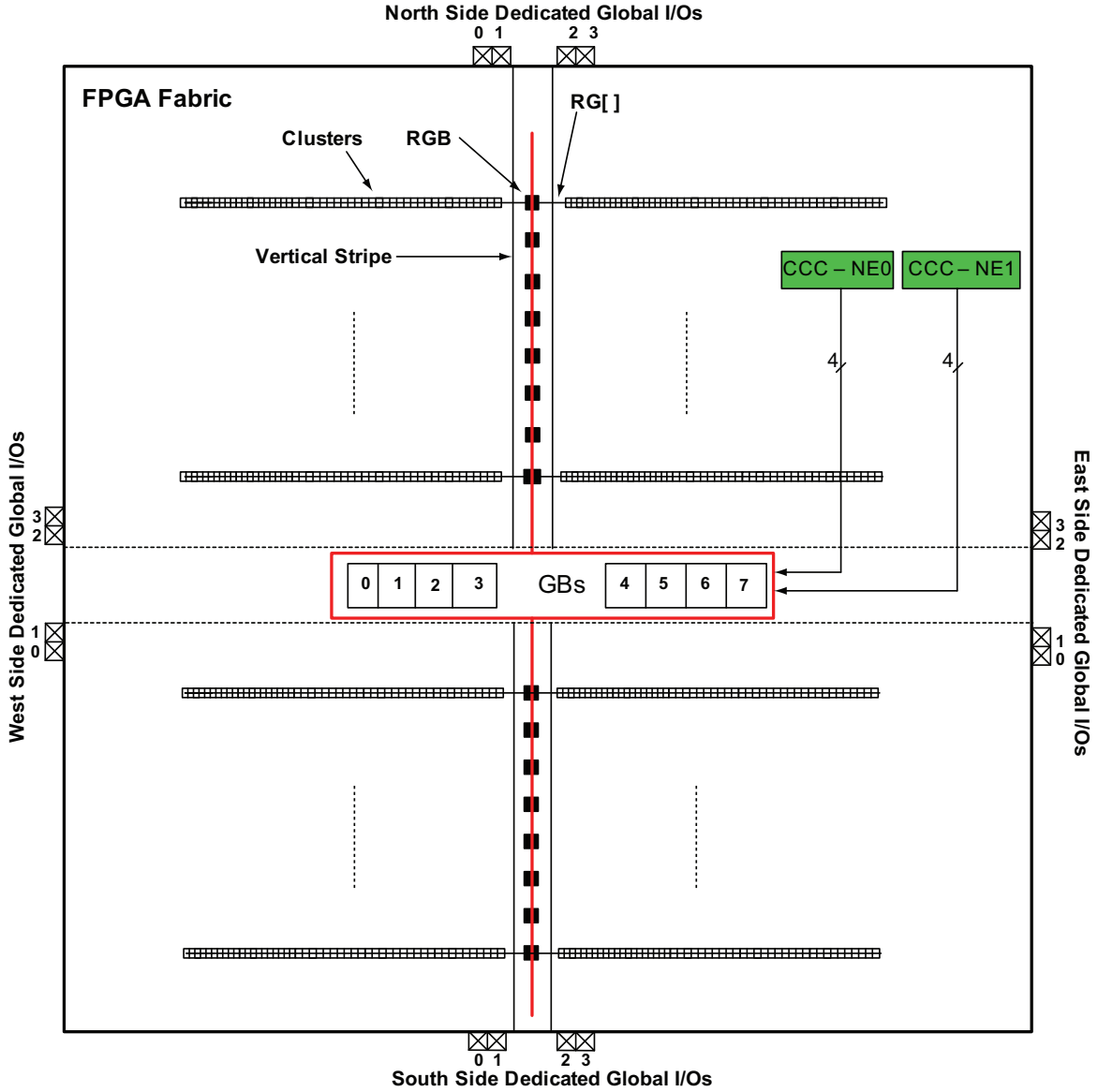


Figure 6 • **Global Network Architecture for SmartFusion 2 and IGLOO 2 M2S005/M2GL005 and M2S010/M2GL010 Devices**



3.1.1 Dedicated Global I/Os

SmartFusion 2 and IGLOO 2 user I/Os are grouped into multi-standard I/Os (MSIO and MSIOD) and DDRIOs. Some of these user I/Os, referred to as dedicated global I/Os, are dual-use I/Os which are capable of driving the global routing network or local routing network. Dedicated global I/Os can be used to bring in external clock signals as inputs to the FPGA fabric. Dedicated global I/Os can be used as regular I/Os, as either input or output for any design signal, if they are not utilized for clocking. Dedicated global I/Os are located on each of the four sides of the FPGA fabric.

Some of the dedicated global I/Os have direct access to GBs, whereas others have to go through either VCCCs or CCCs to reach GBs. Each fabric CCC has four dedicated global I/Os as inputs and can drive up to four GBs. The dedicated global I/Os connect to fabric CCCs, VCCCs, and GBs through a hardwired connection.

3.1.1.1 Naming Conventions for Dedicated Global I/Os

Due to the comprehensive and flexible nature of dedicated global I/Os, a naming scheme is used to provide the detailed information on each I/O.

The dedicated global I/O uses the generic name **IOxyBz/GBn/CCCtype_xyz_CLKIw**, where

- IO represents the type of I/O: MSIO, MSIOD, or DDRIO.
- x represents the I/O pair number.
- y represents differential I/O polarity – P (positive) or N (negative).
- Bz represents the bank number.
- GBn represents dedicated global I/Os that drive the GBs directly, where n is 0 to 15 or 0 to 7.
- CCC_xyz_CLKIw represents dedicated global I/Os that drive GBs through CCCs and VCCC_xyz_CLKI represents dedicated global I/Os that drive GBs through VCCCs, where:
  - xy represents the location – NE, SW, SE, or NW.
  - z represents the CCC or VCCC number – 0 or 1.
  - I represents the clock input
  - w represents one of the four dedicated global I/Os associated with each CCC: CLKI0, CLKI1, CLKI2, or CLKI3.

Some of the dedicated global I/Os are multiplexed with hard IP blocks such as MSS_MMUART, MSS_I2C, MSS_USB, MSS_GPIO, MDDR, and FDDR. These multiplexed or multi-function dedicated global I/Os act as regular I/Os and cannot be used for accessing the global network or CCCs when the associated hard IP block is enabled in the design. See *DS0115: SmartFusion2 Pin Descriptions Datasheet* and *DS0124: IGLOO2 Pin Descriptions Datasheet* for more information on the functions supported by the dedicated global I/Os. The name of a pin shows the functionalities for which that pin can be configured and used.

Example pin name: **DDRIO76PB0/GB12/CCC_NE1_CLKI2/MDDR_DQ12**

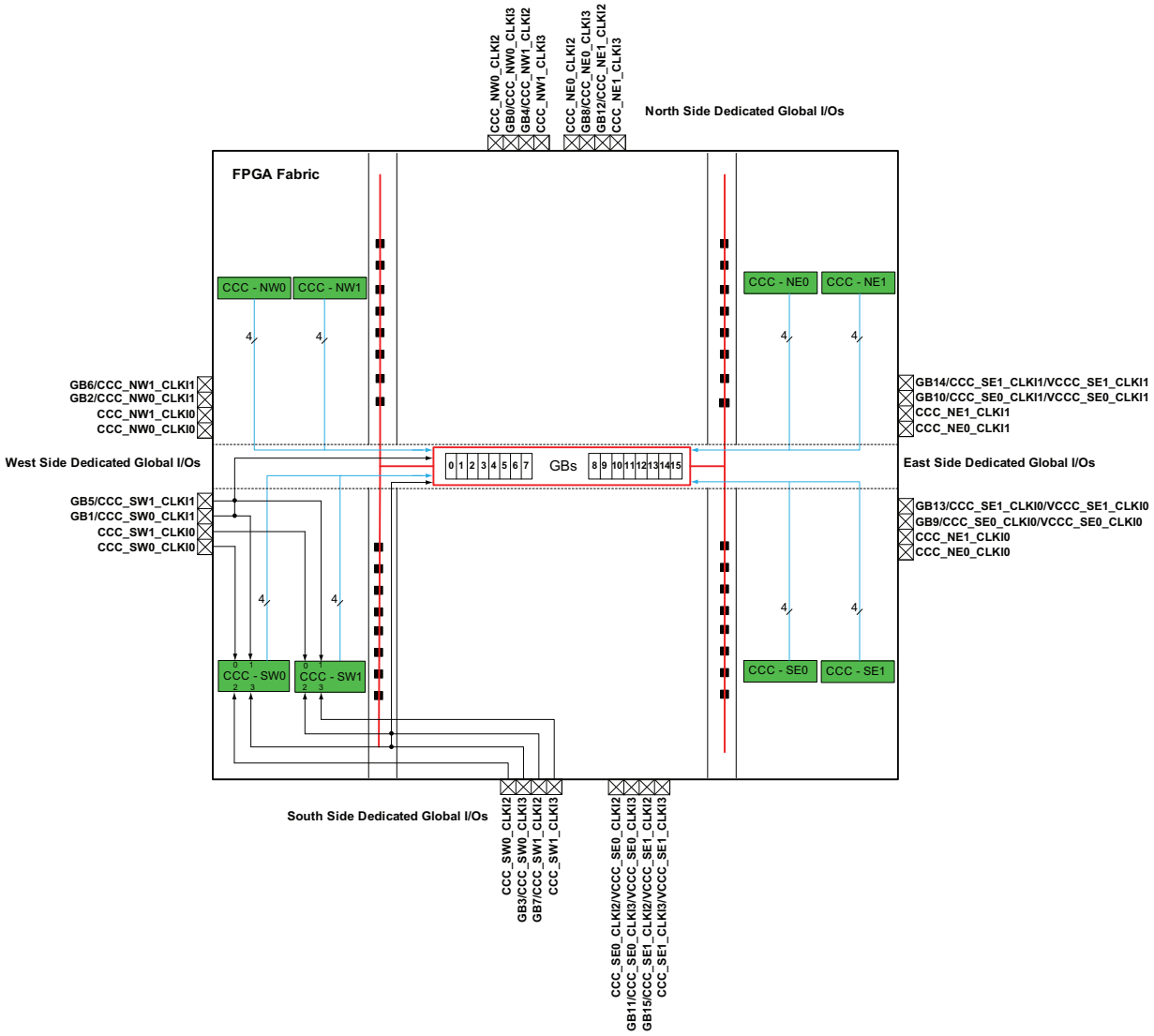
The above example pin is a multi-purpose I/O, which is configured as an MDDR I/O (MDDR_DQ12) when the MDDR is enabled. If MDDR is not used in the design, this I/O can be configured as dedicated global I/O which can drive the CCC_NE1_CLKI2 input port of the CCC_NE1 or GB 12 directly.

Dedicated global I/Os are routed to GBs by the Libero SoC routing software as follows:

- If a dedicated global I/O that has direct access to GBs is selected, the routing software directly routes the dedicated global I/O input to an associated GB.
- If a dedicated global I/O in the east or south side having access to the VCCCs is selected, the routing software routes the dedicated global I/O through the VCCCs to an associated GB.
- If a dedicated global I/O that has direct access to a fabric CCC (*CCC* macro instantiated in the design with input source selected as dedicated global I/O) is selected, the routing software selects an associated fabric CCC. In this case, the outputs of the fabric CCC reach the associated GBs.

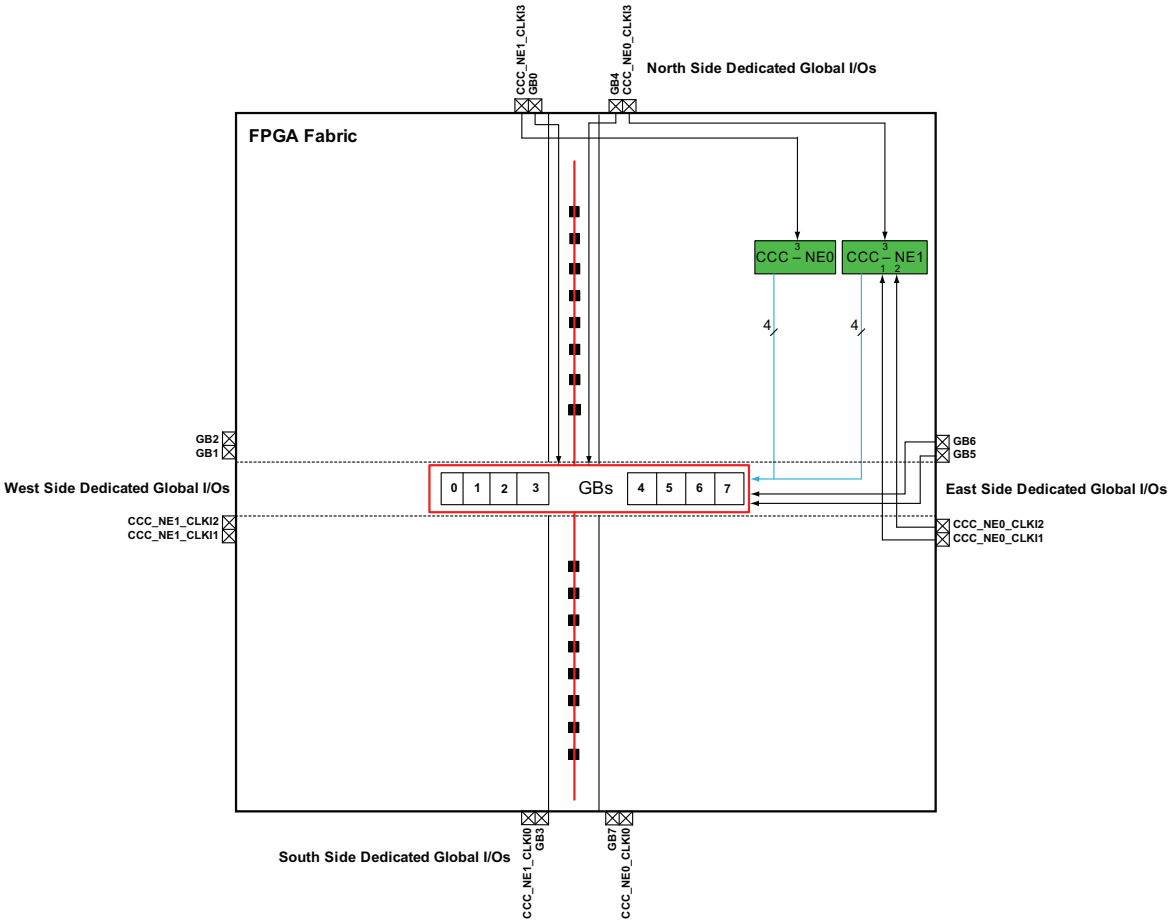
The 2VS16 global network architecture has 16 dedicated global I/Os (4 on each side), which can access GBs directly or through CCCs and the other 16 dedicated global I/Os access GBs through CCCs. The following figure shows the dedicated global I/Os connection for the lower left quadrant of the FPGA fabric in the 2VS16 global network architecture. The rest of the dedicated global I/Os are assigned to CCCs and/or GBs according to the naming convention shown in the figure. The CCC pair present in the southeast corner is replaced by VCCCs for M2S025/M2GL025, M2S050/M2GL050, M2S060/M2GL060, and M2S090/M2GL090 devices.

Figure 7 • **Dedicated Global I/Os (South-West Quadrant) Assignment in M2S025/M2GL025, M2S050/M2GL050, M2S060/M2GL060, M2S090/M2GL090, and M2S150/M2GL150 Devices**



The 1VS8 global network architecture has eight dedicated global I/Os (two on each side), which can access GBs directly or through CCCs and the other 8 dedicated global I/Os access GBs through CCCs. The following figure shows the dedicated global I/Os (located on the north and east side) assignment in the 1VS8 global network architecture. The rest of the dedicated global I/Os are assigned to CCCs or GBs as per the naming convention shown in the figure.

Figure 8 • **Dedicated Global I/Os (North-East Quadrant) Assignment in M2S005/M2GL005, and M2S010/M2GL010 Devices**



3.1.1.2 Dedicated Global I/O Voltage Standards

Dedicated global I/Os are located in different I/O banks with each bank having its own supply and ground pins. The voltage standards supported by dedicated global I/Os are based on the I/O bank it is located in. The voltage standard for a dedicated global I/O can be set using I/O Attribute Editor available in the Libero SoC software. See the "Supported Voltage Standards" table in the *UG0445: IGLOO2 FPGA and SmartFusion2 SoC FPGA Fabric User Guide* for I/O standards supported by each I/O bank. Dedicated global I/Os can be configured in single-ended mode or differential mode. Differential mode is implemented with a fixed I/O pair and cannot be split with adjacent I/Os. According to the naming convention, differential I/O pairs are denoted with an I/O pair number and their polarity (P and N). In Single-ended mode, the I/O pair operates as two independent I/Os. All the configuration and data inputs/outputs are separate and use names ending in P and N to differentiate between the I/Os.

3.1.1.3 Unused Dedicated Global I/O Configuration

Unused dedicated global I/Os behave similarly to unused regular User I/Os. When regular User I/Os (MSIO, MSIOD, DDRIO) are not used, Libero configures the I/O as input buffer disabled, output buffer tristated with weak pull-up.

3.1.2 Fabric CCC

Fabric CCCs enable flexible clocking schemes to the logic implemented in the FPGA fabric, and can also provide the base clock for on-chip hard IP blocks—MSS/HPMS, FDDR, and SERDESIF. Each fabric CCC operates with a dedicated PLL and generates clock signals of varying frequency and phase. Each fabric CCC generates up to four different global clocks (GL0, GL1, GL2, and GL3) and four core clocks (Y0, Y1, Y2, and Y3). The generated global clocks drive GBs and core clocks drive the local routing resources in the FPGA fabric. Fabric CCC's core clock outputs (Yx) can be used to drive internal logic without using global network resources. Core clocks (Yx) introduce additional delay because of FPGA fabric routing. Core clocks are useful when global network resources must be conserved and utilized for other timing-critical paths. The GBs associated with the global clock outputs (GLx) are available to user logic, if the global clock outputs are disabled. Each fabric CCC has four dedicated global I/Os as inputs. Each fabric CCC output (GLx/Yx), as well as the reference clock, can be driven from any one of these four dedicated global I/Os.

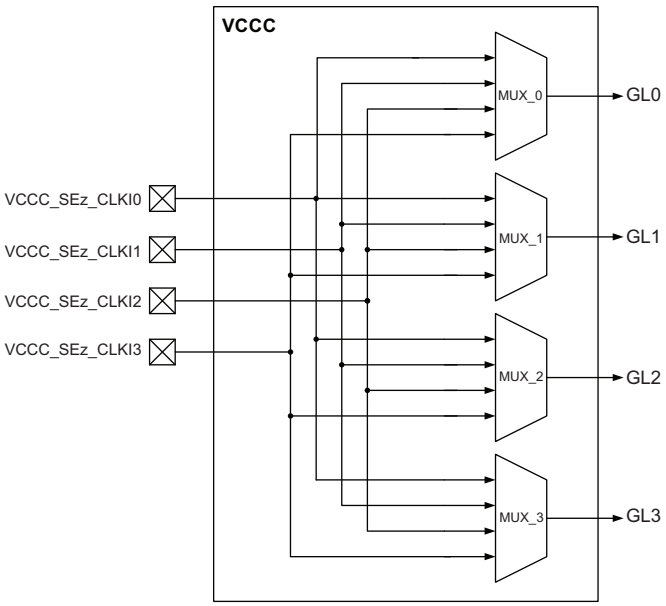
Fabric CCCs are labeled according to their location in the FPGA fabric floor plan. For instance, the fabric CCCs located in the northeast corner are labeled CCC-NE0 and CCC-NE1. See the Fabric Clock Conditioning Circuitry, page 40 for more information on fabric CCCs.

3.1.3 VCCC

A VCCC is a four-input/four-output crossbar switch (as shown in the following figure). Each VCCC has four inputs that come from the dedicated global I/O pads and can drive up to four GBs. VCCC blocks do not have fabric inputs or outputs and they cannot be connected to each other.

VCCCs are automatically instantiated by the Libero SoC place-and-route tool, based on the dedicated global I/Os utilized to facilitate routing and cannot be instantiated manually into a design. VCCCs are configured with flash bits sets by the Libero SoC software at the time of FPGA fabric programming and cannot be dynamically configured.

Figure 9 • **VCCC Functional Block Diagram**



3.1.4 Global Buffer

The global buffer (GB) is a multiplexer that generates an independent global signal. The GBs can be driven from multiple sources such as dedicated global I/Os, fabric CCCs, VCCCs, and fabric routing. For example, an input signal connected to a dedicated global I/O can route directly to a GB, through a CCC or VCCC. Input signals connected to regular I/Os or any internal logic module can also be connected to the GB through fabric routing. The following figures show sources feeding into GBs, which finally feed into RGBs through G[0-15] to span the complete fabric area.

GBs can also be fed through regular I/Os, in which case the signals are first routed to fabric routing and then reach the GBs. Any signal generated from logic modules can reach GBs through fabric routing.

Figure 10 • **Various Sources Feeding Global Buffers for 2VS16 Global Network Architecture**

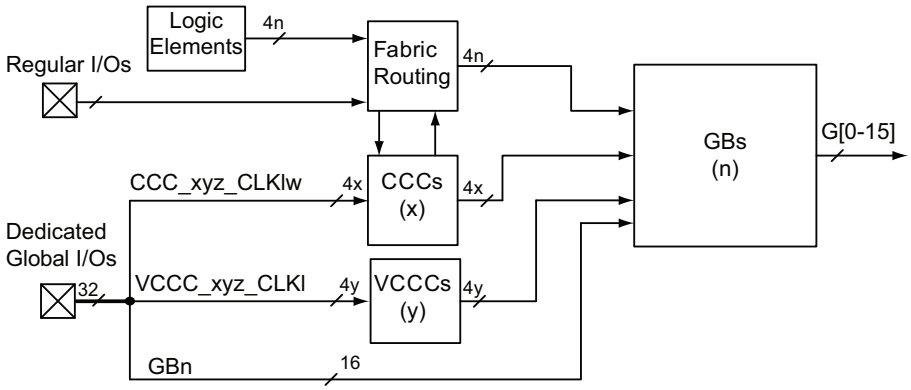
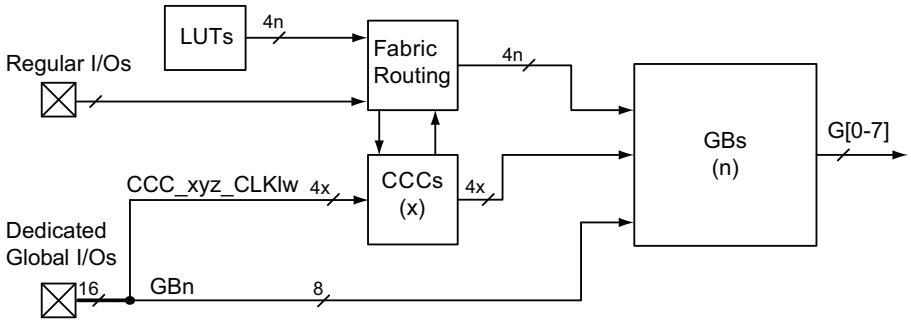


Figure 11 • **Various Sources Feeding Global Buffers for 1VS8 Global Network Architecture**



The following table shows the assignment of fabric CCC global outputs (GLx, x = 0 to 3) to the global buffers in the 2VS16 global network architecture. The global outputs of the fabric CCCs, which are present in the west side of the FPGA fabric are mapped to global buffers 0 to 7. The global outputs of the fabric CCCs, which are present in the east side are mapped to global buffers 8 to 15. See Figure 4, page 10 through Figure 6, page 12 for global buffer assignment. Each global output of the fabric CCC is associated with two global buffers. Each global buffer is also shared between one of the four global outputs of four fabric CCCs. For instance, GB0 can be accessed from GL0 of one of the four fabric CCCs present on the west side.

The Libero SoC place-and-route software performs the global buffer assignment based on the fabric CCC global outputs (GL0 - GL3) selected in the design.

Table 3 • **Global Buffers Assignment to Fabric CCC Global Outputs in M2S025/M2GL025, M2S050/M2GL050, M2S060/M2GL060, M2S090/M2GL090, and M2S150 Devices**

CCC-SW0	CCC-SW1	CCC-NW0	CCC-NW1	CCC-SE0	CCC-SE1	CCC-NE0	CCC-NE1	GB Number
GL0								0
	GL0							
		GL0						
			GL0					
GL1								1
	GL1							
		GL1						
			GL1					
GL2								2
	GL2							
		GL2						
			GL2					
GL3								3
	GL3							
		GL3						
			GL3					
GL0								4
	GL0							
		GL0						
			GL0					
GL1								5
	GL1							
		GL1						
			GL1					
GL2								6
	GL2							
		GL2						
			GL2					
GL3								7
	GL3							
		GL3						
			GL3					

Table 3 • **Global Buffers Assignment to Fabric CCC Global Outputs in M2S025/M2GL025, M2S050/M2GL050, M2S060/M2GL060, M2S090/M2GL090, and M2S150 Devices (continued)**

CCC-SW0	CCC-SW1	CCC-NW0	CCC-NW1	CCC-SE0	CCC-SE1	CCC-NE0	CCC-NE1	GB Number
				GL0				8
					GL0			
						GL0		
							GL0	
				GL1				9
					GL1			
						GL1		
							GL1	
				GL2				10
					GL2			
						GL2		
							GL2	
				GL3				11
					GL3			
						GL3		
							GL3	
				GL0				12
					GL0			
						GL0		
							GL0	
				GL1				13
					GL1			
						GL1		
							GL1	
				GL2				14
					GL2			
						GL2		
							GL2	
				GL3				15
					GL3			
						GL3		
							GL3	

The following table shows the assignment of fabric CCC global outputs (GLx, x = 0 to 3) to the global buffers in the 1VS8 global network architecture. There is a dedicated global buffer available for each global output of the fabric CCCs.

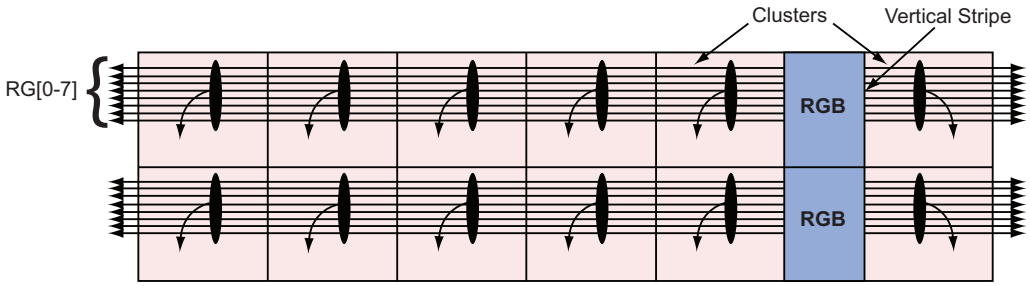
Table 4 • **Global Buffers Assignment to Fabric CCC Global Outputs in M2S005/M2GL005 and M2S010/M2GL010 Devices**

CCC- NE1	CCC- NE0	GB Number
GL0		0
GL1		1
GL2		2
GL3		3
	GL0	4
	GL1	5
	GL2	6
	GL3	7

3.1.5 Row Global Buffers

RGBs are situated on the vertical stripes of the global network architecture inside the FPGA fabric. As shown in the following figure, each RGB drives logic clusters, consisting of 12 logic modules, located on left and right branches using Row Globals (RGs). Each cluster consists of 12 logic modules. The global signals from the GBs are routed to RGBs, which are then fed into the clusters through RGs. Each GB has access to all RGBs available on the vertical stripes since the global network is segmented. Each RGB is independent and can be driven by fabric routing in addition to being driven by GBs. This facilitates the use of RGBs to drive regional clocks spanning a small fabric area.

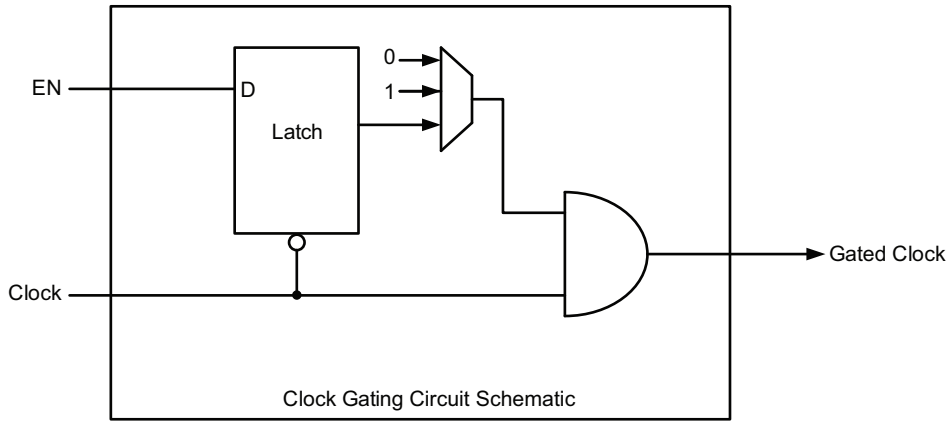
Figure 12 • **Row Global Signals Driving Clusters**



3.1.6 Clock Gating

The global network has clock gating capability built in to the GBs and RGBs to save power. The clock gating can be enabled by instantiating a global clock buffer macro (GCLKBUF, GCLKBIBUF, GCLKINT, or RGCLKINT) that has a clock input, enable input, and gated clock output or through the advanced tab on the CCC configurator. The following figure shows the clock gating circuit schematic. The place-and-route software maps the instantiated clock buffers to appropriate GBs or RGBs and enables the clock gating capability of the GBs or RGBs. See the Global Macros, page 22 for more information on clock buffer macros available in the IP Catalog of the Libero SoC software.

Figure 13 • **Clock Gating Circuit Schematic**

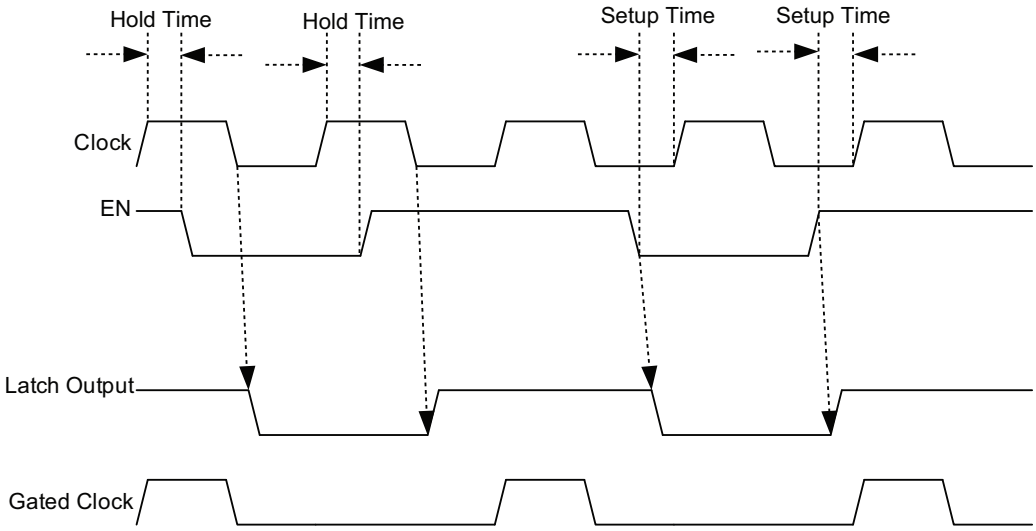


Clock gating is achieved using a latch and enable (EN), which is driven by the user logic implemented in the FPGA fabric. The latch is transparent when the clock input is in Low phase. The latch is in Hold state when the clock is in High phase. The AND gate at the output allows enabling or disabling of the clock based on the latch output.

- When the EN signal is High, clock is active.
- When the EN signal is Low, clock is gated off and drives Low.

The following figure shows the timing waveforms for the clock gating enabled clock buffers. See *DS0128: SmartFusion2 and IGLOO2 Datasheet* for the minimum setup and hold time for clock gating enable signal.

Figure 14 • **Timing Waveforms for the Clock Gating Circuitry**



- If the EN signal changes during the clock High phase and the minimum hold time is met with respect to the prior rising clock edge, the latch output changes after the falling clock edge.
- If the EN signal changes during the clock Low phase and the minimum setup time is met with respect to the next rising clock edge, the latch output will change immediately.
- If the EN signal violates either the setup or hold time with respect to the rising clock edge, the output behavior is unknown.

The CCCs can also produce gated clock output, which can then be fed into the global routing network. Unused global resources such as RGBs and GBs are tied-off automatically to reduce dynamic power consumption.

3.2 Design Recommendations

Microchip recommends that all clocks in a design must be routed using the global routing resources in the device. This ensures that clock skew is minimized throughout the design. Furthermore, global routing resource usage ensures that predictable levels of clock jitter can be accounted for during static timing analysis. This guideline must be applied while selecting clock input pin assignments, selecting CCC reference clock assignments, and generating clocks inside the FPGA. The following sections provide recommendations for using the global network in a design.

3.2.1 Global Macros

Global macros can be used for assigning signals to the global network. The CLKBUF, CLKBIBUF, GCLKBUF, and GCLKBIBUF macros allow you to route the clock coming from the dedicated global I/Os to GBs. Selecting the dedicated global I/Os, which are having direct connection to the GBs provide less delay. Dedicated global I/Os, which have direct access to GBs or through VCCCs are available as input pads to these macros. The CLKINT and GCLKINT macros route the internal clock signals or regular I/O signals to GBs through FPGA fabric routing. The RCLKINT and RGCLKINT macros route the internal clock signals to RGBs through FPGA fabric routing. The gated clock buffer macros (GCLKBUF, GCLKBIBUF, GCLKINT, and RGCLKINT) have an FPGA fabric routed control input (EN) to gate off the global clock.

In addition to these global macros, the CCC macro's global outputs (GLx) also drive the GBs. Dedicated global I/Os and their I/O standard can be assigned using I/O Attribute Editor, part of the Libero SoC software. The Libero place-and-route software runs the pre-layout checker and checks the validity of dedicated global I/O assignment.

The following table lists the global macros available for SmartFusion 2 and IGLOO 2 devices.

Table 5 • **Global Macros**

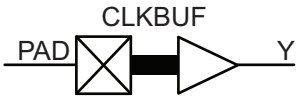
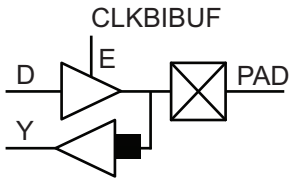
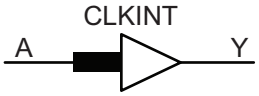
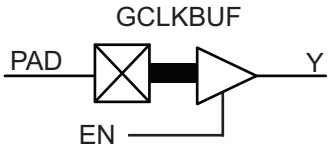
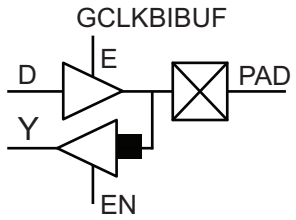
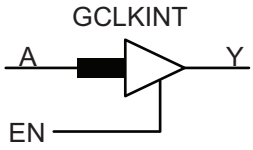
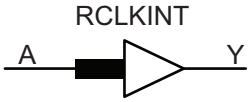
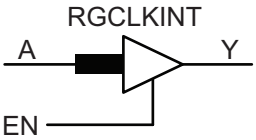
Macro Name	Description	Functional Symbol
CLKBUF	Macro to drive a clock signal coming from input pad to a global buffer	
CLKBIBUF	Macro to drive a clock signal coming from bidirectional I/O pad to a global buffer	
CLKINT	Macro to drive a global buffer from FPGA fabric routing	
GCLKBUF	Input macro with control input (EN) to drive or gate off the input clock to a chip-level global net. The clock signal comes from input pad.	

Table 5 • **Global Macros (continued)**

Macro Name	Description	Functional Symbol
GCLKBIBUF	Bidirectional macro with control input (EN) to drive or gate off the input clock to a chip-level global net. The clock signal comes from bidirectional I/O pad.	
GCLKINT	Macro with control input (EN) to drive or gate off the input clock to a chip-level global net; input clock is routed through FPGA fabric.	
RCLKINT	Macro to drive a row global net from FPGA fabric.	
RGCLKINT	Macro with control input (EN) to drive or gate off the input clock to a row global net; input clock is routed through FPGA fabric.	

3.2.2 Managing Global Signals

Assigning high fan-out nets to the global clock network is an effective way of reducing routing congestion and minimizing skew. Due to its high propagation delays, the global clock network is not recommended for use in timing-critical data paths.

The clock macros can be used for assigning signals to the global clock network:

- The CLKBUF and CLKBIBUF macro connects a dedicated global I/O to GB. Dedicated global I/Os have direct hardwired routing to GBs.
- The CLKINT macro connects fabric routed signal to GB. The CLKINT macro must be used to connect a regular I/O to GB through the FPGA fabric.
- The RCLKINT macro connects a fabric routed signal to RGB.

The CCCs and transceivers drive GBs through hardwired routing.

The Libero SoC software supports automated global buffer allocation to minimize the user intervention. The allocation strategy for global buffers employs the following priority:

- User inserted global macros
- Very high fan-out nets
- Clock nets
- Asynchronous reset/set nets

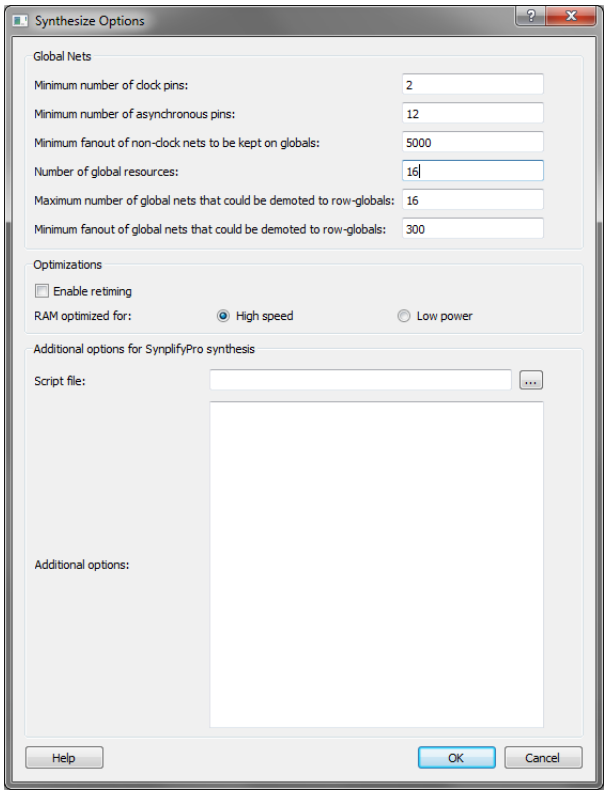
In Libero tool, the default fan-out threshold for global net promotion is larger for data pins (pins involved in register-to-register paths) than asynchronous logic pins (pins involved in register-to-asynchronous paths).

Due to this, the automated design flow is more likely to employ global nets on register-to-asynchronous paths than register-to-register paths. The reasoning for this is that asynchronous pins are not normally timing-critical, and routing them on global nets reduce routing congestion. However, register-to asynchronous paths are functionally equivalent to register-to-register paths from the perspective of achieving timing closure. As a result, when designing register-to-asynchronous paths, ensure that timing critical connections do not unnecessarily employ global nets.

If a design contains a failing register-to-asynchronous timing path, check if the path drives a global net in SmartTime. This is done by examining the path and looking for a GB between its launching and latching registers. If a GB is present, you may be able to improve the likelihood of timing closure by demoting the net to a fabric-routed net. An asynchronous net can be demoted by increasing the fan-out threshold of asynchronous pins above the fan-out of the asynchronous net. Alternatively, you can manually adjust the RTL by moving timing-critical pins from high-fanout asynchronous nets to lower fan-out nets.

Users have the option of setting the minimum fan-out for automatic global assignment. The fan-out threshold values are set in the Libero tool to automate clock pin promotion to global nets. In the Libero Design Flow window, expand Implement Design, right-click Synthesize, and choose Configure Options. This opens the Synthesize Options dialog box, as shown in the following figure.

Figure 15 • **Synthesize Options Dialog Box**



The following options specify the threshold value for nets promotion or demotion:

- **Minimum number of clock pins** – Specifies the fan-out threshold value for clock pin promotion. The default value is 2.
- **Minimum number of asynchronous pins** – Specifies the fan-out threshold value for Asynchronous pin promotion. The default value is 12.
- **Minimum fan-out of non-clock nets to be kept on globals** – Specifies the fan-out threshold value for data pin promotion to global resources. It is the minimum fan-out of non-clock (data) nets to be kept on global nets (no demotion). The default value is 5,000 (must be between 1,000 and 200,000). If you run out of global resources for your design, increase this number. If a CLKINT net with fan-out less than this threshold value has data pins along with some clock or asynchronous reset/set pin, move all the data pins to the CLKINT driver net.
- **Number of global resources** – Specifies the number of global resources to be used in the design. The default value is 16.
- **Maximum number of global nets that could be demoted to row-globals** – Specifies the maximum number of global nets that could be demoted to RGB resources. The default value is 16.
- **Minimum fan-out of global nets that could be demoted to row-globals** – Specifies the minimum fan-out of global nets that could be demoted to RGB resources. The default value is 300. It is

undesirable to have high fan-out clock nets demoted using RGB resources because it may result in high skew. If you run out of global resources for your design, reduce this number to allow more globals to be demoted to RGB resources.

Note: Hardwired connections to global resources, such as connections from CCCs and Dedicated Global I/Os cannot be controlled by synthesize options.

After synthesis, the compiler tool performs the following steps to assign nets to global buffers:

1. Sorting all CLKINT nets in the following priority order.
   - Fan-out, only if fan-out threshold value specified by minimum fanout of non-clock nets to be kept on globals
   - Number of clock pins
   - Number of asynchronous reset/set pins
   - Number of data pins
2. Determining the number of GB resources available for CLKINT nets after allocating them to CLKBUF and CLKBIBUF.
3. Demoting CLKINT nets from the sorted list that are beyond the limit specified by the number of global resources.
   - If such a net has at least the number of pins specified by minimum fanout of global nets that could be demoted to row-globals, replace the CLKINT with an RCLKINT macro. Limit the number of nets demoted to RCLKINT to the count specified by maximum number of global nets that could be demoted to row-globals.
   - Otherwise, merge the net with the driver of the CLKINT.

The HDL source file or SmartDesign schematic is the preferred place for defining which signals must be assigned to a global network using global macro instantiation. A signal with high-fanout may have logic replication, if it is not promoted to a global during synthesis.

3.2.3 Place and Route

Place and Route step aligns placement of I/Os, CCC, and GB resources while preserving any user locked placement, finding feasible I/O bank technology solution and optimizing the number of hardwired connections to reduce clock insertion delays. An I/O may be hardwired to a GB as well as route directly to data pins in the fabric.

The RCLKINT/RGCLKINT nets are mapped to RGB resources with routed inputs and the placement algorithm resolves the RGB bandwidth per row. The algorithm contains a cost structure to penalize over subscription of RGB resources per row beyond the limit of 8. Increasing the number of globals and local clock nets therefore adversely impact the runtime.

The power-driven option optimizes the number of rows occupied by a global net or a local clock net, resulting drop in performance (on average).

After performing the place and route, a report of global nets is generated. User can open the report tab to view the global net report. The user must review the generated global net report to understand and minimize clocks that are routed on FPGA fabric routing instead of global routing resources. The following figures show an example of global net report.

Figure 16 • **Example of Global Net Report 1**

Global Nets Information

	From	GB Location	Net Name	Fanout
1	GB[1]	(435, 108)	CLK_18_ibuf_RNI7605/U0_YWn	2266
2	GB[4]	(438, 108)	RESETN_ibuf_RNIM8N5/U0_YWn	2189
3	GB[12]	(450, 108)	CLK_12_ibuf_RNI1837/U0_Ywn	623

I/O to GB Connections

	Port Name	Pin #	I/O Function	From	From Location	To
1	CLK_18	R1	MSIOD119PB7/GB1	CLK_18_ibuf/U0/U_IOIN:Y	West IO #2 (3, 100)	GB[1]
2	RESETN	B9	DDRIO91NB0/GB4	RESETN_ibuf/U0/U_IOIN:Y	North IO #2(285, 205)	GB[4]
3	CLK_12	A17	DDRIO76PB0/GB12	CLK_12_ibuf/U0/U_IOPAD:Y	North IO #6(564, 205)	GB[12]

Fabric to GB Connections

	From	From Location	To	Net name	Net type	Fanout
1	I6/I0/sys_rst_bar:Q	(413, 190)	GB[3]	I6/I0/sys_rst_bar	ROUTED	1
2	I7/I0/I4/vcxclk_1_5m:Q	(361, 61)	GB[11]	I7/I0/I4/vcxclk_1_5m	ROUTED	2
3	I7/I0/I4/clk_3_088m:Q	(368, 61)	GB[7]	I7/I0/I4/clk_3_088m_0	ROUTED	3

Figure 17 • **Example of Global Net Report 2**

CCC to GB Connections

	From	From Location	To	Net Name	Net Type	Fanout
1	M2S090_EVAL_KIT_MTD_0/CCC_0/CCC_INST/INST_CCC_IP:GL0	CCC-NW0 (0, 254)	GB[7]	M2S090_EVAL_KIT_MTD_0/CCC_0/GL0_net	HARDWIRED	1
2	FCCC_2/CCC_INST/INST_CCC_IP:GL0	CCC-NW1 (18, 254)	GB[0]	FCCC_2/GL0_net	HARDWIRED	1
3	FCCC_2/CCC_INST/INST_CCC_IP:GL1	CCC-NW1 (18, 254)	GB[3]	FCCC_2/GL1_net	HARDWIRED	1

CCC Input Connections

	From	From Location	To	CCC Location	Net Name	Net Type	Fanout
1	SERDES_IF2_0/SERDESIF_INST/INST_SERDESIF_IP:EPCS_RXCLK[1]	(12, 2)	FCCC_2/CCC_INST/INST_CCC_IP:CLK0	CCC-NW1 (18, 254)	SERDES_IF2_0_EPCS_3_RX_CLK	ROUTED	1
2	SERDES_IF2_0/SERDESIF_INST/INST_SERDESIF_IP:EPCS_RXCLK_1	(12, 2)	FCCC_0/CCC_INST/INST_CCC_IP:CLK0	CCC-NE1 (1014, 254)	SERDES_IF2_0_EPCS_1_RX_CLK	ROUTED	1

Figure 18 • **Example of Global Net Report 3**

Local Clock Nets to RGB Connections

	Port Name	Pin Number	I/O Function	From	From Location	Net Name	Fanout	RGB Location	Local Fanout
1	RSTN	F19	MSIO24PB1/MMUART_0_CTS/GPIO_19_B/USB_DATA7_C	RSTN_ibuf/U0/U_IOIN:Y	(270, 103)	RSTN_ibuf	9	(218, 15)	3
								(218, 18)	6
2	-	-	-	SYSCLK/PROC_IF/reset_ctl_1/reset_bits_RNIPLVN[0]:Y	(229, 36)	SYSCLK/PROC_IF/reset_ctl_1/reset_bits_RNIPLVN[0]	37	(219, 3)	2
								(219, 6)	4
								(219, 9)	31

4 On-Chip Oscillators

SmartFusion 2 and IGLOO 2 devices have the following on-chip oscillators for generating free-running clocks:

- 1 MHz RC Oscillator
- 50 MHz RC Oscillator
- Main Crystal Oscillator
- Auxiliary Crystal Oscillator

Figure 19, page 28 and Figure 20, page 29 show clock sourcing capabilities of the SmartFusion 2 and IGLOO 2 on-chip oscillators. On-chip oscillators can provide reference clock input to the following resources:

- Fabric CCCs
- User logic in the FPGA fabric through fabric routing
- MSS RTC (available only in the SmartFusion 2 device)
- MSS and HPMS during Flash*Freeze mode

The auxiliary crystal oscillator is available only in SmartFusion 2 devices and is dedicated for MSS RTC clocking as an alternative clock source. SmartFusion 2 M2S050 devices do not have an auxiliary crystal oscillator.

The on-chip oscillators also supply clocks to hard IP blocks such as the system controller and POR circuitry. This chapter describes the on-chip oscillator's clock sourcing capabilities and use models. See the "On-Chip Oscillators" section in the *DS0128: SmartFusion2 and IGLOO2 Datasheet* for the electrical characteristics of on-chip oscillators.

Figure 19 • **SmartFusion 2 On-Chip Oscillators Clock Sourcing Capabilities**

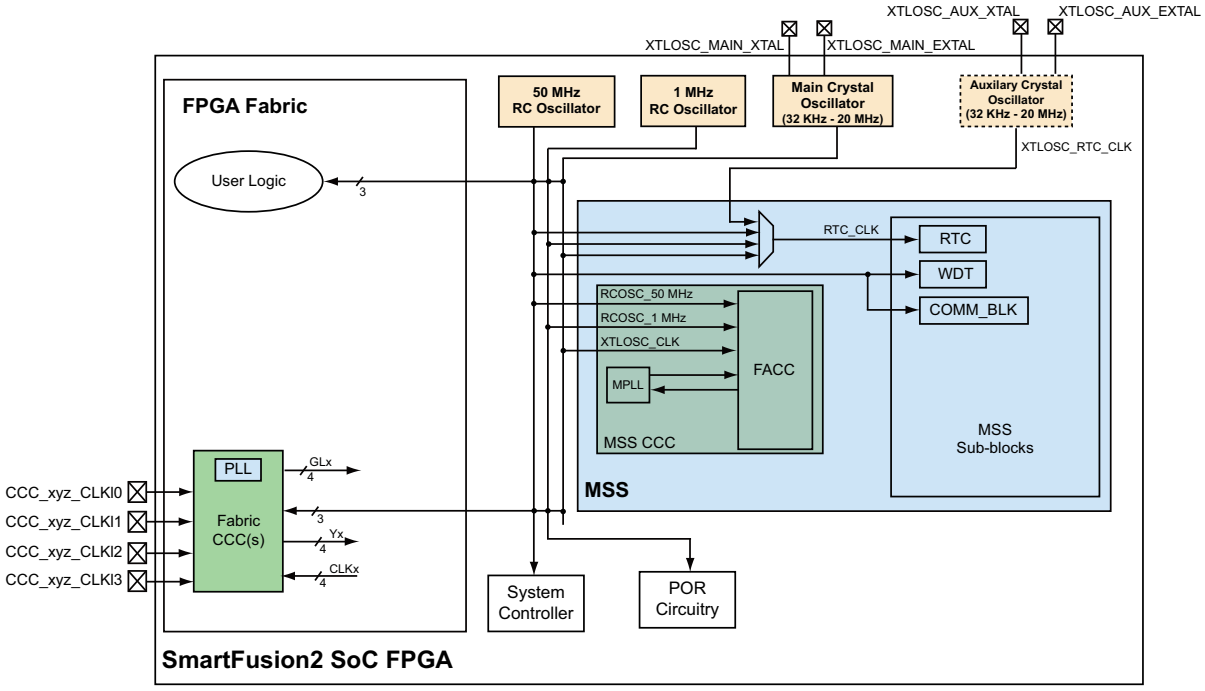
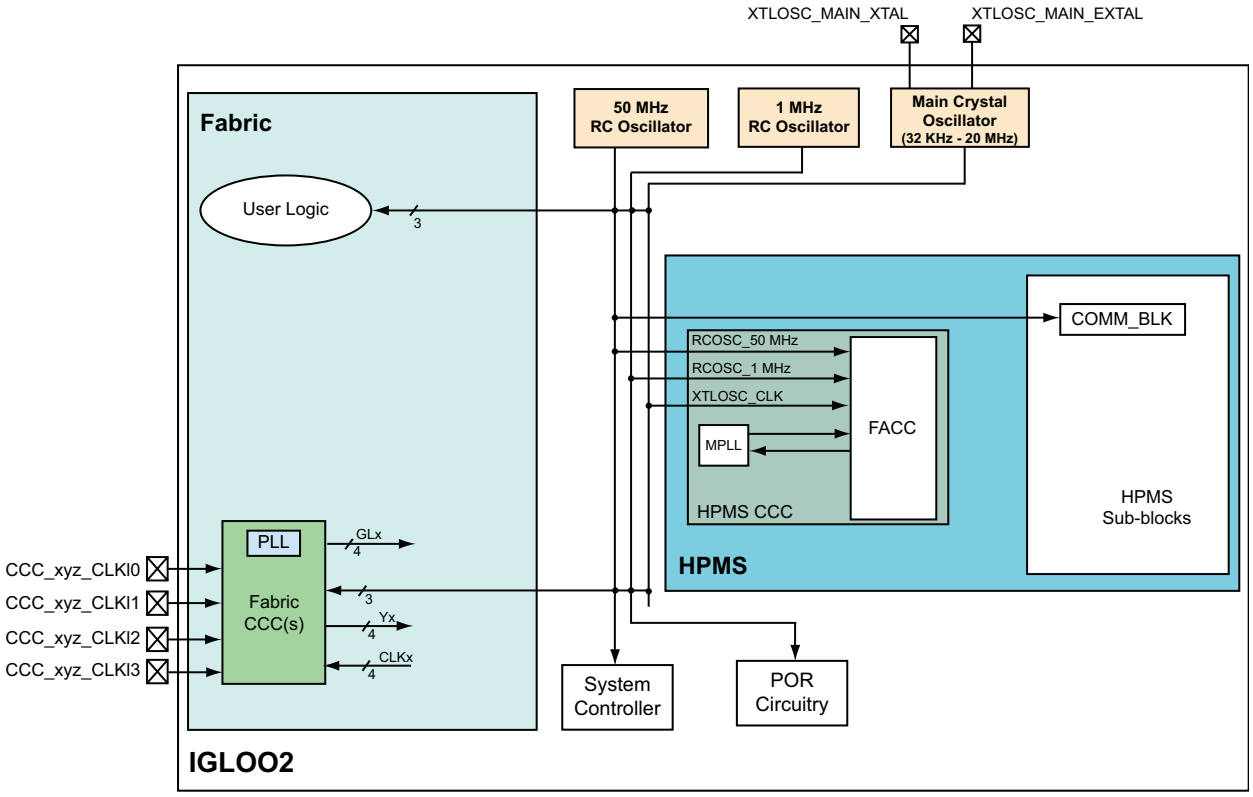


Figure 20 • **IGLOO 2 On-Chip Oscillators Clock Sourcing Capabilities**



4.1 Functional Description

The outputs of on-chip oscillators are connected to the fabric CCCs, MSS/HPMS CCC, and hard IP blocks through dedicated hardwired connections (not routed through FPGA fabric). The on-chip oscillators can be configured by Chip Oscillators macro in the Libero SoC software to drive user logic and/or fabric CCCs. Unused on-chip oscillators are automatically disabled by the Libero SoC design software if neither of their outputs are enabled. The oscillator is turned off synchronously without generating runt clock pulses.

4.1.1 1 MHz RC Oscillator

The 1 MHz RC oscillator generates a nominal 1 MHz digital clock signal. It is powered by the device core supply VDD and does not require external components for operation.

The 1 MHz RC oscillator can be configured as a clock source to the following:

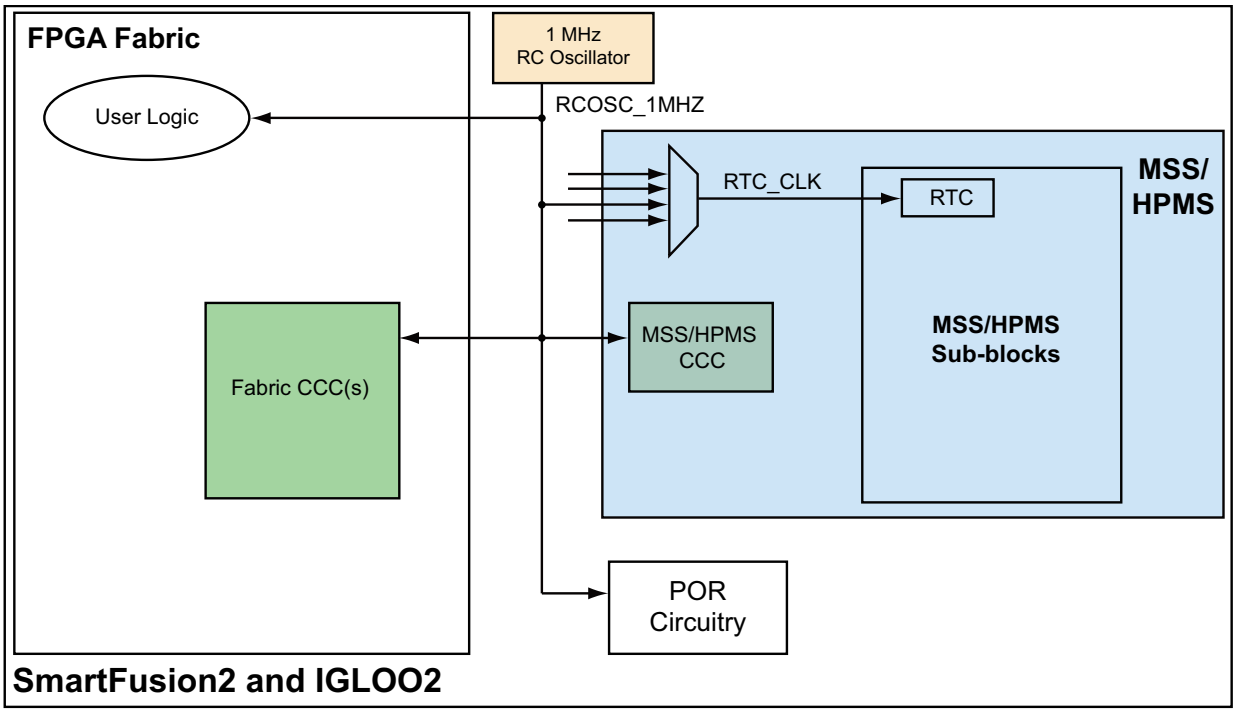
- Fabric CCCs
- User logic in the FPGA fabric through fabric routing
- MSS RTC (SmartFusion 2 only)
- MSS and HPMS during Flash*Freeze mode

During device power-up, it provides the clock to the POR circuitry through a dedicated connection to determine the duration of POR signal assertion.

Note: 1 MHz Oscillator should not be used as a reference input source to CCC. It is recommended to use the 50 MHz RC Oscillator as a reference input to CCC.

The following figure shows the 1 MHz RC oscillator clock sourcing capabilities.

Figure 21 • **1 MHz RC Oscillator Clock Sourcing Capabilities**



Note: Real-time clock (RTC) is available only in the SmartFusion2 device.

4.1.2 50 MHz RC Oscillator

The 50 MHz RC oscillator generates a nominal 50 MHz digital clock signal. It is powered by the device core supply VDD and does not require external components for operation.

The 50 MHz RC oscillator can be configured as a clock source to the following:

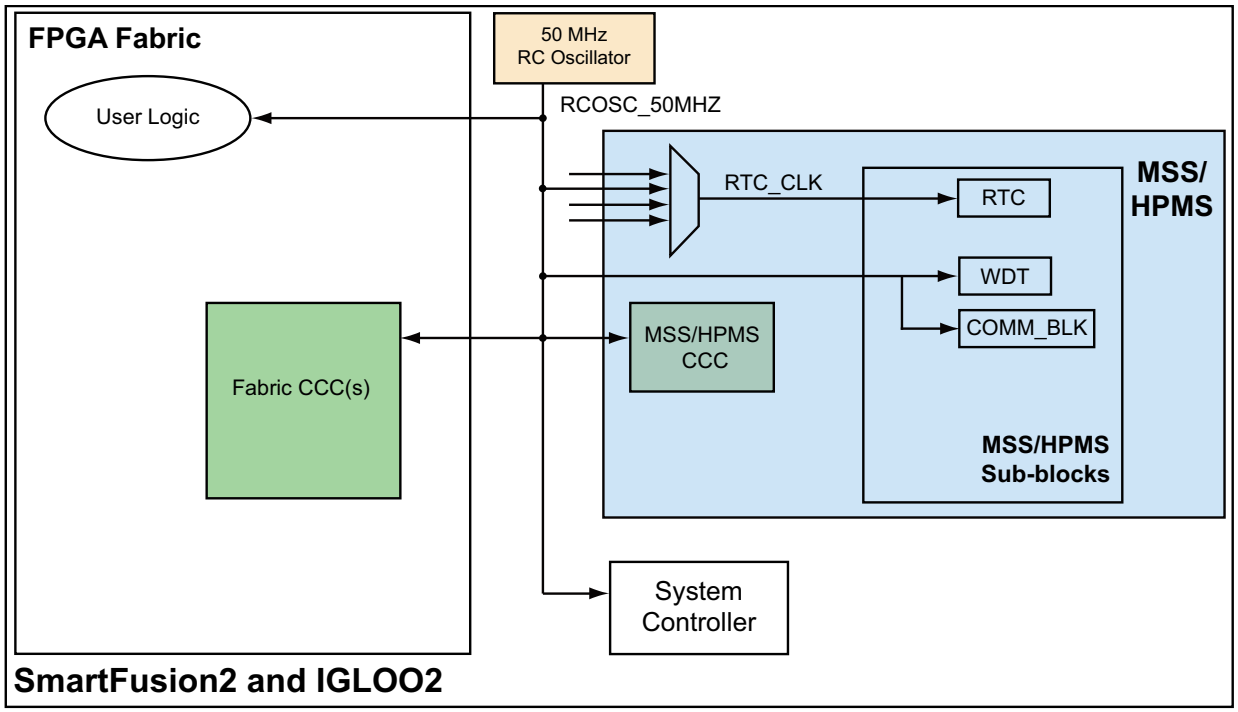
- Fabric CCCs
- User logic in the FPGA fabric through fabric routing
- MSS RTC (SmartFusion 2 only)
- MSS and HPMS during Flash*Freeze mode

The 50 MHz RC oscillator also supplies a clock through dedicated connections to the following hard IP blocks:

- System controller
- MSS and HPMS after power-on reset for system initialization
- MSS Watchdog 32-bit counter (SmartFusion 2 only)
- Communication block (COMM_BLK) core logic

The following figure shows the 50 MHz RC oscillator clock sourcing capabilities.

Figure 22 • **50 MHz RC Oscillator Clock Sourcing Capabilities**



Note: RTC and Watchdog counter are available only in the SmartFusion2 device.

4.1.3 Main Crystal Oscillator

The main crystal oscillator works with an external crystal, ceramic resonator, or a resistor-capacitor (RC) network to generate a high-precision clock in the range of 32 KHz to 20 MHz. The main crystal oscillator has two I/O pads, XTLOSC_MAIN_EXTAL and XTLOSC_MAIN_XTAL, which can be connected to a crystal, a ceramic resonator, or an RC circuit. If the main crystal oscillator is not used, the XTLOSC_MAIN_EXTAL and XTLOSC_MAIN_XTAL pins can be left floating. The following table shows the output frequency range of the main crystal oscillator with different possible sources.

Table 6 • **Main Crystal Oscillator Output Frequency Range**

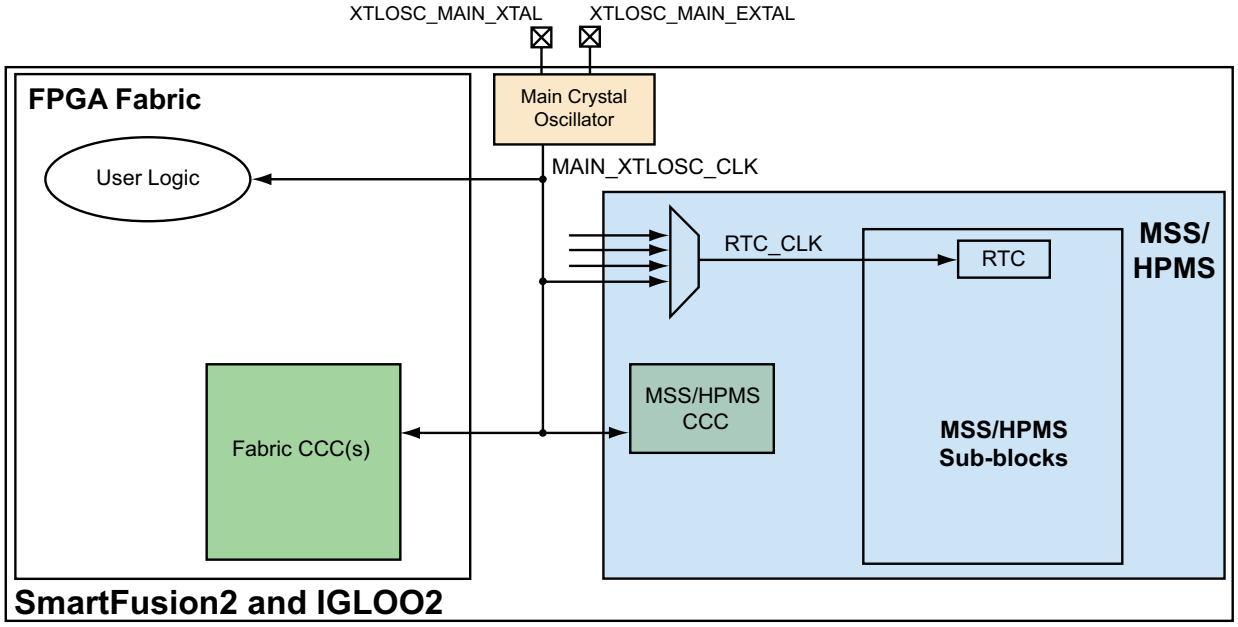
Source	Output Frequency Range
Crystal	32 KHz to 20 MHz
RC Circuit	32 KHz to 4 MHz
Ceramic Resonator	500 KHz to 4 MHz

The main crystal oscillator can be configured as a clock source to the following:

- Fabric CCCs
- User logic in the FPGA fabric through fabric routing
- MSS RTC (SmartFusion 2 only)
  - To generate a real-time clock using the MSS RTC, Microchip recommends using the main crystal oscillator or auxiliary crystal oscillator (if present) with an external crystal as the MSS RTC clock source. RTC is available only in the SmartFusion 2 device.

The following figure shows the main crystal oscillator clock sourcing capabilities.

Figure 23 • **Main Crystal Oscillator Clock Sourcing Capabilities**



Note: RTC is available only in the SmartFusion2 device.

The main crystal oscillator supports the four modes of operation as defined in the following table. The operating mode of the main crystal oscillator is automatically selected by the Chip Oscillators macro available in the IP Catalog of the Libero SoC design software based on the *oscillator* source and output frequency.

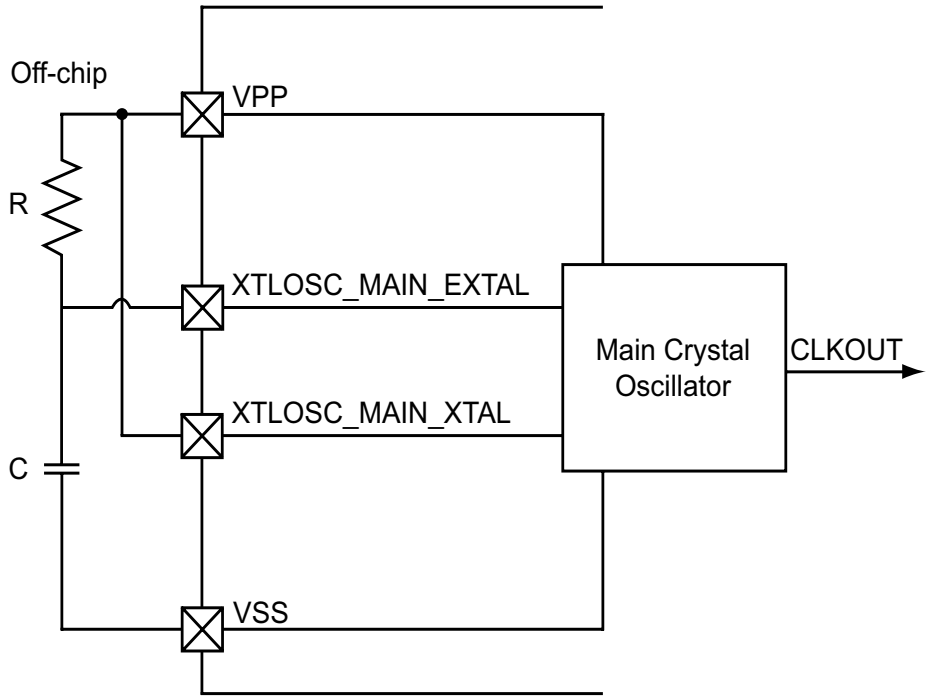
Table 7 • **Main Crystal Oscillator Operational Modes**

Oscillator Mode	Frequency Range	Oscillator Source	Description
RC network	32 KHz to 4 MHz	RC network	RC network mode. R and C are connected at XTLOSC_MAIN_EXTAL pad.
Low gain	32 KHz to 75 KHz	Crystal	Low power or Low frequency mode. Oscillator consumes the least current of the three modes with a crystal connected to I/O pads.
Medium gain	75 KHz to 2 MHz	Crystal or ceramic resonator	Medium power or Medium frequency mode of the three modes with a crystal or ceramic resonator connected to I/O pads.
High gain	2 MHz to 20 MHz	Crystal	High power or High frequency mode. Consumes the maximum current of the three modes with a crystal connected to I/O pads.

4.1.3.1 RC Network Mode

The frequency generated by the main crystal oscillator in RC network mode is determined by the RC time constant of the selected R and C components. The R and C components are connected to the XTLOSC_MAIN_EXTAL pin, with the XTLOSC_MAIN_XTAL pin tied to the power pin, VPP, as shown in the following figure.

Figure 24 • **Main Crystal Oscillator in RC Network Mode**

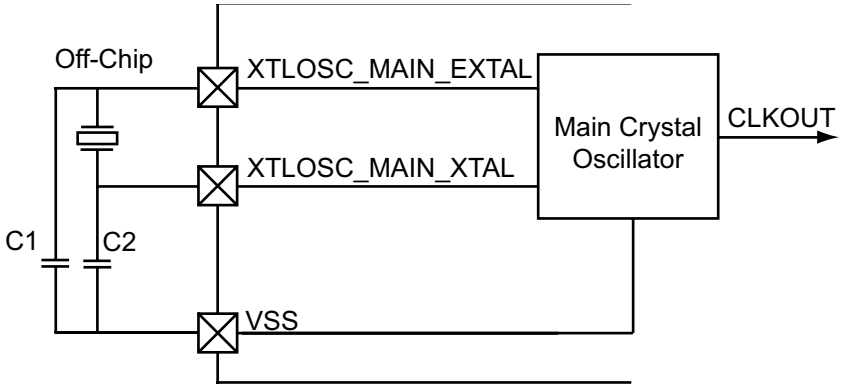


4.1.3.2 Crystal/Ceramic Resonator Modes

In crystal/ceramic resonator modes, the main crystal oscillator is configured to support an external crystal or ceramic resonator connected between the XTLOSC_MAIN_EXTAL and XTLOSC_MAIN_XTAL pins. The main crystal oscillator can operate in Low Gain, Medium Gain, or High Gain mode when a crystal is connected between the XTLOSC_MAIN_EXTAL and XTLOSC_MAIN_XTAL pins. The main crystal oscillator operates in Medium Gain mode when a ceramic resonator is connected between the XTLOSC_MAIN_EXTAL and XTLOSC_MAIN_XTAL pins. Additionally, a capacitor is required on both XTLOSC_MAIN_EXTAL and XTLOSC_MAIN_XTAL pins to ground, as shown in the following figure. These capacitance values are specific to crystal/ceramic and are vendor specific. Most of the commercially available crystals/ceramic resonators operate in parallel resonant frequency mode. In such cases, the vendor also specifies a load capacitance value which must be connected at XTLOSC_MAIN_EXTAL and XTLOSC_MAIN_XTAL pins (C1 and C2).

The following figure shows a main crystal oscillator in ceramic or crystal resonator mode.

Figure 25 • **Main Crystal Oscillator in Ceramic Resonator or Crystal Mode**



Typically, designers choose the values of capacitors C_1 and C_2 to match the crystal C_L using the below equation:

$$C_L = \frac{C_1 \times C_2}{C_1 \times C_2} + C_S$$

C_L is the load capacitance provided in manufacturer datasheet.

C_S is stray capacitance on the PCB and this can be assumed to be in the range of 2 to 5 pF.

Usually C_1 and C_2 are selected such that they are equal.

Note: The preceding equation is only a guideline, and selection of Capacitors depends on design requirements like cost, availability, frequency accuracy, PPM, and type of application etc.

Large values of C_1 and C_2, increases the frequency stability but decrease the loop gain and may cause oscillator startup problems. The basic rule of thumb is “value of C_1 and C_2 is twice as C_L“. However, the values of C_1 and C_2 depends on design requirements.

Table 8 • **Example Crystal Oscillators**

Part Number	Manufacturer	Description
CM519-32.768KEZF-UT	Citizen	CRYSTAL 32.768 kHz 12.5 pF SMD
ECS-32.7-12.5-34B-C-TR	ECS Inc	
ABS10-32.768KHz-1-T	Abracon LLC	

See *DS0128: SmartFusion2 and IGLOO2 Datasheet* for more information on external component selection for the main crystal oscillator.

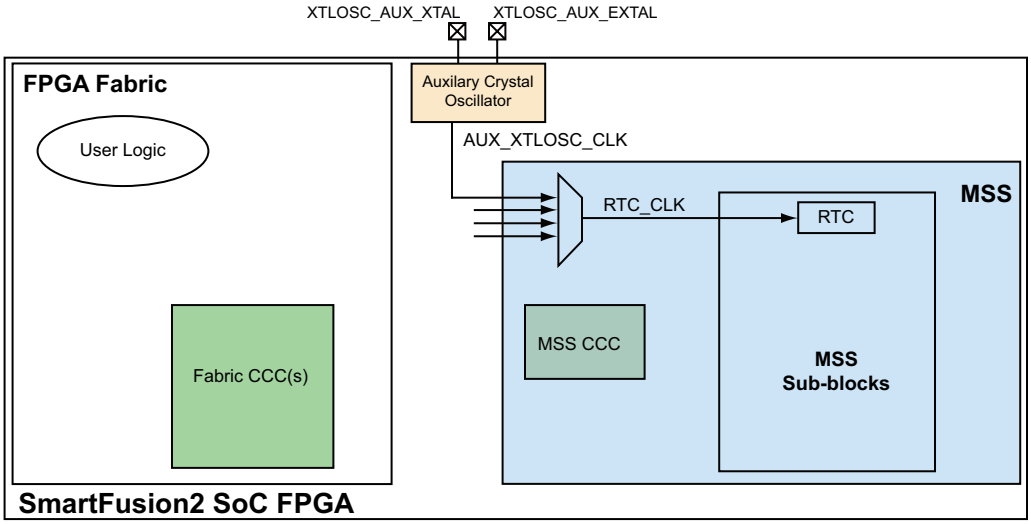
4.1.4 Auxiliary Crystal Oscillator

Auxiliary crystal oscillator is available only in the SmartFusion 2 device. SmartFusion 2 devices, except M2S050, have an auxiliary crystal oscillator dedicated to MSS RTC clocking as an alternative clock source. The MSS RTC generates a real-time clock by taking input from the auxiliary crystal oscillator when the main crystal oscillator is being used for another purpose.

Similar to the main crystal oscillator, the auxiliary crystal oscillator works with an external crystal, ceramic resonator, or an RC circuit to generate a high-precision clock in the range of 32 KHz to 20 MHz. The auxiliary crystal oscillator has two I/O pads, XTLOSC_AUX_EXTAL and XTLOSC_AUX_XTAL, which can be connected to a crystal, a ceramic resonator, or an RC circuit. If the main crystal oscillator is not used, the XTLOSC_MAIN_EXTAL and XTLOSC_MAIN_XTAL pins can be left floating. The output frequency range, operating modes, and characteristics of the auxiliary crystal oscillator are the same as those of the main crystal oscillator. For more information, see Table 6, page 31 and Table 7, page 32.

The following figure shows the auxiliary crystal oscillator clock sourcing capabilities.

Figure 26 • **Auxiliary Crystal Oscillator Clock Sourcing Capabilities in SmartFusion 2 Devices**



4.2 How to Use On-Chip Oscillators

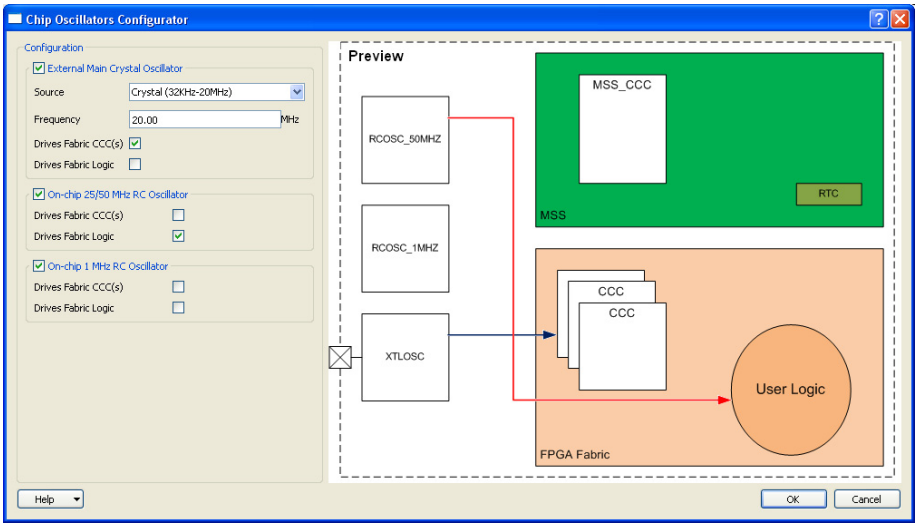
On-chip oscillators are configured statically using the Chip Oscillators macro available in the IP Catalog of the Libero SoC design software. Using the Chip Oscillators macro, oscillators can be configured to clock the fabric CCC(s) and user logic in the FPGA fabric through fabric routing. It is also possible to configure the operating mode of the main crystal oscillator using the Chip Oscillators macro. The Chip Oscillators macro must be instantiated in the design in order to configure the on-chip oscillators. The Chip Oscillators macro need not to be instantiated in the design for MSS Watchdog and RTC operation since they have dedicated hardwired connections from the oscillators.

The following figure shows the Chip Oscillators macro configurator. The Chip Oscillators macro configurator dialog box is organized as follows:

The Configuration pane displays all the configuration options.

- The Preview pane displays a high-level block diagram of how the current configuration relates to the various chip components driven by the various oscillators selected for the design.

Figure 27 • **Chip Oscillators Macro Configurator**

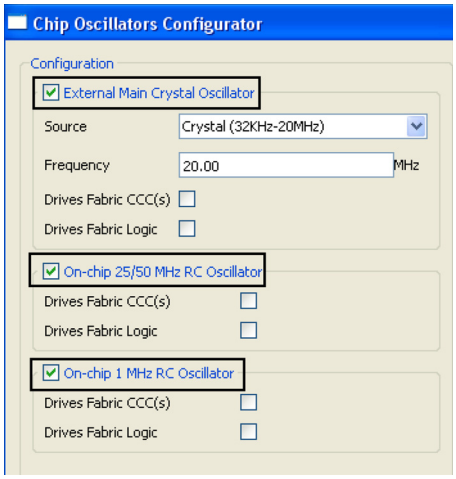


Note: MSS_CCC in the **Chip Oscillator Configurator** window actually applicable for both MSS CCC and HPMS CCC. This is a limitation of the Libero SoC software because different devices use the same core.

The following steps must be followed for each oscillator to use in the design:

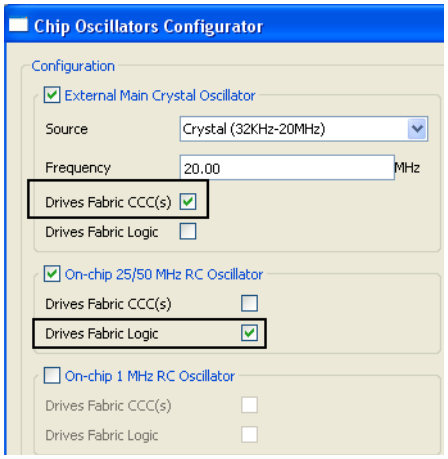
1. Select the oscillator(s) by selecting the appropriate check box.

Figure 28 • **Select the Oscillator(s)**



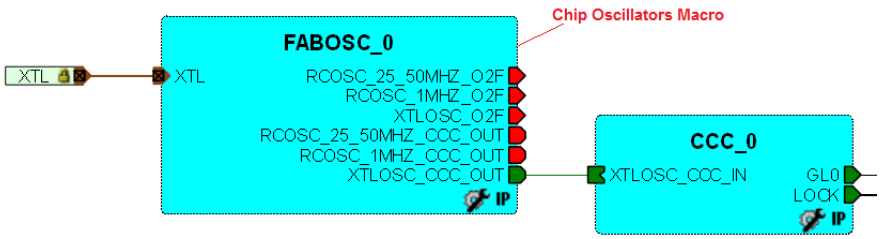
2. If **Main Crystal Oscillator** is selected, its source and frequency must be configured.
3. Select the resource type—Fabric CCC and/or Fabric Logic—to which the selected oscillator(s) will be connected.

Figure 29 • **Select the On-chip Oscillator(s) Resource Type**



For each type of oscillator configured, a port will be exposed on the instance of the Chip Oscillators macro. Each exposed port can only be connected in SmartDesign to a port that matches the type of connection (oscillator and connectivity type) implied by the configuration made in the Chip Oscillators macro. For instance, if a fabric CCC is configured to use the main crystal oscillator, the input port exposed on the CCC for the main crystal oscillator—XTLOSC_CCC_IN—can only be driven by the output port—XTLOSC_CCC_OUT—of the Chip Oscillators macro.

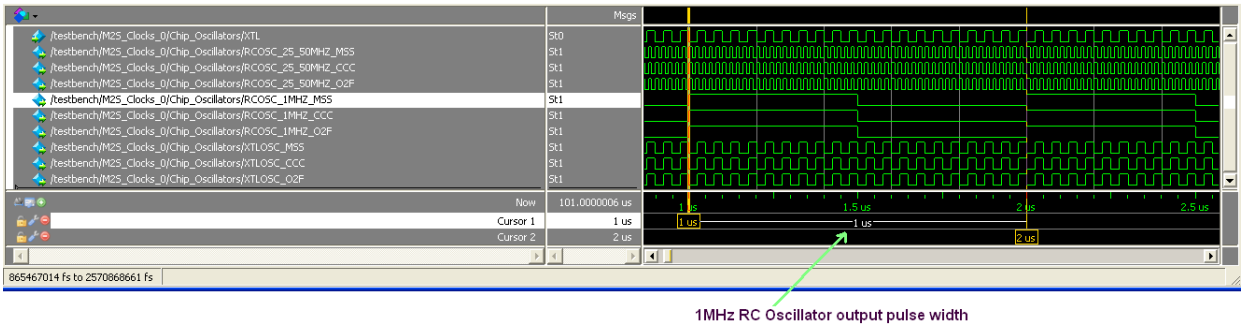
Figure 30 • **Main Crystal Oscillator and Fabric CCC Connectivity**



4.2.1 Simulation Support

Microchip Libero SoC design software provides pre-compiled simulation models for the Chip Oscillators macro to show functional behavior of on-chip oscillators. The simulation steps include generating the top-level component, which instantiates the Chip Oscillators macro, performing simulation for verification with the ModelSim tool, and performing static timing analysis with SmartTime in the Libero SoC software. The following figure shows the simulation results of the Chip Oscillator macro.

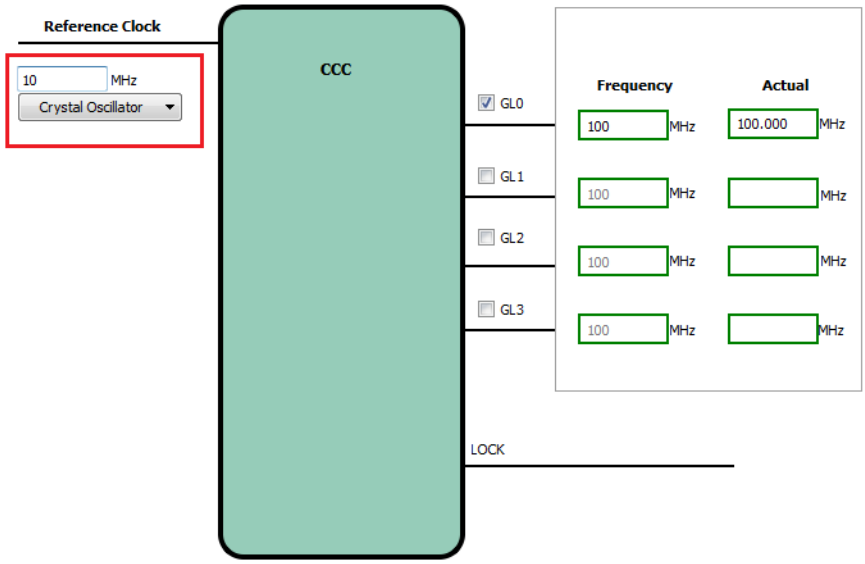
Figure 31 • **Chip Oscillators Simulation Results**



4.2.2 On-Chip Oscillator Driving Fabric CCC(s)

This section describes the on-chip oscillator as a clock source to the fabric CCC(s). This requires instantiation of both the Chip Oscillators and CCC macros in the design. To drive the fabric CCC reference clock input, the CCC macro must be configured to select a particular on-chip oscillator as the reference clock source, as shown in the following figure.

Figure 32 • **Fabric CCC Reference Clock Selection**

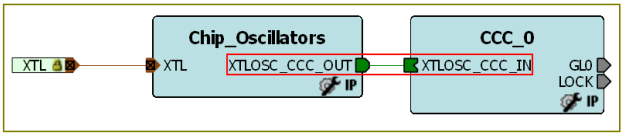


Note: During Libero project creation, PLL analog supply voltage must be configured to either 2.5 V or 3.3 V to match the supply provided on the board.

The same on-chip oscillator must be enabled in the Chip Oscillators macro and must select Fabric CCC as its resource type to connect each other. For instance, if a fabric CCC is configured to use the main crystal oscillator as clock source, the input port exposed on the CCC for the main crystal oscillator—XTLOSC_CCC_IN—can only be driven by the output port—XTLOSC_CCC_OUT—of the Chip Oscillators macro. Make the connection, as shown in the following figure.

The same main crystal oscillator can also source the reference clock to multiple fabric CCCs instantiated in the design.

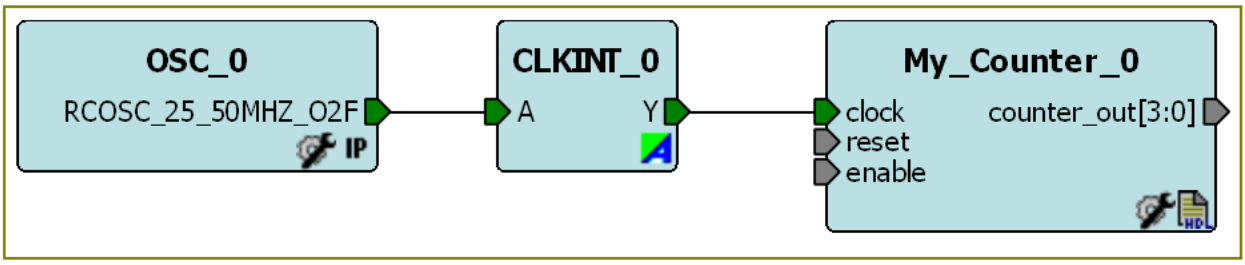
Figure 33 • **Main Crystal Oscillator Driving Fabric CCC**



4.2.3 On-Chip Oscillator Driving User Logic

This section describes the on-chip oscillator as a clock source to the user logic implemented in the FPGA fabric. This requires instantiation of the Chip Oscillators macro in the design. To drive user logic clock input directly, the Chip Oscillators macro must be configured to enable required on-chip oscillator(s) and must select Fabric Logic as the resource type. For instance, if you configure the Chip Oscillators macro to use the 50 MHz RC oscillator as a clock source for the user logic, the clock port exposed on the user logic can be driven by the output port—RCOSC_25_50MHz _O2F—of the Chip Oscillators macro, either directly or through the CLKINT macro. Microchip recommends connecting the selected on-chip oscillator's output through a CLKINT clock macro to route the clock over the global network. Make the connections, as shown in the following figure.

Figure 34 • **50 MHz RC Oscillator Driving User Logic**



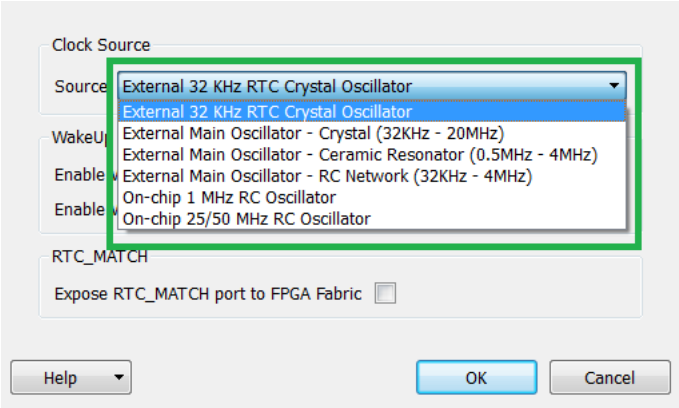
4.2.4 MSS RTC Clock Source Selection

This section describes the selection of the MSS RTC clock source from one of the on-chip oscillators. This requires instantiation of the MSS component in the design. The MSS_RTC instance in the MSS configurator must be configured to select the clock source for the MSS RTC, as shown in the following figure.

Note: MSS RTC is available only in the SmartFusion 2 device.

Note: To generate a real-time clock using the MSS RTC, Microchip recommends using either the main crystal oscillator or auxiliary crystal oscillator (if present) with an external crystal as the MSS RTC clock source. Irrespective of external main oscillator or RTC crystal oscillator selection, the auxiliary external oscillator is used as RTC clock source when an external oscillator is used to drive the user logic.

Figure 35 • **MSS RTC Configurator – MSS RTC Clock Source Selection**



5 Fabric Clock Conditioning Circuitry

SmartFusion 2 and IGLOO 2 devices have two, six, or eight fabric CCCs. Each fabric CCC enables flexible clocking schemes for the logic implemented in the FPGA fabric and can provide the base clock for on-chip hard IP blocks—MSS/HPMS, FDDR, and SERDESIF. Each fabric CCC operates with a dedicated PLL and can generate clock signals of different frequency and phase. The associated PLL can be bypassed if it is not required. This chapter describes the fabric CCC features, configuration, and use models.The following table lists the number of fabric CCCs available in SmartFusion 2 and IGLOO 2 devices.

Table 9 • **Number of Fabric CCCs in SmartFusion 2 and IGLOO 2 Devices**

Resource	SmartFusion® 2 and IGLOO® 2 Device Part Number						
	M2S005 and M2GL005	**M2S010 and M2GL010**	**M2SL025 and M2GL025**	**M2SL050 and M2GL050**	**M2SL060 and M2GL060**	**M2S090 and M2GL090**	**M2S150 and M2GL150**
Fabric CCCs[1]	2	2	6	6	6	6	8

1. Each fabric CCC has a dedicated PLL for clock synchronization and clock synthesis.

5.1 Features

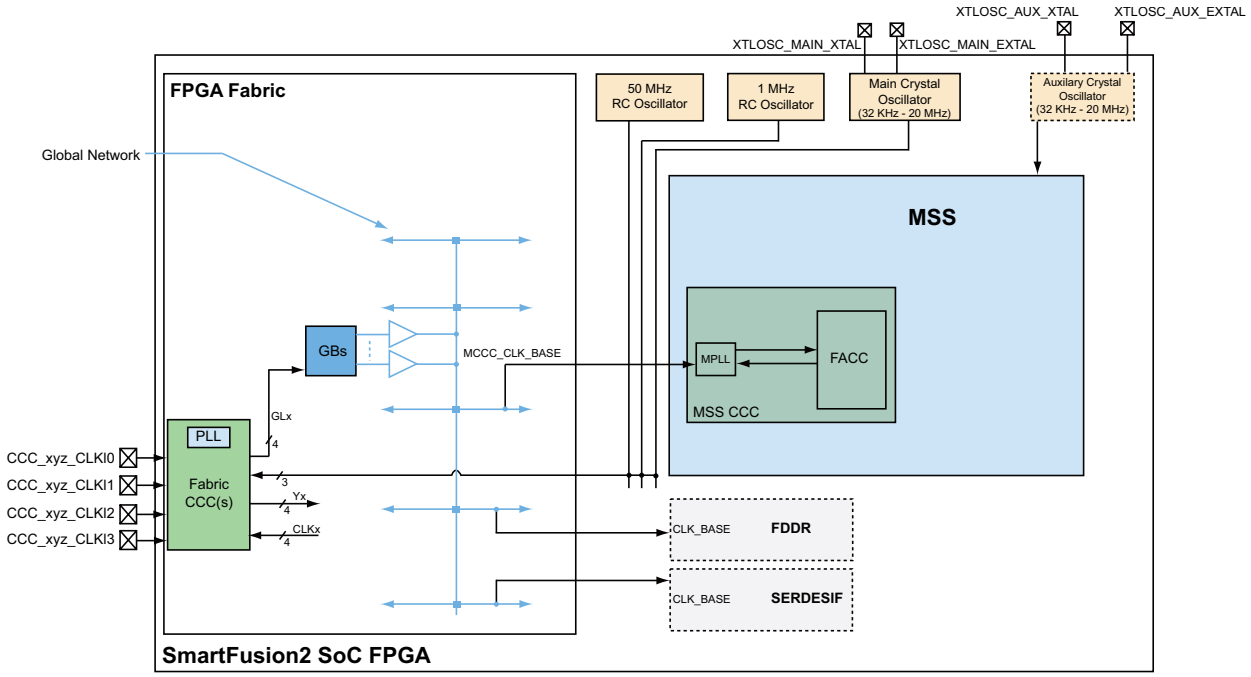
Each fabric CCC supports the following features:

- Internal and external clock sources
- Generation of four different clocks with a maximum frequency up to 400 MHz
  - Drives global network and/or the local routing network
- Automatic output resynchronization after PLL lock
- Configurable phase error window for PLL lock assertion
- Configurable PLL lock delay
- Spread spectrum clock generation (SSCG)
- Dynamic clock switching without any glitches
- Clock delay or clock advancement
- Clock phase shifting
- Clock inversion
- Clock holding
- Clock gating

5.2 System-Level Block Diagram

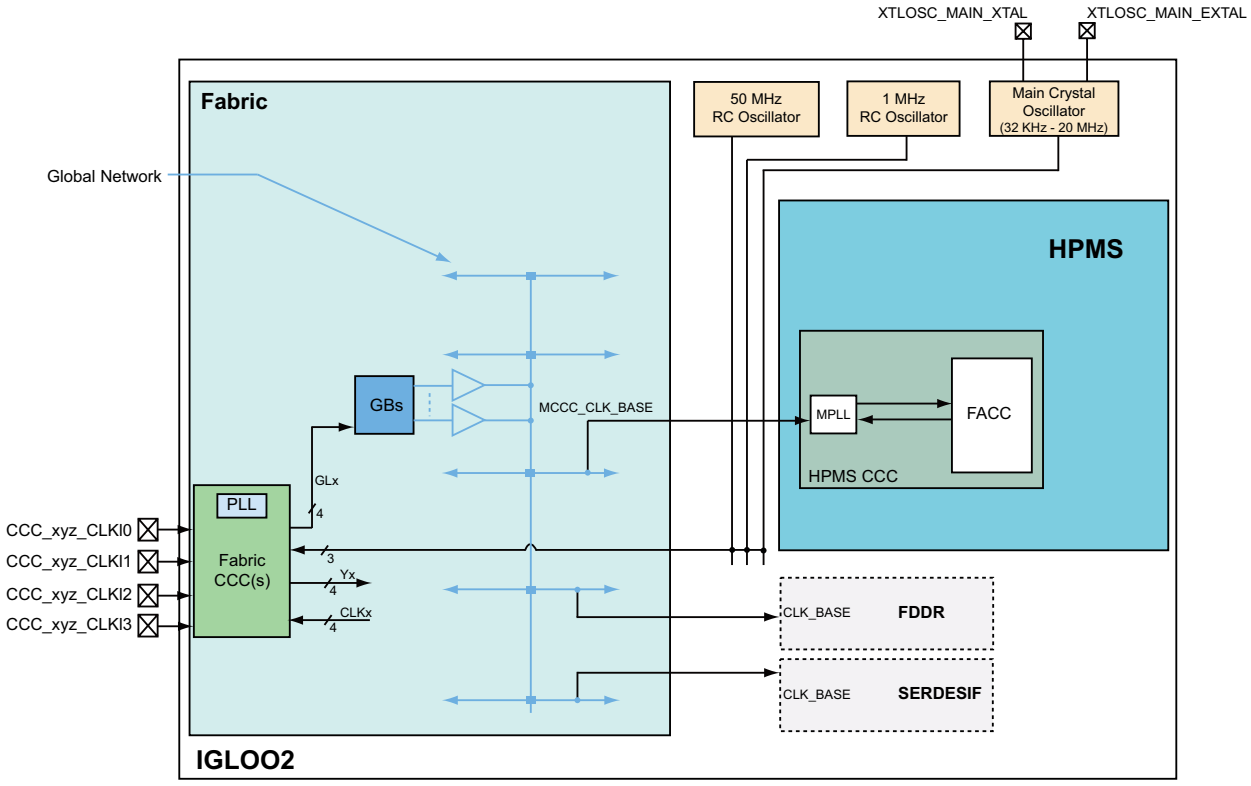
The following figures show the system-level block diagram of the fabric CCC. The figures show the inputs and outputs for one fabric CCC; each fabric CCC has a similar set of inputs and outputs.

Figure 36 • **Fabric CCC System-Level Block Diagram of SmartFusion 2**



Note: The blocks marked with dotted lines are available only in selected devices of SmartFusion2 SoC FPGA family.

Figure 37 • **Fabric CCC System-Level Block Diagram of IGLOO 2**



Note: The blocks marked with dotted lines are available only in selected devices of IGLOO2 FPGA family.

5.3 Fabric CCC Locations

The fabric CCCs are labeled according to their locations in the FPGA fabric floor plan. For instance, the fabric CCCs located in the northeast corner are labeled as CCC-NE0 and CCC-NE1. All the available fabric CCCs are associated with a dedicated PLL (not shown in the figures). The PLL associated with the CCC-NE1 is capable of accepting 32 KHz reference clock input whereas others are capable of accepting reference clock input between 1 MHz and 200 MHz. See the Fabric PLL Circuitry, page 50 for more information on PLLs.

The following figures show the locations of fabric CCCs in the FPGA fabric of SmartFusion 2 and IGLOO 2 devices. As seen in the figures, fabric CCC global clock outputs are connected to the GBs.

Figure 38 • **Fabric CCC Locations in M2S005/M2GL005 and M2S010/M2GL010 Devices**

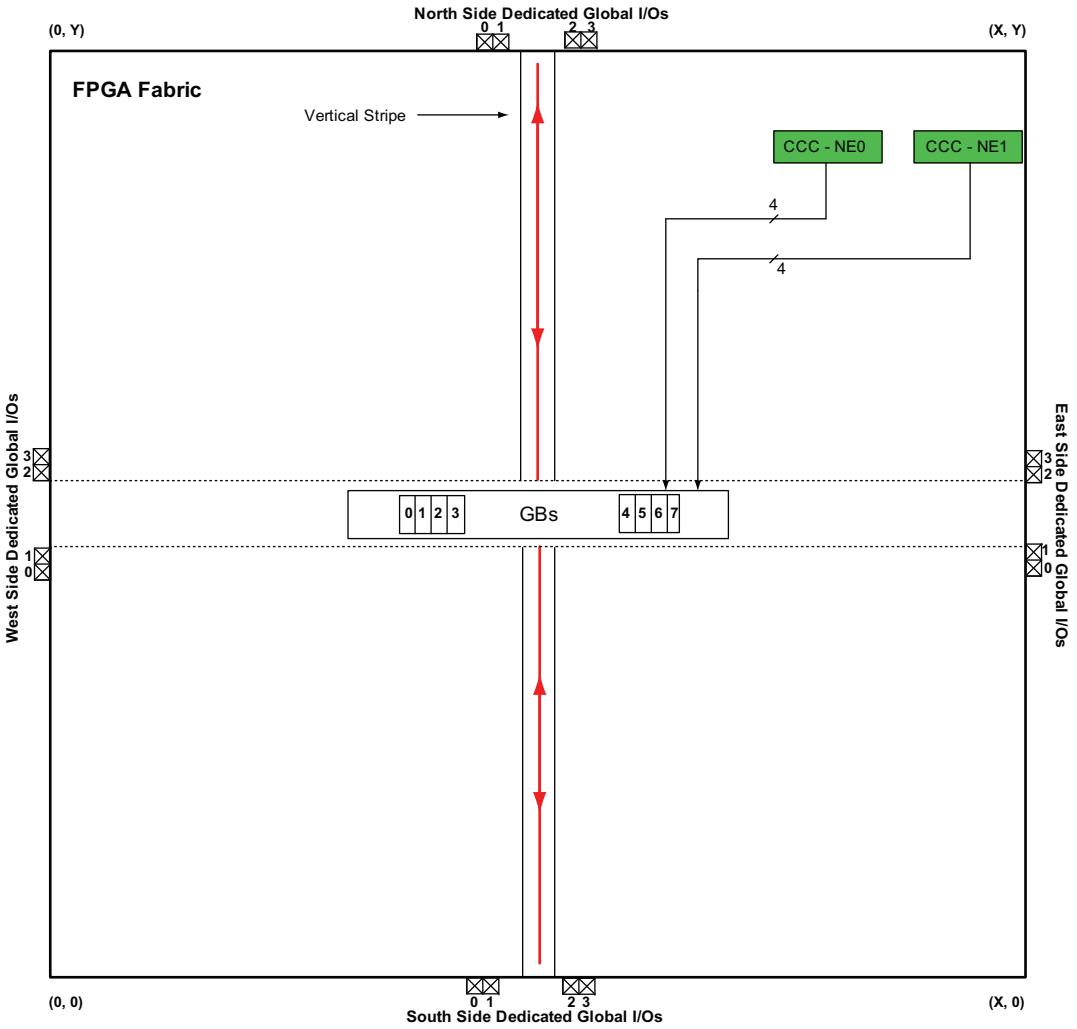


Figure 39 • **Fabric CCC Locations in M2S025/M2GL025, M2S050/M2GL050, M2S060/M2GL060, and M2S090/M2GL090 Devices**

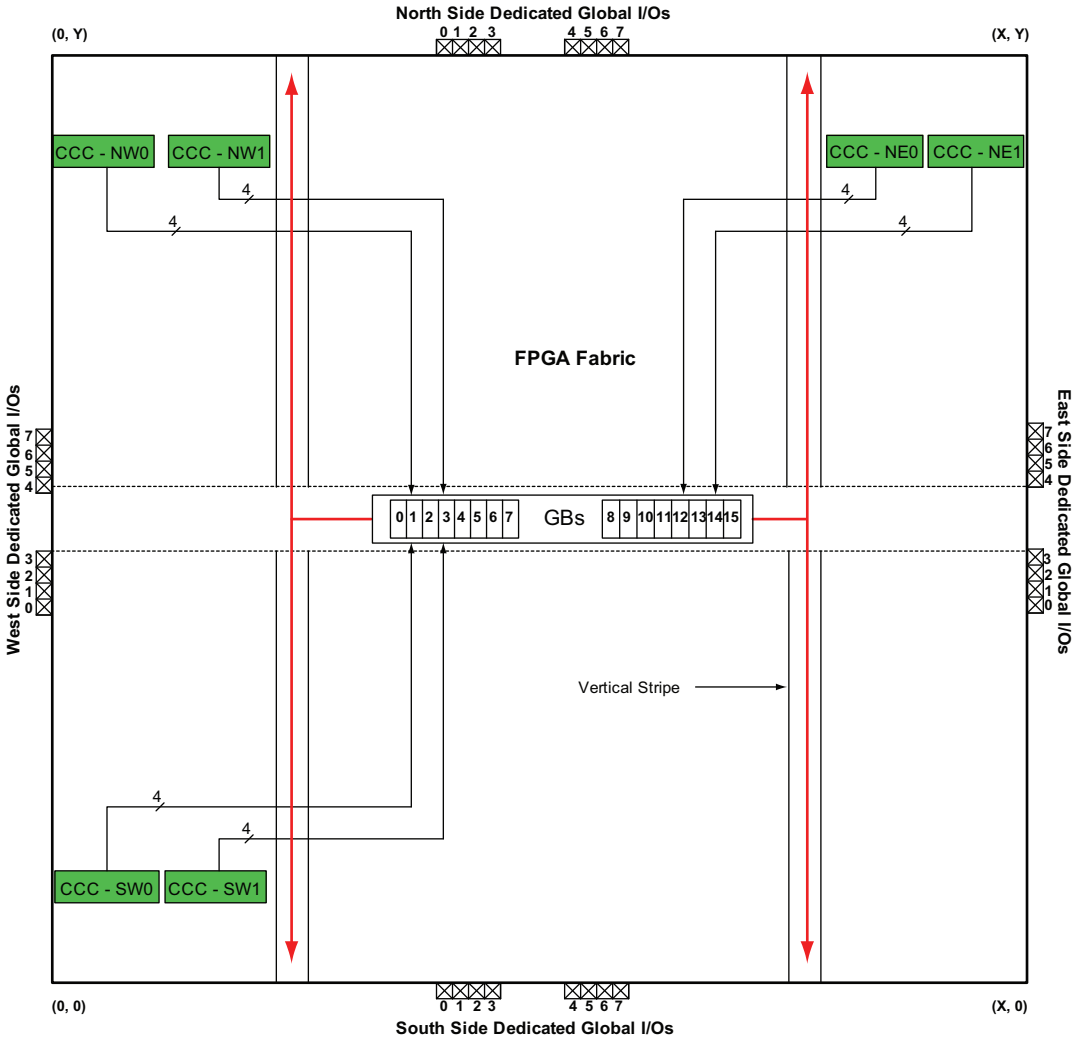
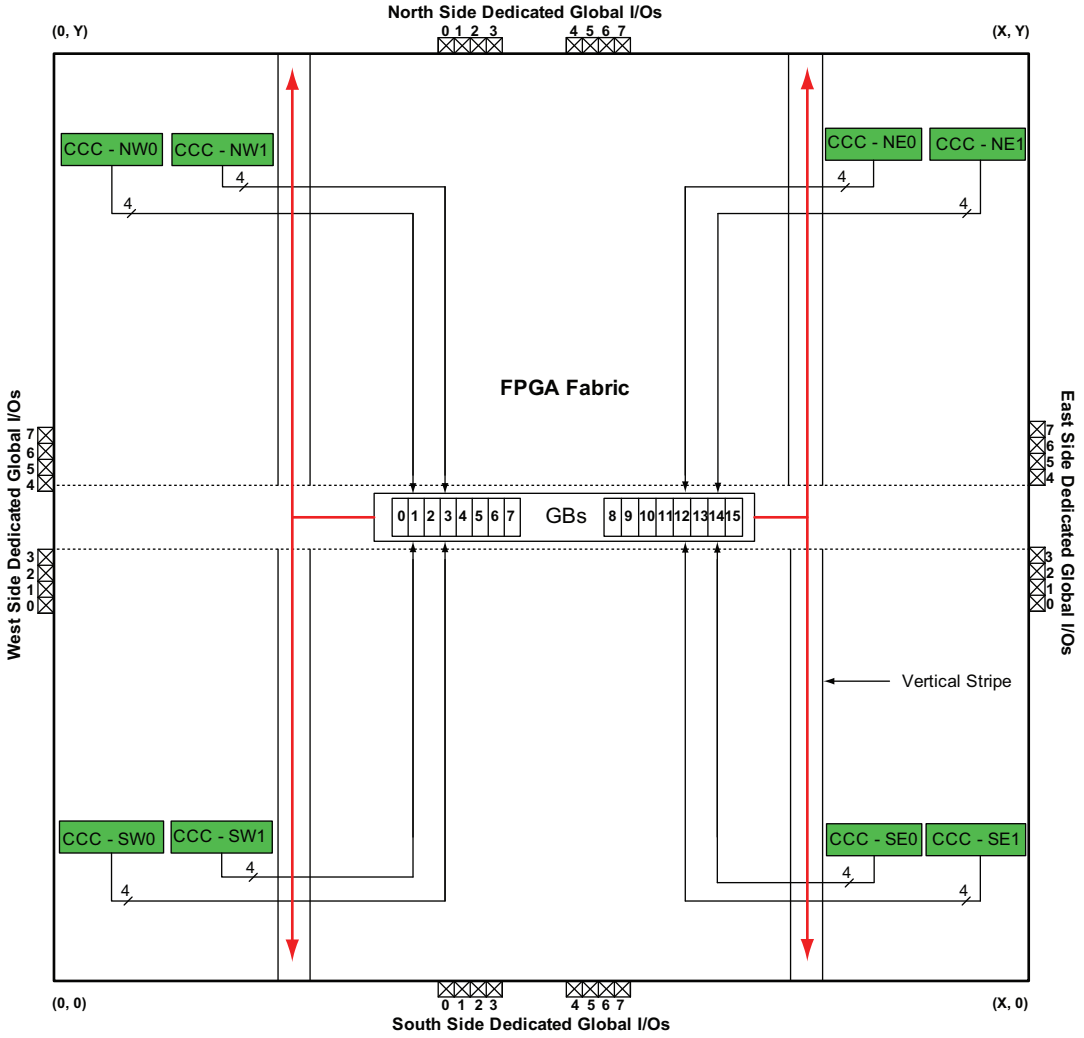


Figure 40 • **Fabric CCC Locations in M2S150/M2GL150 Devices**



In order to use CCC from specific location, set CCC location in the `.pdc` file in the Libero. The following table lists the X, Y location.

Table 10 • **Fabric CCC Locations**

	CCC Locations (X,Y)							
Device Part Number	**SW0**	**SW1**	**SE0**	**SE1**	**NW0**	**NW1**	**NE0**	**NE1**
M2S005/M2GL005							(372, 44)	(390, 44)
M2S010/M2GL010							(372, 92)	(390, 92)
M2SL025/M2GL025	(0, 11)	(18, 11)			(0, 134)	(18, 134)	(600, 134)	(618, 134)
M2SL050/M2GL050	(0, 11)	(18, 11)			(0, 194)	(18, 194)	(852, 194)	(870, 194)
M2SL060/M2GL060	(0, 11)	(18, 11)			(0, 194)	(18, 194)	(852, 194)	(870, 194)
M2S090/M2GL090	(0, 11)	(18, 11)			(0, 254)	(18, 254)	(996, 254)	(1014, 254)
M2S150/M2GL150	(0, 11)	(18, 11)	(1428, 11)	(1446, 11)	(0, 302)	(18, 302)	(1428,3 02)	(1446, 302)

Use the following syntax in PDC file to use CCC from a specific location.

```
set_location macro_name -fixed value x y
```

- `macro_name` – Specifies the name of the macro in the netlist to assign to a particular location on the chip.
- `-fixed value` – Sets whether the location of this instance is fixed (that is, locked). Locked instances are not moved during layout. If default is Yes, then the location of this instance is locked. If default is No, then the location of this instance is unlocked.
- `x y` – The x and y coordinates specify where to place the macro on the chip.

5.4 Functional Description

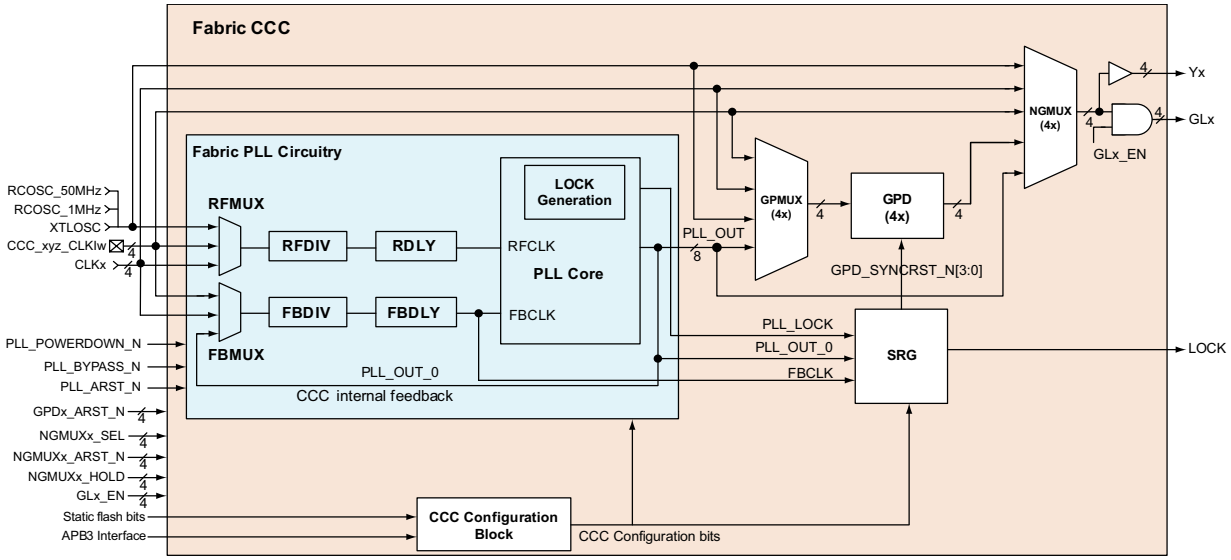
Each fabric CCC block consists of the following components:

- Fabric PLL Circuitry
- GPMUX
- GPD
- SRG
- NGMUX

The fabric CCC features are statically configured by the *CCC* macro, which is available in the IP Catalog of the Libero design software. Fabric CCCs can also be configured dynamically through an APB3 bus interface. See the Fabric CCC Configuration, page 60 for more information.

The following figure shows the functional block diagram of the fabric CCC.

Figure 41 • **Fabric CCC Block Diagram**



The following table lists fabric CCC ports and their descriptions.

Note: Access and visibility to these ports is controlled according to user configuration. For more information, see the How to Use Fabric CCC(s), page 72.

Table 11 • **Fabric CCC Port Description**

Port Name	Direction	Polarity	Description
CCC_xyz_CLKI0	Input		Input clock from dedicated input pad 0. The xy portion refers to the CCC location (NE, SW, SE, or NW) and z represents the CCC number (0 or 1).
CCC_xyz_CLKI1	Input		Input clock from dedicated input pad 1. The xy portion refers to the CCC location (NE, SW, SE, or NW) and z represents the CCC number (0 or 1).

Table 11 • **Fabric CCC Port Description (continued)**

Port Name	Direction	Polarity	Description
CCC_xyz_CLKI2	Input		Input clock from dedicated input pad 2. The xy portion refers to the CCC location (NE, SW, SE, or NW) and z represents the CCC number (0 or 1).
CCC_xyz_CLKI3	Input		Input clock from dedicated input pad 3. The xy portion refers to the CCC location (NE, SW, SE, or NW) and z represents the CCC number (0 or 1).
CLK0	Input		Input clock from the FPGA fabric
CLK1	Input		Input clock from the FPGA fabric
CLK2	Input		Input clock from the FPGA fabric
CLK3	Input		Input clock from the FPGA fabric
RCOSC_50MHz	Input		Input clock from the 50 MHz RC oscillator
RCOSC_1MHz	Input		Input clock from the 1 MHz RC oscillator
XTLOSC	Input		Input clock from the main crystal oscillator
GL0	Output		Global clock output 0 (NGMUX0 output), drives a global buffer.
GL1	Output		Global clock output 1 (NGMUX1 output), drives a global buffer.
GL2	Output		Global clock output 2 (NGMUX2 output), drives a global buffer.
GL3	Output		Global clock output 3 (NGMUX3 output), drives a global buffer.
GL0_EN	Input	High	Enable signal for GL0
GL1_EN	Input	High	Enable signal for GL1
GL2_EN	Input	High	Enable signal for GL2
GL3_EN	Input	High	Enable signal for GL3
Y0	Output		Core clock output 0 (NGMUX0 output), drives the FPGA fabric local routing resources.
Y1	Output		Core clock output 1 (NGMUX1 output), drives the FPGA fabric local routing resources.
Y2	Output		Core clock output 2 (NGMUX2 output), drives the FPGA fabric local routing resources.
Y3	Output		Core clock output 3 (NGMUX3 output), drives the FPGA fabric local routing resources.
LOCK	Output	High	PLL Lock indicator signal.
PLL_BYPASS_N	Input	Low	Powers-down the PLL core and bypasses it such that PLL_OUT tracks RFCLK.
PLL_ARST_N	Input	Low	Powers-down the PLL core and asynchronously reset all its internal digital blocks such that the PLL outputs are driven Low.
PLL_POWERDOWN_N	Input	Low	Powers-down the PLL for the lowest quiescent current and the PLL outputs are Low.
NGMUX0_SEL	Input		NGMUX0 output selection signal. Switches between the primary and secondary clock without any glitches. 0: Primary clock 1: Secondary clock

Table 11 • **Fabric CCC Port Description (continued)**

Port Name	Direction	Polarity	Description
NGMUX1_SEL	Input		NGMUX1 output selection signal. Switches between the primary and secondary clock without any glitches. 0: Primary clock 1: Secondary clock
NGMUX2_SEL	Input		NGMUX2 output selection signal. Switches between the primary and secondary clock without any glitches. 0: Primary clock 1: Secondary clock
NGMUX3_SEL	Input		NGMUX3 output selection signal. Switches between the primary and secondary clock without any glitches. 0: Primary clock 1: Secondary clock
NGMUX0_HOLD_N	Input	Low	NGMUX0 hold signal. Synchronously sets the NGMUX0 output low when 0.
NGMUX1_HOLD_N	Input	Low	NGMUX1 hold signal. Synchronously sets the NGMUX1 output low when 0.
NGMUX2_HOLD_N	Input	Low	NGMUX2 hold signal. Synchronously sets the NGMUX2 output low when 0.
NGMUX3_HOLD_N	Input	Low	NGMUX3 hold signal. Synchronously sets the NGMUX3 output low when 0.
NGMUX0_ARST_N	Input	Low	NGMUX0 asynchronous reset signal. Resets the NGMUX0 when 0.
NGMUX1_ARST_N	Input	Low	NGMUX1 asynchronous reset signal. Resets the NGMUX1 when 0.
NGMUX2_ARST_N	Input	Low	NGMUX2 asynchronous reset signal. Resets the NGMUX2 when 0.
NGMUX3_ARST_N	Input	Low	NGMUX3 asynchronous reset signal. Resets the NGMUX3 when 0.
GPD0_ARST_N	Input	Low	GPD0 asynchronous reset signal. The GPD0 is held in reset when 0.
GPD1_ARST_N	Input	Low	GPD1 asynchronous reset signal. The GPD1 is held in reset when 0.
GPD2_ARST_N	Input	Low	GPD2 asynchronous reset signal. The GPD2 is held in reset when 0.
GPD3_ARST_N	Input	Low	GPD3 asynchronous reset signal. The GPD3 is held in reset when 0.
Fabric CCC Configuration Bus Interface Signals – APB3 Bus Interface			
PCLK	Input		APB3 interface clock signal.
PRESET_N	Input	Low	APB3 interface active low reset signal.
PADDR[7:2]	Input		APB3 interface address bus. This port is used to address fabric CCC internal registers.
PSEL	Input		APB3 interface slave select signal.
PENABLE	Input		APB3 interface strobe signal to indicate the second cycle of an APB3 transfer.

Table 11 • **Fabric CCC Port Description (continued)**

Port Name	Direction	Polarity	Description
PWRITE	Input		APB3 interface read/write control signal. When High, this signal indicates an APB3 write access and when Low, a read access.
PWDATA[7:0]	Input		APB3 interface write data bus.
PRDATA[7:0]	Output		APB3 interface read data bus.

5.4.1 Fabric CCC Output Clocks

Each fabric CCC generates up to four different global clocks (GL0, GL1, GL2, and GL3) and four core clocks (Y0, Y1, Y2, and Y3) with a maximum frequency up to 400 MHz. The generated global clocks drive the FPGA fabric global networks and core clocks drive the local routing resources in the FPGA fabric. Fabric CCC's core clock outputs (Yx) can be used to drive internal logic without using the global network resources and introduce additional delay because of FPGA fabric routing. Core clocks are useful when global network resources must be conserved and utilized for other timing-critical paths. When using core clocks, timing analysis must be performed to ensure that there are no skew issues on these paths. The frequencies and phases of the core clocks are the same as those of the associated global clocks. The global buffers associated with the global clock outputs (GLx) are available to the user logic when the global clock outputs are not enabled. The fabric CCCs can be connected to each other or cascaded through the FPGA fabric.

Each of the four output clocks (GLx/Yx) can be driven by the following sources:

- PLL outputs
- General purpose dividers (GPDs) outputs
- Dedicated global I/Os
- On-chip oscillators
- FPGA fabric internal clock signals

5.4.2 Fabric CCC Clock Sources

The following clock sources can be used to drive the fabric CCC clock input as well as the fabric CCC outputs (GLx/Yx) directly:

- 1 MHz RC Oscillator
- 50 MHz RC Oscillator
- Main Crystal Oscillator
- Dedicated Global I/Os
- FPGA Fabric Clock Sources

The input clock frequency range for the fabric CCCs depends on the PLL usage for the output clocks generation.

- When the PLL is used, the PLL reference clock frequency must be between 1 MHz and 200 MHz.
  - CCC-NE1 has 32 KHz reference clock support.
- When the PLL is bypassed, the fabric CCC input clock frequency can be up to 400 MHz.

5.4.2.1 On-Chip Oscillators

On-chip oscillators (1 MHz RC oscillator, 50 MHz RC oscillator, and main crystal oscillator) have direct access to the fabric CCC(s) through hardwired connections. To use a particular on-chip oscillator as a clock source, the Chip Oscillators macro must be instantiated in the design and its output connected to the associated fabric CCC input.

5.4.2.2 Dedicated Global I/Os

The dedicated global I/Os are the primary source for bringing external clock inputs into the SmartFusion 2 or IGLOO 2 device. Some of the dedicated global I/Os have direct access to the GBs, whereas others have to go through either fabric CCC or VCCC to reach GBs.

Each fabric CCC has four dedicated global I/Os (CCC_xyz_CLKIw, w = 0 to 3) as inputs. These dedicated global I/Os have direct access to the fabric CCCs and are not routed through the FPGA fabric. Each fabric CCC output (GLx/Yx), as well as the PLL reference clock, can be driven from any one of the four dedicated global I/Os. Table 12, page 49 lists the dedicated global I/Os associated with each fabric CCC for the SmartFusion 2 M2S050-FG896. As listed in the M2S050 FG896 Pin Function column of the table, some dedicated global I/Os are shared with the hard IP blocks such as MSS_MMUART, MSS_I2C, MSS_USB, MSS_GPIO, MDDR and FDDR. The shared dedicated global I/Os are not available to the fabric CCCs when the associated hard IP blocks are enabled and configured to use the shared dedicated global I/Os.

For other SmartFusion 2 and IGLOO 2 devices, see *DS0115: SmartFusion2 Pin Descriptions Datasheet* and *DS0124: IGLOO2 Pin Descriptions Datasheet* and the associated pin outs to identify the dedicated global I/Os associated with fabric CCCs.

See the Dedicated Global I/Os, page 12 for more information on dedicated global I/Os.

Table 12 • **Dedicated Global I/Os Connections to the Fabric CCC/PLLs**

CCC Location	Dedicated Global I/O Pin Name	M2S050 FG896 Pin Number	M2S050 FG896 Pin Function
CCC-NE1	CCC_NE1_CLKI0	V26	MSIO11NB3/I2C_1_SCL/GPIO_1_A/USB_DATA4_A
	CCC_NE1_CLKI1	M25	MSIO40PB1/MMUART_1_RI/GPIO_15_B
	CCC_NE1_CLKI2	A17	DDRIO76PB0/GB12/MDDR_DQ12
	CCC_NE1_CLKI3	E17	DDRIO75PB0/MDDR_DQ14
CCC-NE0	CCC_NE0_CLKI0	V23	MSIO11PB3/I2C_1_SDA/GPIO_0_A/USB_DATA3_A
	CCC_NE0_CLKI1	N23	MSIO39PB1/MMUART_1_CTS/GPIO_13_B
	CCC_NE0_CLKI2	D15	DDRIO79PB0/MDDR_DQ10
	CCC_NE0_CLKI3	A16	DDRIO78PB0/GB8/MDDR_DQS1
CCC-SW1	CCC_NW1_CLKI0	T7	MSIOD120PB7
	CCC_NW1_CLKI1	R3	MSIOD118PB7/GB5
	CCC_NW1_CLKI2	AG11	DDRIO152NB5/GB7/FDDR_DQ_ECC3
	CCC_NW1_CLKI3	AK15	DDRIO159PB5/FDDR_DQ8
CCC-SW0	CCC_NW0_CLKI0	U7	MSIOD121PB7
	CCC_NW0_CLKI1	R1	MSIOD119PB7/GB1
	CCC_NW0_CLKI2	AG9	DDRIO147NB5
	CCC_NW0_CLKI3	AF11	DDRIO152PB5/GB3/FDDR_DQ_ECC2
CCC-NW1	CCC_NW1_CLKI0	P9	MSIO116PB8
	CCC_NW1_CLKI1	P3	MSIO114PB8/GB6
	CCC_NW1_CLKI2	B9	DDRIO91NB0/GB4
	CCC_NW1_CLKI3	E11	DDRIO87PB0/MDDR_DQ_ECC2
CCC-NW0	CCC_NW0_CLKI0	R9	MSIO117PB8
	CCC_NW0_CLKI1	P1	MSIO115PB8/GB2
	CCC_NW0_CLKI2	D9	DDRIO92NB0
	CCC_NW0_CLKI3	A9	DDRIO91PB0/GB0

5.4.2.3 FPGA Fabric Clock Sources

FPGA fabric clock sources are those clocks coming through FPGA fabric routed nets. Each fabric CCC has four clock inputs (CLKx, x = 0 to 3) from the FPGA fabric. Each fabric CCC output (GLx/Yx), as well as the reference clock, can be driven from any one of the four fabric clock inputs.

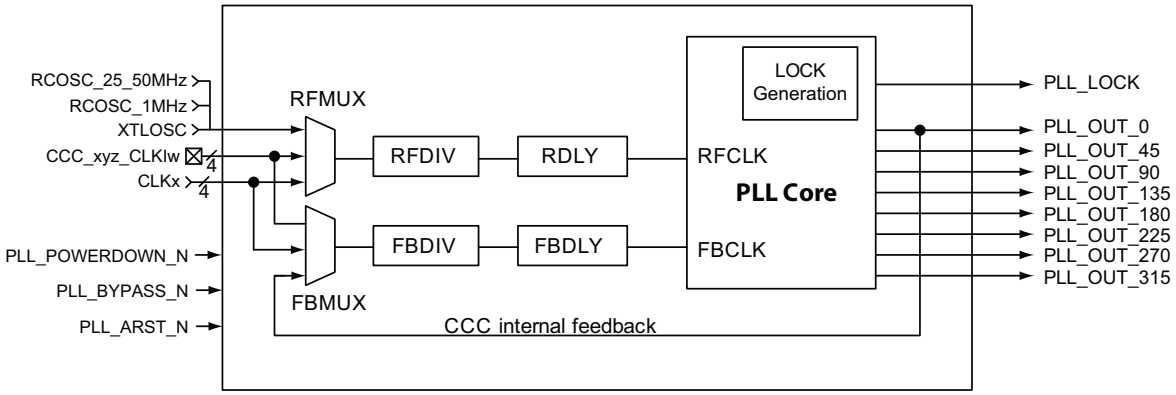
It is also possible to assign the fabric clock input to any of the regular I/O pins. Choosing this option provides the flexibility of selecting any regular I/O location but introduces additional delay due to FPGA fabric routing.

5.4.3 Fabric PLL Circuitry

Each fabric CCC includes components for operating the PLL associated with it. Clocks requiring frequency synthesis and phase adjustments can utilize the PLL before connecting to the global or local routing networks. The following figure shows the block diagram of the fabric PLL circuitry. Fabric PLL circuitry includes the following components:

- PLL Core
- RFMUX
- FBMUX
- RFDIV and FBDIV
- RDLY and FBDLY

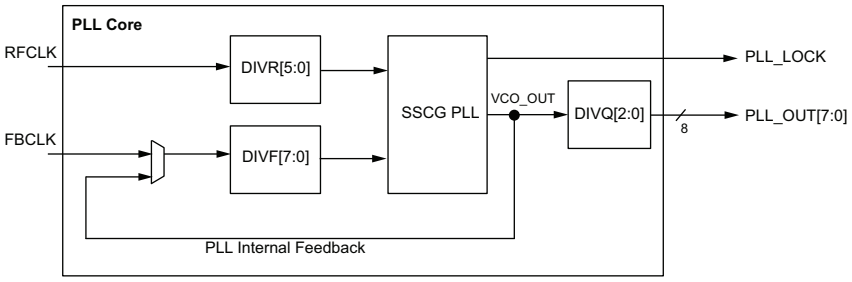
Figure 42 • **Fabric PLL Circuitry**



5.4.3.1 PLL Core

SmartFusion 2 and IGLOO 2 devices have two, six, or eight PLLs, depending on the size of the device (see Table 9, page 40). The PLL associated with the northeast corner fabric CCC (CCC-NE1), is capable of accepting 32 KHz reference clock input and generating an output frequency up to 1000 MHz. All other PLLs can accept a reference clock input ranging from 1 MHz to 200 MHz. The minimum output frequency of the PLL is 20 MHz. Each PLL includes a phase shifter and the PLL output clock is available in eight phases with a 45° phase difference (0°, 45°, 90°, 135°, 180°, 225°, 270°, and 315°). The VCO in the PLL Core has 500 MHz to 1000 MHz frequency range. The reference clock divider and feedback clock divider (DIVR and DIVF) must be configured such that the VCO operates with in its range. The output divider (DIVQ) divides the VCO output frequency such that PLL generate frequencies in 20 MHz to 1000 MHz range. The PLL output is passed through GPD and it's division value must be configured such that the overall CCC output frequency must be less than or equal to 400 MHz. The following figure shows the PLL Core internal circuitry.

Figure 43 • **PLL Core Internal Circuitry**



The fabric PLL features are configured using static flash bits set by the *CCC* macro available in the Libero SoC design software. The Libero SoC place-and-route software automatically uses the 32 KHz PLL when the PLL reference frequency is set to 32 KHz in the *CCC* macro configurator.

The supplies required to power the PLL are VDD for the digital section and an analog supply (CCC_xyz_PLL_VDDA, where xyz represents the CCC location and number) for the analog section. Each fabric PLL has its own analog supply pin. To achieve a reasonable level of long term jitter performance, it is vital to deliver a low noise power supply to the PLL. Typically an R-C or R-L-C filter is used, with the C being composed of multiple devices to achieve a wide spectrum of noise absorption. See *DS0115: SmartFusion2 Pin Descriptions datasheet* and *DS0124: IGLOO2 Pin Descriptions datasheet* for more information on the analog power supply pins of PLLs. The PLL analog supply voltage (CCC_xyz_PLL_VDDA) can be either 2.5 V or 3.3 V, based on the power supply availability on the board. User can only use either 3.3 V or 2.5 V PLL supply for the entire design. Libero does not support using both 2.5 V and 3.3 V power supplies for the PLLs used in a design. The analog power supply does not impact the PLL output frequency range. The PLL analog supply voltage is internally regulated, which provides independence from the external supply. See *DS0128: SmartFusion2 and IGLOO2 Datasheet* for the PLL operational range and characteristics.

Note: The unused PLLs are configured to be held in power-down mode so that they consume lowest quiescent current.

Note: If REFCK (Reference clock) is disconnected from the PLL then the VCO keeps running at a low-end frequency.
If FBCK (Feedback clock) is disconnected from the PLL when in use then the VCO keeps running at a high-end frequency.

5.4.3.2 RFMUX

As shown in Figure 42, page 50, the reference clock multiplexer (RFMUX) allows selecting the PLL reference clock from the available fabric CCC clock sources.

5.4.3.3 FBMUX

As shown in Figure 42, page 50, the feedback clock multiplexer (FBMUX) allows selecting the PLL feedback clock source from an internal or an external feedback loop, depending on the system requirements. The internal feedback loop can be PLL internal feedback (not shown in Figure 42, page 50 because it is internal to the PLL core) or CCC internal feedback. The external feedback loop is from one of the CCC global clocks (GLx) through the FPGA fabric or externally to the chip through dedicated global I/O pads.

Minimum jitter and phase error can be achieved, if the PLL internal feedback loop is used. It is necessary to operate the PLL with internal feedback when its reference clock is set to 32 KHz. When PLL internal feedback is used, the programmable delay and GPD synchronization are not available.

The RFMUX and FBMUX have a feature to invert the clock inputs. Since the PLL aligns reference and feedback clocks on the rising clock edge, the inversion feature allows aligning them on the falling reference and feedback clocks.

5.4.3.4 RFDIV and FBDIV

The PLL reference clock divider (RFDIV) is an 8-bit programmable divider. The feedback clock divider (FBDIV) is a 14-bit programmable frequency divider. RFDIV and FBDIV generate PLL reference clock

(RFCLK) and feedback clock (FBCLK). The function of RFDIV and FBDIV is described in the following equations:

RFCLK = (Input clock frequency) / (RFDIV [7:0] + 1)

FBCLK = (Input clock frequency) / (FBDIV [13:0] + 1)

The visual CCC macro configurator derives the necessary divider values based on the input frequency, desired output frequencies, and the selected feedback loop. These dividers are configured statically by flash bits when the FPGA fabric is programmed.

5.4.3.5 RDLY and FBDLY

As shown in Figure 42, page 50, each fabric CCC has a programmable delay element in the PLL reference clock path (RDLY) and feedback loop path (FBDLY). For PLLs, adding delay in the reference clock path enables clock delay and adding delay in the feedback clock path enables clock advancement with respect to the PLL reference clock. A typical application for clock advancement is to advance the on-chip clock in order to cancel the global network delay and align the internal clock to an external clock on the board.

Each fabric CCC can be configured with a delay in the reference path or feedback path. The delay time of the programmable delay elements can be varied between 0 and 6.3 ns with a 100 ps step size. The programmable delay elements are bypassed when the PLL delay is set to zero or the PLL internal feedback loop is used.

5.4.3.6 Lock Generation Circuit

A lock signal, PLL_LOCK, is provided to indicate that the PLL has locked onto the incoming signal. The lock signal asserts High to indicate that the frequency and phase lock is achieved. The precision of the lock discrimination can be adjusted using the lock window controls.

The lock window represents the phase error window for lock assertion as a fraction of the divided reference clock period. The lock window can be adjusted between 500 parts per million (ppm) and 64,000 ppm in powers of 2. The integration of the lock period can be adjusted using the lock counter. The lock counter or lock delay indicates the number of divided reference clock cycles to wait after the PLL is locked before asserting the lock signal. The lock delay is useful for avoiding false toggling of the lock signal. The lock counter can be configured between 32 and 10,48,576 cycles in multiples of 2.

For SmartFusion 2 and IGLOO 2 devices, if the fabric PLL reference clock is sourced from an internal RC oscillator, the minimum PLL lock window setting should be 64000 ppm and 32000 ppm for 1 MHz RC oscillator and 50 MHz RC oscillator, respectively. If the RFDIV divisor value is 2 or less and a wrong PLL lock window setting is used, then the PLL lock may get toggled.

Note: The combination of high register utilization and toggling rate results high PLL jitter and subsequently the PLL may come out of lock. Use higher PLL lock window for stable PLL operation.

5.4.3.7 Fabric PLL Control Signals

A reset control signal, PLL_ARST_N, is provided to power-down the PLL core and asynchronously reset all its internal digital blocks. When the PLL_ARST_N signal is asserted Low, all the PLL outputs are driven Low. A bypass signal, PLL_BYPASS_N, is provided which powers-down the PLL core and bypasses it such that the PLL output tracks the reference clock. All the output clocks have the same phase in PLL Bypass mode. The PLL_BYPASS_N signal has precedence over the PLL_ARST_N signal.

A fabric PLL can be held in Power-down mode for the lowest quiescent current by asserting its power-down control signal, PLL_POWERDOWN_N. When the PLL_POWERDOWN_N signal is asserted Low, the PLL powers-down and its outputs are held Low. The PLL_POWERDOWN_N signal has precedence over all other control signals. The following table shows the control signal values for PLL output and power states. Each PLL has its own control signals available to drive from the user logic in the FPGA fabric.

Table 13 • **Control Signals for PLL Output and Power State**

PLL_POWERDOWN_N	PLL_BYPASS_N	PLL_ARST_N	PLL_OUT	PLL POWER
0	X	X	Low	OFF

Table 13 • **Control Signals for PLL Output and Power State (continued)**

PLL_POWERDOWN_N	PLL_BYPASS_N	PLL_ARST_N	PLL_OUT	PLL POWER
1	0	0	Low	OFF
1	0	1	Tracks RFCLK input	OFF
1	1	0	Low	OFF
1	1	1	Normal Operation	ON

5.4.3.8 SSCG

Each fabric PLL is integrated with a spread spectrum clock function for SSCG. The spread spectrum clocking feature spreads the fundamental clock signal energy to a wide band of frequencies to reduce the EMI.

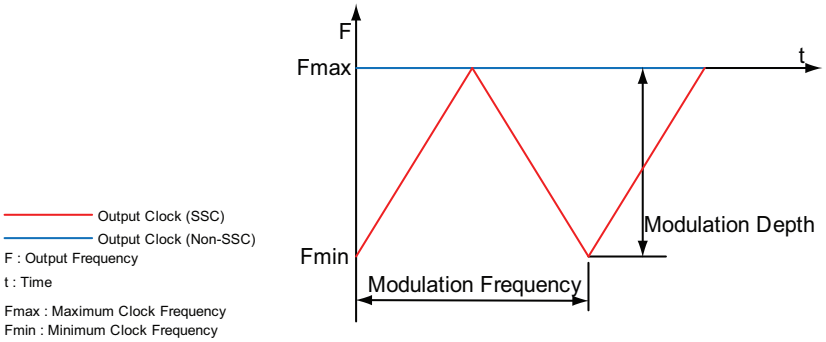
As shown in the following figure, the spread spectrum function generates a down-spreading triangular modulation profile. The spread spectrum programmable options allow to control the modulation frequency and depth.

The modulation depth is represented by percentage spread which defines the frequency range of the modulated clock resulting from the spread spectrum modulation. For instance, with a down-spread percentage of 1.5% and nominal PLL output frequency of 100 MHz, the spread spectrum modulated output frequency is swept between 98.5 MHz and 100 MHz. See *DS0128: SmartFusion2 and IGLOO2 FPGA Datasheet* for the spread spectrum modulation characteristics of the fabric PLL.

The SSCG functionality is disabled if the down-spread percentage is set to 0%. More reduction in EMI can be achieved by spreading the clock energy over a wide band of frequencies. The modulation frequency is the rate at which the spreading signal sweeps from the minimum to the maximum PLL output frequency.

Note: The spread spectrum functionality of the PLL cannot be used with the 32 KHz reference clock input or when external PLL feedback input is used.

Figure 44 • **Spread Spectrum Modulation**

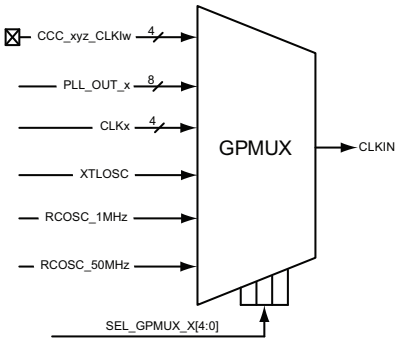


5.4.4 GPMUX

The clock source for generating the output clocks through GPDs is selected using a general purpose multiplexer (GPMUX). The output of the GPMUX is the clock input to the GPD. Each GPD has its own GPMUX that allows to choose the clock source independently for each GPD. Each GPMUX selects the clock source from the fabric CCC clock sources or PLL outputs, as shown in the following figure. The fabric PLL is bypassed if none of the fabric CCC outputs uses PLL outputs.

Each GPMUX has a feature to invert the output of the GPMUX that can be controlled by writing to appropriate fabric CCC configuration registers (FCCC_GPMUXx_CR, x = 0 to 3) through APB3 interface. See the Fabric CCC Configuration, page 60 for more information.

Figure 45 • **GPMUX Input Clock Sources**



5.4.5 GPD

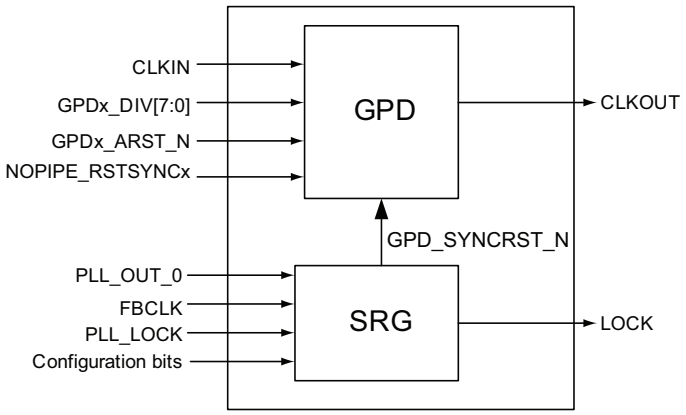
GPDs divide the frequency of the source clock selected by the GPMUX by a factor of 1 to 255 to generate the desired fabric CCC output frequencies as shown in the following equations. Each fabric CCC has four GPDs. The output of the GPDs can be used as the source for any of the four fabric CCC outputs. For instance, GPD0 can be used on a path to the GL1 output. The *CCC* macro configurator configures the GPD values to achieve the closest possible match to the desired output frequencies, based on the input frequency and the selected feedback loop.

CLKOUT = CLKIN / GPDx_DIV[7:0], for GPDx_DIV[7:0] = 1 TO 255

CLKOUT = CLKIN, for GPDx_DIV[7:0] = 0

The following figure shows a block diagram for a GPD block.

Figure 46 • **GPD Block Diagram**



The GPD's behavior is outlined as follows:

- The duty cycle of the GPD output clock (CLKOUT) will be 50% if the duty cycle of the input clock (CLKIN) is 50%.
- The rising edge of CLKOUT always occurs as a result of a rising edge inputs on CLKIN.
- The falling edge of CLKOUT always occurs as a result of a rising edge on CLKIN, if the divider value is an even number > 2.
- The falling edge of CLKOUT always occurs as a result of a falling edge on CLKIN, if the divider value is an odd number or if divisor = 0 or 1.

The GPDs divide the phase of the non-zero phase PLL output by the division value. In other words, the actual phase shift is not the same as the PLL phase shift selected, if the GPD division value is not one. The actual output phase shift of the output is shown in the fabric CCC configurator. For more information See the Output Clock Phase Adjustment, page 79.

Actual Phase Shift = (Selected PLL output phase shift) / (GPD division value)

As shown in Figure 46, page 54, each GPD has an asynchronous reset (GPD_ARST_N) and a synchronous reset (GPD_SYNCRST_N) for resetting the GPD. Both the resets are active Low. The GPD output clock (CLKOUT) is reset to Low when a reset signal is asserted. The asynchronous reset comes from the FPGA fabric and synchronous reset comes from the synchronous reset generator (SRG).

5.4.5.1 SRG

The SRG block sends synchronous reset signals to the GPDs when the PLL locks or on request by the System Controller. The purpose of the SRG is to synchronize (or re-align) the outputs of the GPDs with the PLL input clock after the PLL lock is achieved. This block also produces a LOCK signal which is sent out to the FPGA fabric.

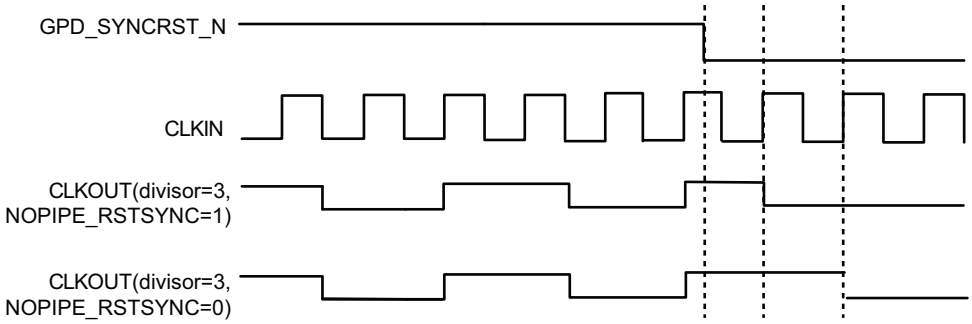
The SRG has an internal reset pipeline stage to get the correct phase alignment at the GPD output while using non-zero phase PLL output as the GPD input. The *CCC* macro configurator automatically sets the reset pipeline stage control signal based on the GPD input selection.

Whenever GPD_SYNCRST_N changes from High to Low:

- CLKOUT is reset to Low at the next rising edge of CLKIN, if the internal reset pipeline stage is not enabled.
- CLKOUT is reset to Low at the second rising edge of CLKIN, if the internal reset pipeline stage is enabled.

The following figure shows the GPD output clock behavior for synchronous reset assertion.

Figure 47 • **Assertion of GPD Synchronous Reset – GPD_SYNCRST_N**

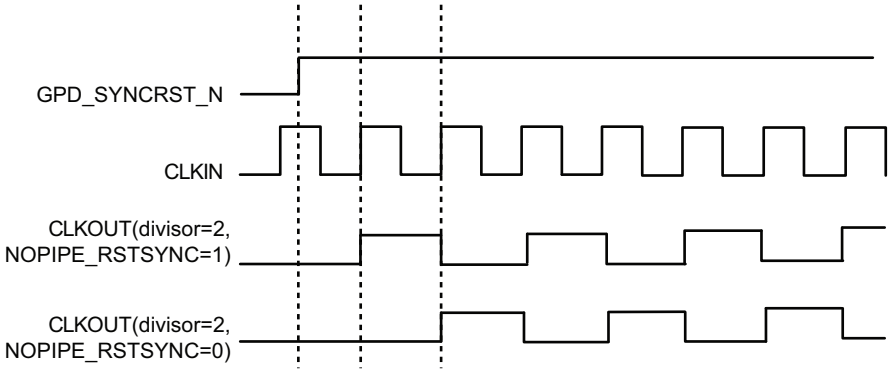


When releasing GPD_SYNCRST_N from Low to High:

- CLKOUT will be clocked to High at the next rising edge of CLKIN, if the reset pipeline stage is not enabled
- CLKOUT will be clocked to High at the second rising edge of CLKIN, if the internal reset pipeline stage is enabled.

The following figure shows the GPD output clock behavior for synchronous reset release.

Figure 48 • **Releasing of GPD Synchronous Reset – GPD_SYNCRST_N**



5.4.5.2 GPD Operating Modes

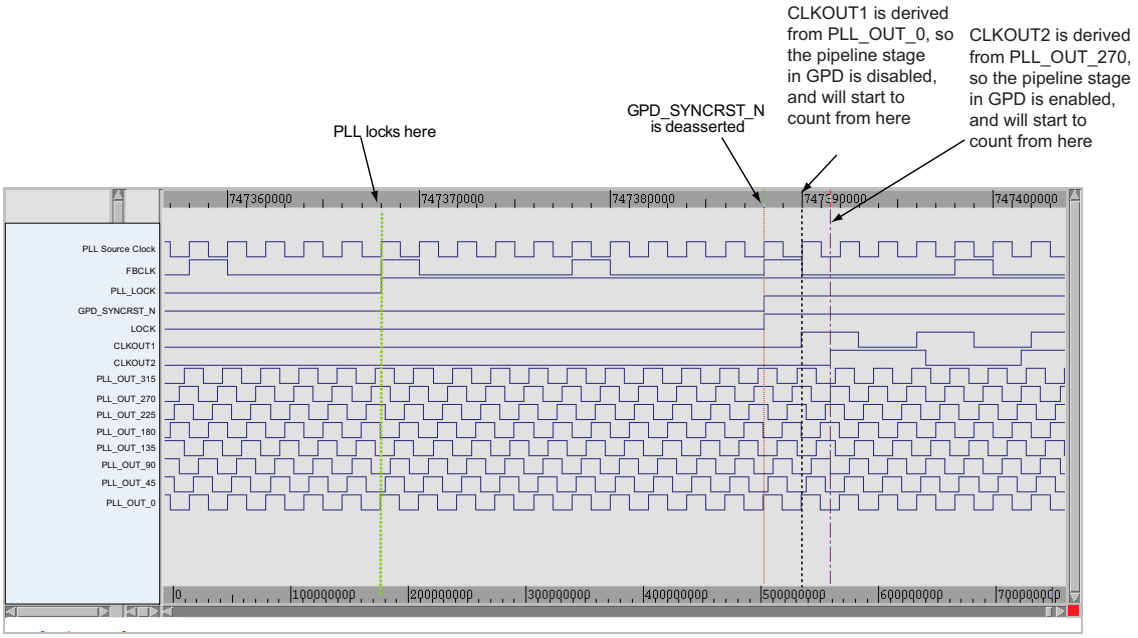
The GPDs support the following operating modes. For the GPDs that are driven by the PLL output clock, Microchip recommends using either the first or second mode described below:

1. GPDs are held in reset after power-up, release and synchronize with the PLL reference clock after the PLL is locked:
   - If enabled, GPDs outputs are held in reset Low after power-up. Hence, the output(s) (GLx/Yx) driven by the GPDs are held Low (or High if inverted) after power-up.
   - After the PLL lock is achieved, GPD's synchronous reset is released synchronously with second FBCLK rising edge.
   - This mode can only be used if GPDs are not on the PLL external feedback path.
   - This mode is not available when the PLL internal feedback path is enabled.

The following figure shows the GPDs outputs (CLKOUT1 and CLKOUT2) synchronization operation for this mode.

- The PLL source clock goes to PLL through RFDIV and the RFDIV division value is set to 4 (divided by 5)
- The FBDIV division value is set to 4 (divided by 5)
- CLKOUT1 is generated by a GPD whose input is PLL_OUT_0 and the GPD divider value is set to 3
- CLKOU T2 is generated by another GPD whose input is PLL_OUT_270 and the GPD divider value is set to 5.

Figure 49 • **GPD_SYNCRST_N Released after PLL Locks**

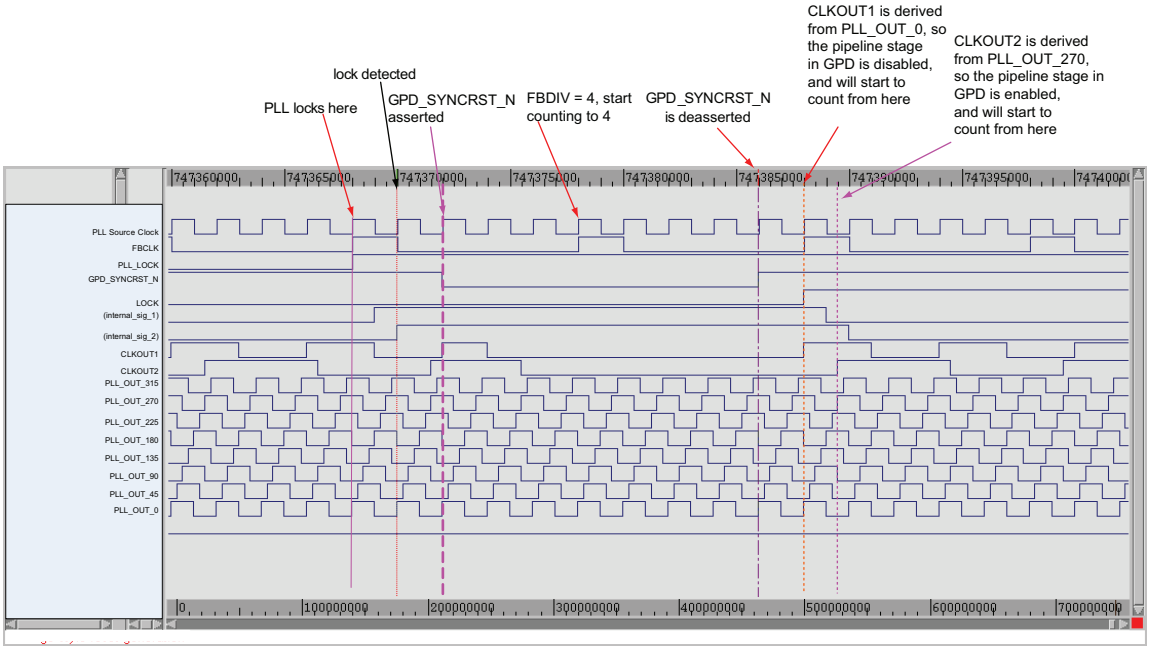


2. GPDs operate after power-up and resynchronize with the PLL reference clock after the PLL is locked:
   - If enabled, GPDs operate after power-up. Hence, the outputs (GLx/Yx) driven by the GPDs are operational after power-up.
   - When the PLL is locked and detected by the rising edge of PLL_OUT_0, the SRG asserts the GPD_SYNCRST_N signal after completion of one PLL_OUT_0 clock cycle.
   - After GPD_SYNCRST_N is asserted, the SRG waits for the first rising edge of FBCLK to trigger the counting of (M – 1) clock edges of PLL_OUT_0 clock, where M = FBDIV [13:0] + 1.
   - The SRG releases GPD_SYNCRST_N on the $(M – 1)^{th}$ rising edge of PLL_OUT_0 clock to let GPD start counting on the Mth clock edge of PLL_OUT_0 clock. In other words, the release of GPD_SYNCRST_N restarts the GPDs on the PLL output clock edge, which correspond to the 2nd rising edge of FBCLK after the lock changes state from 0 to 1.
   - This mode can only be used if GPDs are not on the PLL external feedback path.
   - This mode is not available when the PLL internal feedback path is enabled.

The following figure shows the GPDs outputs (CLKOUT1 and CLKOUT2) synchronization operation for this mode.

- The PLL source clock goes to PLL through RFDIV and the RFDIV division value is set to 4 (divided by 5)
- The FBDIV division value is set to 4 (divided by 5)
- CLKOUT1 is generated by a GPD whose input is PLL_OUT_0 and the GPD divider value is set to 3.
- CLKOUT2 is generated by another GPD whose input is PLL_OUT_270 and the GPD divider value is set to 5.

Figure 50 • **GPD Output Resynchronization after PLL Locks**



3. GPDs operate after power-up, no automatic resynchronization:
   - If enabled, GPDs operate after power-up. Hence the outputs (GLx/Yx) driven by the GPDs are operational after power-up.
   - There is no automatic resynchronization after the PLL lock. In this case, the GLx output phase alignments are not guaranteed. This phase alignment behavior is not captured by the static timing analysis tool. Hence, clock domain crossing analysis between GLx outputs is not guaranteed.

5.4.6 NGMUX

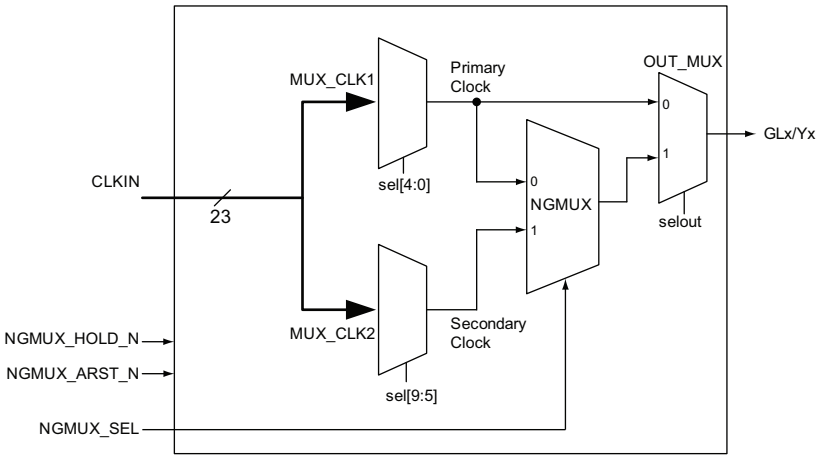
The NGMUX provides a special switching sequence between two asynchronous clocks without generating any unwanted narrow clock pulses on the output clock. There are four NGMUX blocks in a fabric CCC and each NGMUX produces one fabric CCC output clock (GLx/Yx).

Internally, there are two multiplexers to select primary and secondary clocks from the available clock sources. The primary clock source and secondary clock source to NGMUX can come from any of the following clock sources:

- On-chip oscillators
- FPGA fabric
- Dedicated global I/O pads
- 8 PLL outputs
- GPDs outputs

The following figure shows the NGMUX circuit.

Figure 51 • **NGMUX Circuit**



The resulting two internal clocks (primary and secondary) are fed to an NGMUX that allows switching between the two clocks without any glitches. The selection control input for each NGMUX (NGMUX_SEL) comes from the fabric logic and can be changed dynamically. The NGMUX is bypassed when there is no secondary clock source.

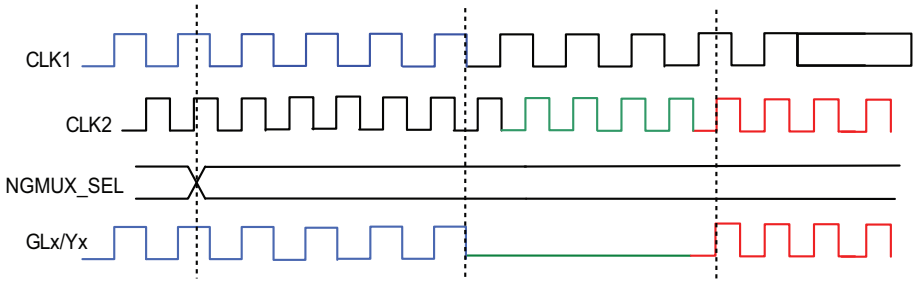
Each NGMUX has an asynchronous active low reset signal (NGMUX_ARST_N) to force the NGMUX output to Low. Asserting the NGMUX_ARST_N may cause a narrow pulse on the NGMUX output, so this must only be used for system reset on power-up or to clear an error condition. During run-time, use NGMUX_HOLD_N control input, if it is required to hold the NGMUX output at Low. The NGMUX_ARST_N and NGMUX_SEL signals are to be managed as follows for a glitch-less transition of clocks using NGMUX. After device power-up, reset using NGMUX_ARST_N signal to ensure that the NGMUX is in safe state. Then trigger a switch using NGMUX_SEL signal when both clock sources are stable and ready.

5.4.6.1 Clock Switching

The sequence of clock switching between the primary clock (CLK1) and the secondary clock (CLK2) is explained in this section. The clock switching specified here is for the selection change from the primary clock to the secondary clock, and the same rule applies when selection changes from the secondary clock to the primary clock.

Upon changing the selection control input (NGMUX_SEL), the NGMUX output (GLx/Yx) follows the original clock waveform for "1 fall and then 4 rise-and-fall" sequence, as shown in the following figure. Then the NGMUX output is grounded and waits for the "1 fall and then 4 rise-and-fall" sequence of the new clock, after which the NGMUX output is driven by the new clock. After the transition is done, the original clock can be stopped. The maximum frequency of the primary and secondary clocks is 400 MHz.

Figure 52 • **Clock Switching Using NGMUX**



The primary clock and secondary clock source must generate free-running clocks without any glitches for glitch free clock switching. If not, the NGMUX may still produce unwanted glitches on the output clock.

The requirements for a successful clock switching are:

1. The primary and second clocks must be toggled during the switching operation.
2. NGMUX_SEL signal must not be switched when the previous switch is still under-going and not yet completed.

The output of the NGMUX is undefined if NGMUX_SEL changes when the previous switch is still in progress. For any clock switching between CLK1 and CLK2 with a relationship other than as described above, the output clock is not guaranteed to be glitch free and it may fail to switch between clock sources. The asynchronous reset, NGMUX_ARST_N, can be used to provide reset if such a condition occurs.

It is possible to read the state of the NGMUX through APB3 configuration bus. The FCCC_NGMUXx_CR1[NGMUXx_BUSY] register field indicates the NGMUX switching status. When NGMUXx_BUSY is '1', the switching is in progress, old clock source must keep toggling until this signal becomes low and no new switching must be started.

5.4.6.2 Clock Holding

Each NGMUX has its own control input (NGMUX_HOLD_N) from the FPGA fabric to hold the clock output at Low. When the NGMUX_HOLD_N control signal is asserted, both corresponding Y and GL outputs will be synchronously set to low. During run-time, the NGMUX_HOLD_N control input can be driven to stop/start the clock without any glitches. The NGMUX_HOLD_N control input has precedence over the NGMUX_SEL. Once the NGMUX_HOLD_N signal is deasserted, the NGMUX synchronously switches to one of the two input clocks based on the NGMUX_SEL value.

5.4.6.3 Clock Inversion

Each NGMUX can be configured to invert the polarity of output clock from its source. If the NGMUX inversion feature is enabled, both the primary and secondary clocks are inverted.

5.4.7 Clock Gating

The global clocks (GLx) generated from the fabric CCCs can be gated dynamically by driving the control signals (GLx_EN, where x ranges from 0 to 3) from the FPGA fabric. See the Clock Gating, page 20 of FPGA Fabric Global Network Architecture chapter for more information. See Figure 36, page 41 and Figure 37, page 41 for more information on the clocking hierarchy.

5.5 Fabric CCC Configuration

Fabric CCC(s) are configurable statically through flash configuration bits set in the programming bitstream or dynamically through an APB3 bus interface. The flash configuration bits are the configuration bits associated with programmed flash switches. The fabric CCC flash configuration bits provide the default operating state of the fabric CCC.

5.5.1 Fabric CCC Static Configuration

The CCC macro available in the IP Catalog of the Libero SoC design software provides a visual configuration wizard for a quick and easy way to configure the fabric CCC with desired the settings.

5.5.2 Fabric CCC Dynamic Configuration

Each fabric CCC has an APB3 bus interface for dynamic configuration of the fabric CCC parameters without reprogramming the device. The fabric CCC configuration is controlled by volatile configuration registers which are loaded with values from the flash configuration bits at power-up. An APB3 master can change the fabric CCC configuration by writing to the appropriate registers. See the Fabric CCC Configuration Registers, page 61 for more information on fabric CCC control registers and their bit definitions.

To meet all the datasheet specifications, there are certain requirements that must be met when configuring the PLL parameters. The Libero CCC configurator implements all these requirements and creates a valid solution for the requested output clock frequencies and phases. Hence, it is

recommended to generate the required configuration using the Libero CCC configurator and use the generated parameters in the dynamic configuration solution. The PLL_POWERDOWN_N input must be asserted before making changes to the PLL configuration parameters. Asserting PLL_POWERDOWN_N signal resets the PLL operation.

5.6 Fabric CCC Configuration Registers

The following table lists the fabric CCC configuration registers and their brief description.

Table 14 • **Fabric CCC Register Map**

Register Name	Address	Description
Reserved	0x00	Reserved
FCCC_RFMUX_CR	0x04	RFMUX configuration register, see Table 15, page 62.
FCCC_RFDIV_CR	0x08	RFDIV configuration register, see Table 16, page 62.
FCCC_FBMUX_CR	0x0C	FBMUX configuration register, see Table 17, page 62.
FCCC_FBDIV_CR0	0x10	FBDIV configuration register 0, see Table 18, page 63.
FCCC_FBDIV_CR1	0x14	FBDIV configuration register 1, see Table 19, page 63.
FCCC_NGMUX0_CR0	0x18	NGMUX0 configuration register 0, see Table 20, page 63.
FCCC_NGMUX0_CR1	0x1C	NGMUX0 configuration register 1, see Table 21, page 63.
FCCC_NGMUX1_CR0	0x20	NGMUX1 configuration register 0, see Table 22, page 63.
FCCC_NGMUX1_CR1	0x24	NGMUX1 configuration register 1, see Table 23, page 64.
FCCC_NGMUX2_CR0	0x28	NGMUX2 configuration register 0, see Table 24, page 64.
FCCC_NGMUX2_CR1	0x2C	NGMUX2 configuration register 1, see Table 25, page 64.
FCCC_NGMUX3_CR0	0x30	NGMUX3 configuration register 0, see Table 26, page 64.
FCCC_NGMUX3_CR1	0x34	NGMUX3 configuration register 1, see Table 27, page 65.
FCCC_GPMUX0_CR	0x38	GPMUX0 configuration register, see Table 28, page 65.
FCCC_GPMUX1_CR	0x3C	GPMUX1 configuration register, see Table 29, page 65.
FCCC_GPMUX2_CR	0x40	GPMUX2 configuration register, see Table 30, page 66.
FCCC_GPMUX3_CR	0x44	GPMUX3 configuration register, see Table 31, page 66.
FCCC_GPD0_CR	0x48	GPD0 configuration register, see Table 32, page 66.
FCCC_GPD1_CR	0x4C	GPD1 configuration register, see Table 33, page 66.
FCCC_GPD2_CR	0x50	GPD2 configuration register, see Table 34, page 66.
FCCC_GPD3_CR	0x54	GPD3 configuration register, see Table 35, page 67.
FCCC_PLL_CR0	0x58	PLL lock window configuration register, see Table 36, page 67.
FCCC_PLL_CR1	0x5C	PLL lock counter configuration and lock status register, see Table 37, page 67.
FCCC_PLL_CR2	0x60	PLL SSCG modulation frequency configuration register, see Table 38, page 67.
FCCC_PLL_CR3	0x64	PLL SSCG modulation depth configuration register, see Table 39, page 67.
FCCC_PLL_CR4	0x68	PLL SSCG functionality enable register, see Table 40, page 67.
FCCC_PLL_CR5	0x6C	PLL operation mode and supply voltages configuration register, see Table 41, page 68.
FCCC_PLL_CR6	0x70	PLL internal or external feedback path selection register, see Table 42, page 68.

Table 14 • **Fabric CCC Register Map (continued)**

Register Name	Address	Description
FCCC_GPDS_SYNC_CR	0x74	GPDs' outputs realignment request configuration register, see Table 43, page 68.
FCCC_PLL_CR7	0x78	PLL's internal output divider configuration register, see Table 44, page 68.
FCCC_PLL_CR8	0x7C	PLL's internal reference clock divider configuration register, see Table 45, page 68.
FCCC_PLL_CR9	0x80	PLL's internal feedback clock divider configuration register, see Table 46, page 69.
FCCC_PLL_CR10	0x84	PLL loop filter range configuration register, see Table 47, page 69.
FCCC_GPD0_SYNC_CR	0x88	GPD0 operating mode configuration register, see Table 48, page 69.
FCCC_GPD1_SYNC_CR	0x8C	GPD1 operating mode configuration register, see Table 49, page 70.
FCCC_GPD2_SYNC_CR	0x90	GPD2 operating mode configuration register, see Table 50, page 70.
FCCC_GPD3_SYNC_CR	0x94	GPD3 operating mode configuration register, see Table 51, page 70.
FCCC_PDLY_CR	0x98	Programmable delay elements configuration register, see Table 52, page 71.

5.6.1 Fabric CCC Configuration Registers Bit Definitions

The following tables list the bit definitions of fabric CCC configuration registers.

Table 15 • **FCCC_RFMUX_CR**

Bit Number	Field Name	R/W	Description
[7:5]	RESERVED	–	Reserved
4	INVRF	R/W	When INVRF = 1, inverts the output of the RFMUX.
[3:0]	SELRF	R/W	RFMUX select lines for reference clock selection. See Table 53, page 71 for reference clock selection.

Table 16 • **FCCC_RFDIV_CR**

Bit Number	Field Name	R/W	Description
[7:0]	RFDIV	R/W	RFDIV division value. RFCLK = (Input clock frequency) / (RFDIV[7:0] + 1)

Table 17 • **FCCC_FBMUX_CR**

Bit Number	Field Name	R/W	Description
[7:5]	RESERVED	–	Reserved
4	INVFB	R/W	When INVFB = 1, inverts the output of the FBMUX
[3:0]	SELFB	R/W	FBMUX selects lines for feedback clock selection. See Table 53, page 71 for feedback clock selection.

Table 18 • FCCC_FBDIV_CR0

Bit Number	Field Name	R/W	Description
[7:0]	FBDIV[7:0]	R/W	FBDIV division value - 8 LSBs FBCLK = (Input clock frequency) / (FBDIV[13:0] + 1)

Table 19 • FCCC_FBDIV_CR1

Bit Number	Field Name	R/W	Description
[7:6]	RESERVED	–	Reserved
[5:0]	FBDIV[13:8]	R/W	FBDIV division value - 6 MSBs FBCLK = (Input clock frequency) / (FBDIV[13:0] + 1)

Table 20 • FCCC_NGMUX0_CR0

Bit Number	Field Name	R/W	Description
[7:5]	RESERVED	–	Reserved
[4:0]	SELGL[4:0]	R/W	NGMUX0 primary clock source selection. NGMUX0 primary clock is the output of NGMUX0_CLK1 multiplexer. See Table 53, page 71 for primary clock source selection.

Table 21 • FCCC_NGMUX0_CR1

Bit Number	Field Name	R/W	Description
7	NGMUX0_BUSY	R	NGMUX0 status. The reset value is '0'. 1: Switching is in progress, old clock source should keep toggling until this signal becomes low and no new switching should be started 0: Switching is not in progress
6	SELOUT_0	R/W	Select or bypass NGMUX0 1: NGMUX0 is used 0: NGMUX0 is bypassed
5	INVGL_0	R/W	NGMUX0 output inversion 1: Invert the NGMUX0 output 0: Non-inverted NGMUX0 output
[4:0]	SELGL[9:5]	R/W	NGMUX0 secondary clock source selection. NGMUX0 secondary clock is the output of NGMUX0_CLK2 multiplexer. See Table 53, page 71 for secondary clock source selection.

Table 22 • FCCC_NGMUX1_CR0

Bit Number	Field Name	R/W	Description
[7:5]	RESERVED	–	Reserved
[4:0]	SELGL[14:10]	R/W	NGMUX1 primary clock source selection. NGMUX1 primary clock is the output of NGMUX1_CLK1 multiplexer. See Table 53, page 71 for primary clock source selection.

Table 23 • FCCC_NGMUX1_CR1

Bit Number	Field Name	R/W	Description
7	GLMUX1_BUSY	R	NGMUX1 status. The reset value is '0'. 1: Switching is in progress, old clock source should keep toggling until this signal becomes low and no new switching should be started 0: Switching is not in progress
6	SELOUT_1	R/W	Select or bypass NGMUX1 1: NGMUX1 is used 0: NGMUX1 is bypassed
5	INVGL_1	R/W	NGMUX1 output inversion 1: Invert the NGMUX1 output 0: Non-inverted NGMUX1 output
[4:0]	SELGL[19:15]	R/W	NGMUX1 secondary clock source selection. NGMUX1 secondary clock is the output of NGMUX1_CLK2 multiplexer. See Table 53, page 71 for secondary clock source selection.

Table 24 • FCCC_NGMUX2_CR0

Bit Number	Field Name	R/W	Description
[7:5]	RESERVED	-	Reserved
[4:0]	SELGL[24:20]	R/W	NGMUX2 primary clock source selection. NGMUX2 primary clock is the output of NGMUX2_CLK1 multiplexer. See Table 53, page 71 for primary clock source selection.

Table 25 • FCCC_NGMUX2_CR1

Bit Number	Field Name	R/W	Description
7	GLMUX2_BUSY	R	NGMUX2 status. The reset value is '0'. 1: Switching is in progress, old clock source must keep toggling until this signal becomes low and no new switching must be started 0: Switching is not in progress
6	SELOUT_2	R/W	Select or bypass NGMUX2 1: NGMUX2 is used 0: NGMUX2 is bypassed
5	INVGL_2	R/W	NGMUX2 output inversion 1: Invert the NGMUX2 output 0: Non-inverted NGMUX2 output
[4:0]	SELGL[29:25]	R/W	NGMUX2 secondary clock source selection. NGMUX2 secondary clock is the output of NGMUX2_CLK2 multiplexer. See Table 53, page 71 for secondary clock source selection.

Table 26 • FCCC_NGMUX3_CR0

Bit Number	Field Name	R/W	Description
[7:5]	RESERVED	–	Reserved

[4:0]	SELGL[34:30]	R/W	NGMUX3 primary clock source selection. NGMUX3 primary clock is the output of NGMUX3_CLK1 multiplexer. See Table 53, page 71 for primary clock source selection.

Table 27 • **FCCC_NGMUX3_CR1**

Bit Number	Field Name	R/W	Description
7	GLMUX3_BUSY	R	NGMUX3 status. The reset value is '0'. 1: Switching is in progress, old clock source must keep toggling until this signal becomes low and no new switching must be started 0: Switching is not in progress
6	SELOUT_3	R/W	Select or bypass NGMUX3 1: NGMUX3 is used 0: NGMUX3 is bypassed
5	INVGL_3	R/W	NGMUX3 output inversion 1: Invert the NGMUX3 output 0: Non-inverted NGMUX3 output
[4:0]	SELGL[39:35]	R/W	NGMUX3 secondary clock source selection NGMUX3 secondary clock is the output of NGMUX3_CLK2 multiplexer. See Table 53, page 71 for secondary clock source selection.

Table 28 • **FCCC_GPMUX0_CR**

Bit Number	Field Name	R/W	Description
7	RESERVED	–	Reserved
6	NOPIPE_SYNCRST0	R/W	1: Disable pipe-line stage for GPD0 synchronous reset is disabled 0: Enable pipe-line stage
5	INV_GPMUX0	R/W	1: Invert the output of the GPMUX0 0: Disable the clock inversion of GPMUX0 The default value of this bit is 0.
[4:0]	SEL_GPMUX0	R/W	Select lines for the GPMUX0. See Table 53, page 71 for GPMUX0 output selection.

Table 29 • **FCCC_GPMUX1_CR**

Bit Number	Field Name	R/W	Description
7	RESERVED	–	Reserved
6	NOPIPE_SYNCRST1	R/W	1: Disable pipe-line stage for GPD1 synchronous reset is disabled 0: Enable pipe-line stage
5	INV_GPMUX1	R/W	1: Invert the output of the GPMUX1 0: Disable the clock inversion of GPMUX1 The default value of this bit is 0.
[4:0]	SEL_GPMUX1	R/W	Select lines for the GPMUX1. See Table 53, page 71 for GPMUX1 output selection.

Table 30 • FCCC_GPMUX2_CR

Bit Number	Field Name	R/W	Description
7	RESERVED	–	Reserved
6	NOPIPE_SYNCRST2	R/W	1: Disable pipe-line stage for GPD2 synchronous reset is disabled 0: Enable pipe-line stage
5	INV_GPMUX2	R/W	1: Invert the output of the GPMUX2 0: Disable the clock inversion of GPMUX2 The default value of this bit is 0.
[4:0]	SEL_GPMUX2	R/W	Select lines for the GPMUX2. See Table 53, page 71 for GPMUX2 output selection.

Table 31 • FCCC_GPMUX3_CR

Bit Number	Field Name	R/W	Description
7	RESERVED	–	Reserved
6	NOPIPE_SYNCRST3	R/W	1: Disable pipe-line stage for GPD3 synchronous reset is disabled 0: Enable pipe-line stage
5	INV_GPMUX3	R/W	1: Invert the output of the GPMUX3 0: Disable the clock inversion of GPMUX3 The default value of this bit is 0.
[4:0]	SEL_GPMUX3	R/W	Select lines for the GPMUX3. See Table 53, page 71 for GPMUX3 output selection.

Table 32 • FCCC_GPD0_CR

Bit Number	Field Name	R/W	Description
[7:0]	GPD0_DIV	R/W	GPD0 division value. clkout = clkin / gpd0_div[7:0], for gpd0_div[7:0] = 1 to 255 clkout = clkin, for gpd0_div[7:0] = 0

Table 33 • FCCC_GPD1_CR

Bit Number	Field Name	R/W	Description
[7:0]	GPD1_DIV	R/W	GPD1 division value. clkout = clkin / gpd1_div[7:0], for gpd1_div[7:0] = 1 to 255 clkout = clkin, for gpd1_div[7:0] = 0

Table 34 • FCCC_GPD2_CR

Bit Number	Field Name	R/W	Description
[7:0]	GPD2_DIV	R/W	GPD2 division value. clkout = clkin / gpd2_div[7:0], for gpd2_div [7:0] = 1 to 255 clkout = clkin, for gpd2_div [7:0] = 0

Table 35 • **FCCC_GPD3_CR**

Bit Number	Field Name	R/W	Description
[7:0]	GPD3_DIV	R/W	GPD3 division value. clkout = clkin / gpd3_div[7:0], for gpd3_div[7:0] = 1 to 255 clkout = clkin, for gpd3_div[7:0] = 0

Table 36 • **FCCC_PLL_CR0**

Bit Number	Field Name	R/W	Description
[7:3]	RESERVED	–	Reserved
[2:0]	LOCKWIN	R/W	PLL lock window parameter setting. 000=500ppm 100=8000ppm 001=1000ppm 101=16000ppm 010=2000ppm 110=32000ppm 011=4000ppm 111=64000ppm

Table 37 • **FCCC_PLL_CR1**

Bit Number	Field Name	R/W	Description
[7:5]	RESERVED	–	Reserved
4	LOCK	R	Indicates PLL is locked 1: PLL is locked 0: PLL is not locked
[3:0]	LOCKCNT	R/W	PLL lock delay counter setting LOCK Delay = (2^(LOCKCNT[3:0]+5)) 0000=32, ...,1111=1048576

Table 38 • **FCCC_PLL_CR2**

Bit Number	Field Name	R/W	Description
[7:5]	RESERVED	–	Reserved
[4:0]	SSMF	R/W	Controls spread spectrum modulation frequency

Table 39 • **FCCC_PLL_CR3**

Bit Number	Field Name	R/W	Description
[7:2]	RESERVED	–	Reserved
[1:0]	SSMD	R/W	Controls spread spectrum modulation depth: 00 = 0% (SSCG off) 10 = 1.0% 01 = 0.5% 11 = 1.5%

Table 40 • **FCCC_PLL_CR4**

Bit Number	Field Name	R/W	Description
[7:1]	RESERVED	–	Reserved
0	SSE	R/W	1: Enable SSCG functionality 0: Disable SSCG functionality SSE is ineffective when FSE = 0 and MODE32K = 1

Table 41 • FCCC_PLL_CR5

Bit Number	Field Name	R/W	Description
[7:3]	RESERVED	–	Reserved
2	MODE_3V3	R/W	PLL analog supply voltage selection 1-3.3 V 0-2.5 V
1	MODE_1V2	R/W	PLL core supply voltage selection 1-1.2 V 0-Not supported
0	MODE32K	R/W	Selects PLL 32 kHz operating mode 1-32 KHz mode selected 0-32 KHz mode is not active, input frequency range is 1 MHz to 200 MHz.

Table 42 • FCCC_PLL_CR6

Bit Number	Field Name	R/W	Description
[7:1]	RESERVED	–	Reserved
0	FSE	R/W	Chooses between internal and external feedback 1: PLL internal feedback path selected 0: External feedback path selected

Table 43 • FCCC_GPDS_SYNC_CR

Bit Number	Field Name	R/W	Description
[7:1]	RESERVED	–	Reserved
0	SW_RESYNC_GPD	R/W	To request realignment of GPDs' output. A pulse of 1 to 0 requests the realignment of GPDs' output, and then the GPDs is resynchronized by resetting.

Table 44 • FCCC_PLL_CR7

Bit Number	Field Name	R/W	Description
[7:3]	RESERVED	–	Reserved
[2:0]	DIVQ	R/W	PLL's internal output divider division value. Valid divider values are 1, 2, 4, 8, 16, and 32: 000 = ÷1 001 = ÷2 010 = ÷4 011 = ÷8 100 = ÷16 101 = ÷32

Table 45 • FCCC_PLL_CR8

Bit Number	Field Name	R/W	Description
[7:6]	RESERVED	–	Reserved
[5:0]	DIVR	R/W	PLL's internal reference divider division value. 000000 = ÷1, …,111111 = ÷64

Table 46 • FCCC_PLL_CR9

Bit Number	Field Name	R/W	Description
[7:0]	DIVF	R/W	PLL's internal feedback divider division value. When SSE = 0, all divider bits DIVF[7:0] are available. When SSE = 1, the 3 MSB bits DIVF[7:5] are disabled and the 5 LSBs DIVF[4:0] are available. If SSE = 0, 00000000 = ÷1, …,11111111 = ÷256 If SSE = 1, 00000 = ÷1, …,11111 = ÷32

Table 47 • FCCC_PLL_CR10

Bit Number	Field Name	R/W	Description
[7:4]	RESERVED	–	Reserved
[3:0]	RANGE	R/W	PLL filter range. These sets the PLL loop filter to work with the post-reference divider frequency. Choose the highest valid range for best jitter performance. 0000 = BYPASS 0001 = 1-1.6 MHz 0010 = 1.6-2.6 MHz 0011 = 2.6-4.1 MHz 0100 = 4.1-6.5 MHz 0101 = 6.5-10.5 MHz 0110 = 10.5-16.8 MHz 0111 = 16.8-26.8 MHz 1000 = 26.8-43 MHz 1001 = 43-69 MHz 1010 = 69-110 MHz 1011 = 110-175 MHz 1100 = 175-200 MHz

Table 48 • FCCC_GPD0_SYNC_CR

Bit Number	Field Name	R/W	Description
[7:3]	RESERVED	–	Reserved
2	GPD_MODE_N[0]	R/W	GPD0 operating mode: 0: GPD0 operates after power-up and resynchronizes with the PLL reference clock after the PLL is locked 1: GPD0 is held in reset after power-up, released, and synchronized with the PLL reference clock after the PLL is locked
1	SRESET_GENEN[0]	R/W	When SRESET_GENEN[0] is set to 1: Generates the GPD0 synchronous reset to reset the GPD0 when SW_RESYNC_GPD is set to 1 Releases GPD0 synchronous reset synchronously when SW_RESYNC_GPD is set to 0 1: Enable GPD0 synchronous reset generation when a pulse is generated on SW_RESYNC_GPD 0: GPD0 does not get reset when a pulse is generated on SW_RESYNC_GPD
0	RESET_GENEN[0]	R/W	GPD0 synchronous reset generation on PLL lock: 1: Enable GPD0 synchronous reset generation on PLL lock 0: Disable GPD0 synchronous reset generation GPD0 synchronous reset remains high after power-up, if synchronous reset generation is disabled.

Table 49 • FCCC_GPD1_SYNC_CR

Bit Number	Field Name	R/W	Description
[7:3]	RESERVED	–	Reserved
2	GPD_MODE_N[1]	R/W	GPD1 operating mode: 0: GPD1 operates after power-up and resynchronizes with the PLL reference clock after the PLL is locked 1: GPD1 is held in reset after power-up, released, and synchronized with the PLL reference clock after the PLL is locked
1	SRESET_GENEN[1]	R/W	When SRESET_GENEN[1] is set to 1: Generates the GPD1 synchronous reset to reset the GPD1 when SW_RESYNC_GPD is set to 1 Releases GPD1 synchronous reset synchronously when SW_RESYNC_GPD is set to 0 1: Enable GPD1 synchronous reset generation when a pulse is generated on SW_RESYNC_GPD 0: GPD1 does not reset when a pulse is generated on SW_RESYNC_GPD
0	RESET_GENEN[1]	R/W	GPD1 synchronous reset generation on PLL lock: 1: Enable GPD1 synchronous reset generation on PLL lock 0: Disable GPD1 synchronous reset generation GPD1 synchronous reset remains high after power-up, if synchronous reset generation is disabled.

Table 50 • FCCC_GPD2_SYNC_CR

Bit Number	Field Name	R/W	Description
[7:3]	RESERVED	–	Reserved
2	GPD_MODE_N[2]	R/W	GPD2 operating mode: 0: GPD2 operates after power-up and resynchronizes with the PLL reference clock after the PLL is locked 1: GPD2 is held in reset after power-up, released, and synchronized with the PLL reference clock after the PLL is locked
1	SRESET_GENEN[2]	R/W	When SRESET_GENEN[2] is set to 1: Generates the GPD2 synchronous reset to reset the GPD2 when SW_RESYNC_GPD is set to 1 Releases GPD2 synchronous reset synchronously when SW_RESYNC_GPD is set to 0 1: Enable GPD2 synchronous reset generation when a pulse is generated on SW_RESYNC_GPD 0: GPD2 does not reset when a pulse is generated on SW_RESYNC_GPD
0	RESET_GENEN[2]	R/W	GPD2 synchronous reset generation on PLL lock: 1: Enable GPD2 synchronous reset generation on PLL lock 0: Disable GPD2 synchronous reset generation GPD2 synchronous reset remains high after power-up, if synchronous reset generation is disabled.

Table 51 • FCCC_GPD3_SYNC_CR

Bit Number	Field Name	R/W	Description
[7:3]	RESERVED	–	Reserved

Table 51 • **FCCC_GPD3_SYNC_CR (continued)**

Bit Number	Field Name	R/W	Description
2	GPD_MODE_N[3]	R/W	GPD3 operating mode: 0: GPD3 operates after power-up and resynchronizes with the PLL reference clock after the PLL is locked 1: GPD3 is held in reset after power-up, released, and synchronized with the PLL reference clock after the PLL is locked
1	SRESET_GENEN[3]	R/W	When SRESET_GENEN[3] is set to 1: • Generates the GPD3 synchronous reset to reset the GPD3 when SW_RESYNC_GPD is set to 1 • Releases GPD3 synchronous reset synchronously when SW_RESYNC_GPD is set to 0 1: Enable GPD3 synchronous reset generation when a pulse is generated on SW_RESYNC_GPD 0: GPD3 does not get reset when a pulse is generated on SW_RESYNC_GPD
0	RESET_GENEN[3]	R/W	GPD3 synchronous reset generation on PLL lock: 1: Enable GPD3 synchronous reset generation on PLL lock 0: Disable GPD3 synchronous reset generation GPD3 synchronous reset remains high after power-up, if synchronous reset generation is disabled.

Table 52 • **FCCC_PDLY_CR**

Bit Number	Field Name	R/W	Description
7	RESERVED	–	Reserved
6	RF_DLINE	R/W	If RF_DLINE = 1, delay setting SEL_PLL_DLINE[5:0] is applied to the delay line in the PLL reference clock path (RDLY). The setting for the feedback clock path delay line (FBDLY) is set to 0. This enables clock-delay. If RF_DLINE = 0, delay setting SEL_PLL_DLINE[5:0] is applied to the delay line in the PLL feedback clock path (FBDLY). The setting for the reference clock path delay line (RDLY) is set to 0. This enables clock-acceleration.
[5:0]	SEL_PLL_DLINE	R/W	Programmable delay lines setting. If set to 6'b000000, the clock delay lines on both reference and feedback paths are bypassed.

Table 53 • **Fabric CCC's Multiplexers Selection Control**

MUX Selection Input	RFMUX Output	FBMUX Output	GPMUXx Output (x = 0 to 3)	NGMUXx_CLK1/NGMUXx_CLK2 Output (x = 0 to 3
0	CCC_xyz_CLKI0	CCC_xyz_CLKI0	PLL_OUT_315	PLL_OUT_315
1	CCC_xyz_CLKI1	CCC_xyz_CLKI1	PLL_OUT_270	PLL_OUT_270
2	CCC_xyz_CLKI2	CCC_xyz_CLKI2	PLL_OUT_225	PLL_OUT_225
3	CCC_xyz_CLKI3	CCC_xyz_CLKI3	PLL_OUT_180	PLL_OUT_180
4	Reserved	Reserved	PLL_OUT_135	PLL_OUT_135
5	Reserved	Reserved	PLL_OUT_90	PLL_OUT_90
6	Reserved	Reserved	PLL_OUT_45	PLL_OUT_45

Table 53 • **Fabric CCC's Multiplexers Selection Control (continued)**

MUX Selection Input	RFMUX Output	FBMUX Output	GPMUXx Output (x = 0 to 3)	NGMUXx_CLK1/NGMUXx_CLK2 Output (x = 0 to 3
7	Reserved	PLL_OUT_0	PLL_OUT_0	PLL_OUT_0
8	Logic - 0	Logic - 0	Reserved	GPD0_CLKOUT
9	XTLOSC	Reserved	Reserved	GPD1_CLKOUT
10	RCOSC_1MHz	Reserved	Reserved	GPD2_CLKOUT
11	RCOSC_50MHz	Reserved	Reserved	GPD3_CLKOUT
12	CLK0	CLK0	Reserved	Reserved
13	CLK1	CLK1	Reserved	Reserved
14	CLK2	CLK2	Reserved	Reserved
15	CLK3	CLK3	Reserved	Reserved
16			CCC_xyz_CLKI0	CCC_xyz_CLKI0
17			CCC_xyz_CLKI1	CCC_xyz_CLKI1
18			CCC_xyz_CLKI2	CCC_xyz_CLKI2
19			CCC_xyz_CLKI3	CCC_xyz_CLKI3
20			CLK0	CLK0
21			CLK1	CLK1
22			CLK2	CLK2
23			CLK3	CLK3
24			Logic - 0	Logic - 0
25			XTLOSC	XTLOSC
26			RCOSC_1MHz	RCOSC_1MHz
27			RCOSC_50MHz	RCOSC_50MHz
28			Reserved	Reserved
29			Reserved	Reserved
30			Reserved	Reserved
31			Reserved	Reserved

5.7 How to Use Fabric CCC(s)

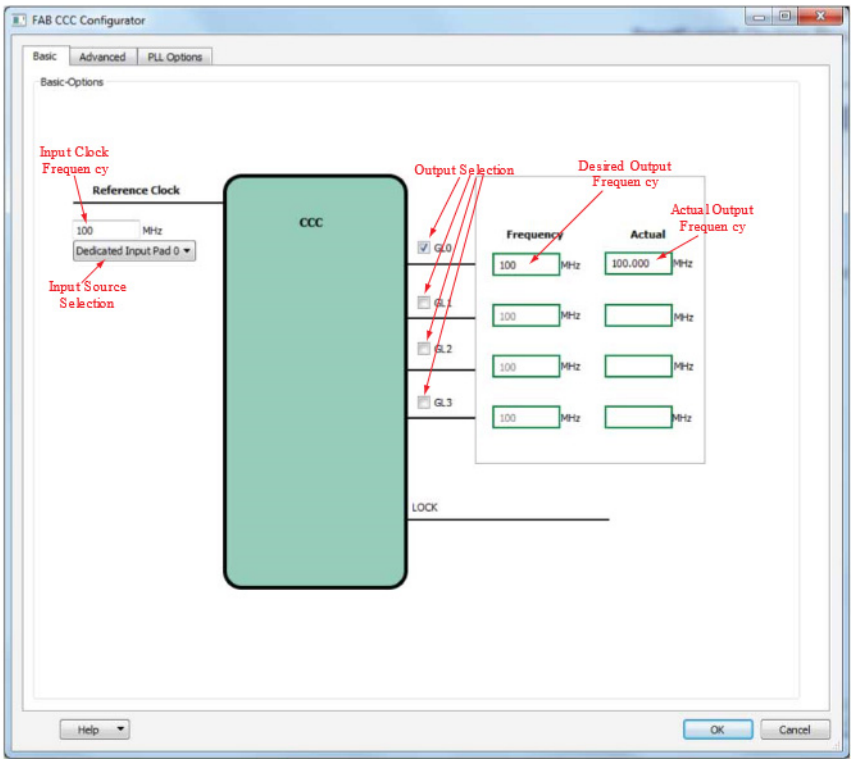
The CCC macro configurator enables static configuration of the CCC/PLL blocks available in SmartFusion 2 and IGLOO 2 devices. The CCC macro must be instantiated into the design in order to use a fabric CCC. Multiple CCC macros must be instantiated in order to use more than one fabric CCC in a design. The CCC configurator includes the following tabs:

- Basic Configuration
- Advanced Configuration
- PLL Options

5.7.1 Basic Configuration

The following figure shows the CCC configurator Basic tab and highlights the configurable options.

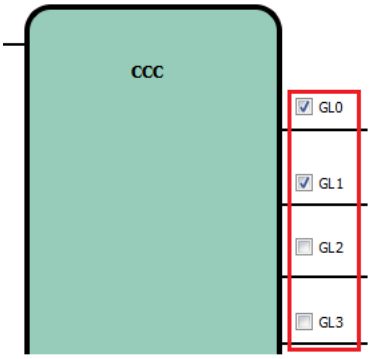
Figure 53 • **CCC Basic Configuration Tab**



Microchip recommends the following flow to configure clocks generated by the CCC for basic use cases. In the **Basic** tab, perform the following:

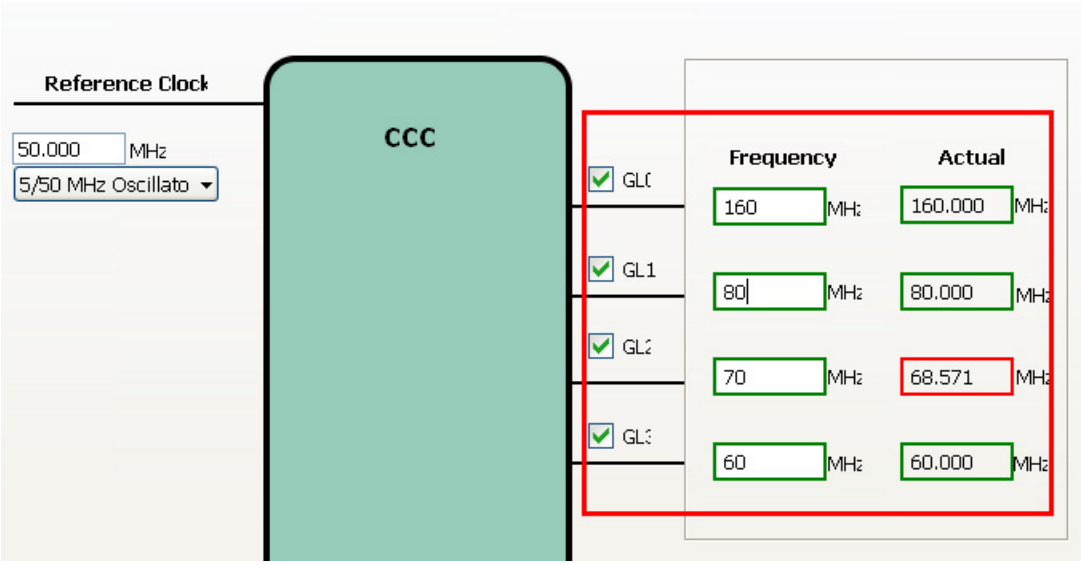
1. Select the number of desired output clocks (up to 4), as shown in the following figure.

Figure 54 • **CCC Output Selection**



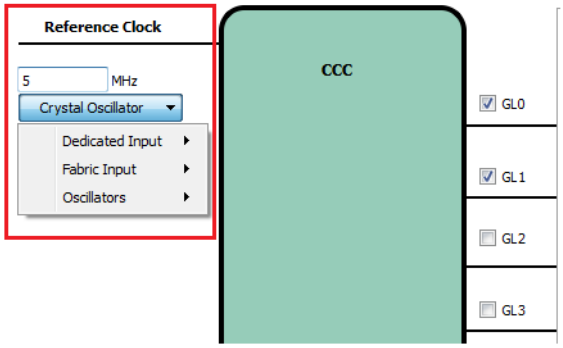
2. For each selected output clock, set the desired output frequency as shown in the following figure. Actual output frequencies achieved by the *CCC* configurator are shown in the **Actual** column of the **Basic** tab.

Figure 55 • **Set the Desired Output Frequency**



3. Configure the CCC reference clock source and frequency, as shown in the following figure.

Figure 56 • **CCC Reference Clock Selection**

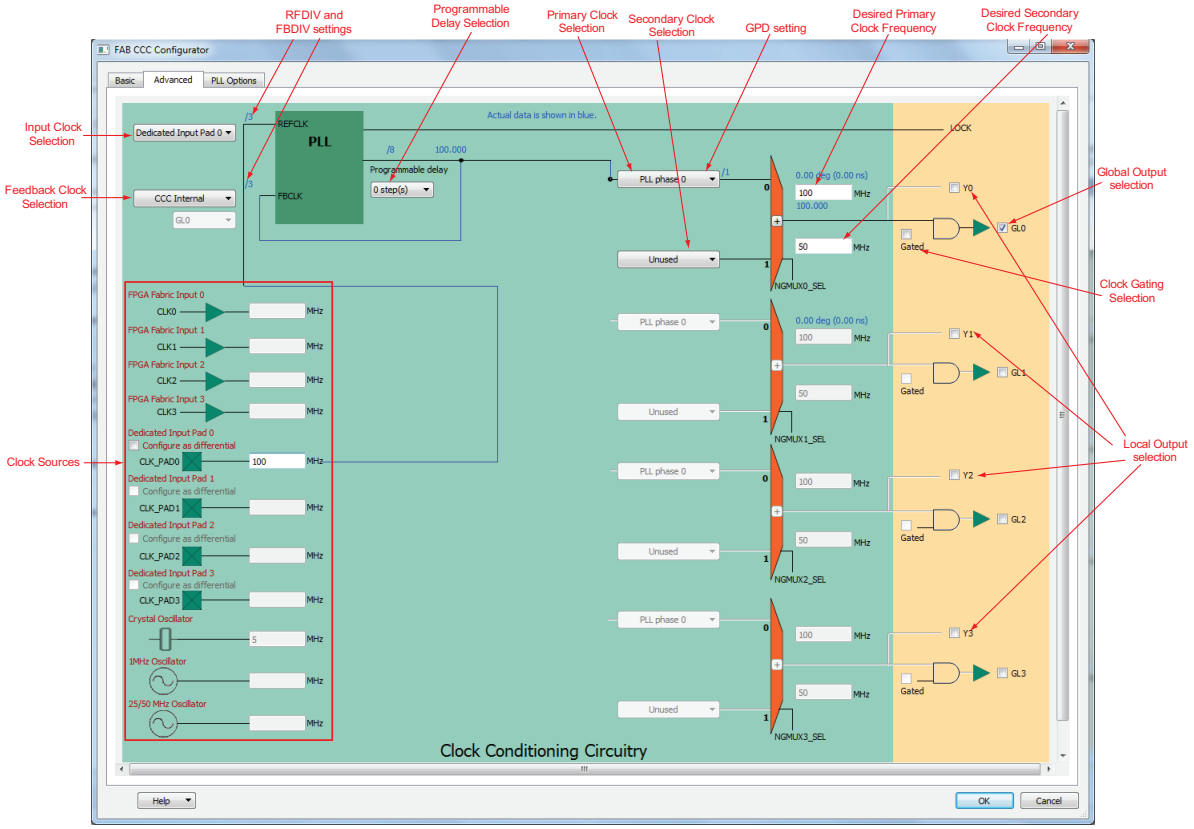


In basic configuration, the PLL is always used to generate the output frequency. The feedback of the PLL is internal to the CCC.

5.7.2 Advanced Configuration

The following figure shows the CCC configurator Advanced tab and highlights the configurable options.

Figure 57 • **CCC Advanced Configuration Tab**

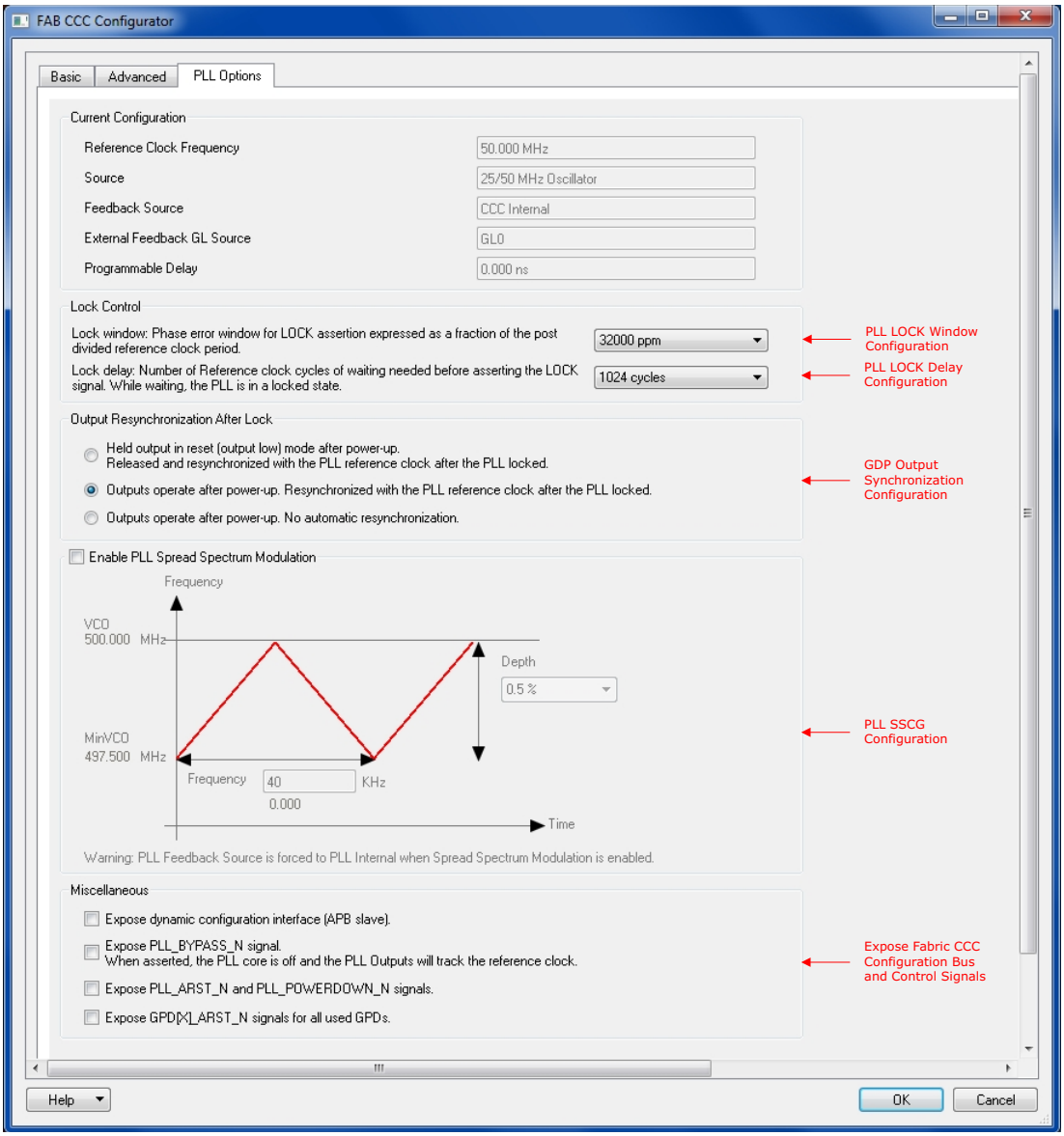


For advanced use cases, Microchip recommends to use the following flow to configure the clocks generated by the CCC (flow proceeds from right to left in the GUI, as shown in Figure 57, page 75):

- Select the number of desired output clocks (up to 4 global and local clocks).
- For each selected output clock, set the desired output frequency.
- For each selected output clock, select the desired reference clock (NGMUX primary clock) from which the output will be derived. It can be either of the following:
  - One of the CCC input clocks (PLL Bypass mode)
  - One of the 8 PLL output phases
- If required, configure the NGMUX secondary clock frequency and reference clock from which it will be derived. It can be either of the following:
  - One of the CCC input clocks (PLL Bypass mode)
  - One of the 8 PLL output phases
- If PLL is used:
  - Select the PLL reference clock source and frequency.
  - Select the PLL feedback clock source.
  - Select the PLL analog supply voltage to either 2.5 V or 3.3 V to match the supply on the board
  - If required, enable the programmable delay elements by configuring the delay.
- Enter the frequency of each selected clock source(s). The configurator uses these frequencies to compute the division factor of the PLL reference and feedback dividers as well as the GPD dividers shown in blue on the GUI.
- Configure Advanced options, such as GPD output synchronization configuration, PLL SSCG, and LOCK control parameters in the **PLL Options** tab, as shown in the following figure.

The CCC configurator automatically tries to compute a configuration that meets the frequency requirements and all the internal CCC/PLL constraints. If the configurator is unable to find an exact solution for all requirements, it finds a configuration that globally minimizes the error between the required and actual frequency. Actual data (divider settings, PLL output frequency, and actual output frequencies) is shown in the **Advanced** tab in blue.

Figure 58 • **CCC PLL Options Configuration Tab**

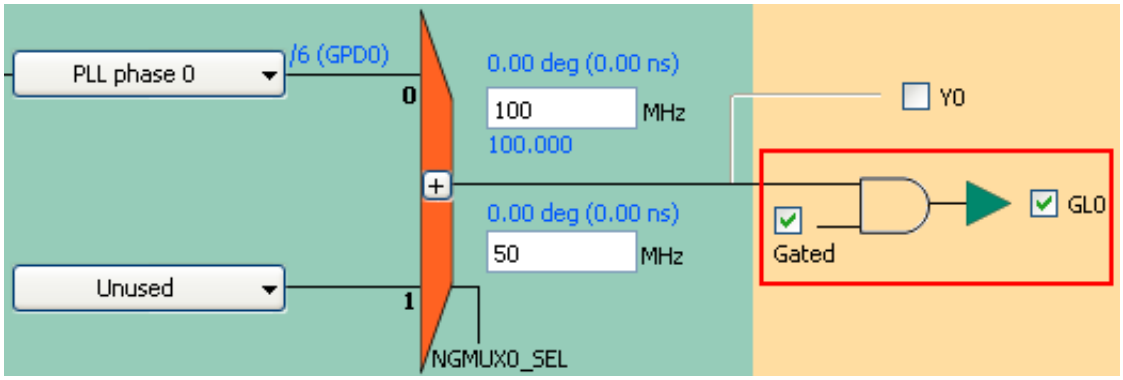


The following subsections explain the advanced configuration options available in the Advanced tab of the CCC configurator. The flow proceeds from right (output) to left (input) in the **Advanced** tab of the CCC configurator.

5.7.2.1 Gated Clock Configuration

For the selected global clock outputs (GLx), you can choose to have enable (GLx_EN) signal that can enable or disable the global clocks, as shown in the following figure.

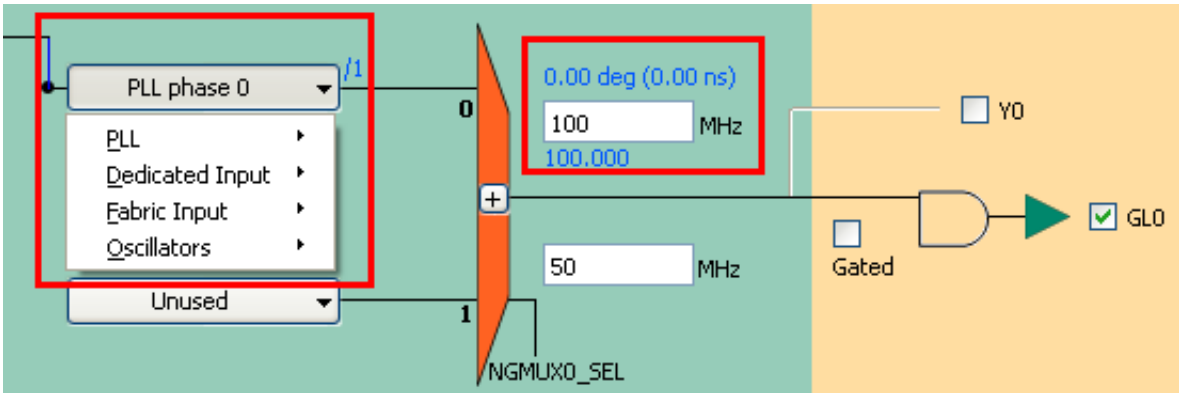
Figure 59 • **Clock Gating**



5.7.2.2 Primary Clock Source Selection

The source of the GLx/Yx clocks can be one of the CCC input clocks or the PLL outputs. You must specify the required primary output clock frequency, as shown in the following figure.

Figure 60 • **Primary Clock Source and Output Frequency Selection**

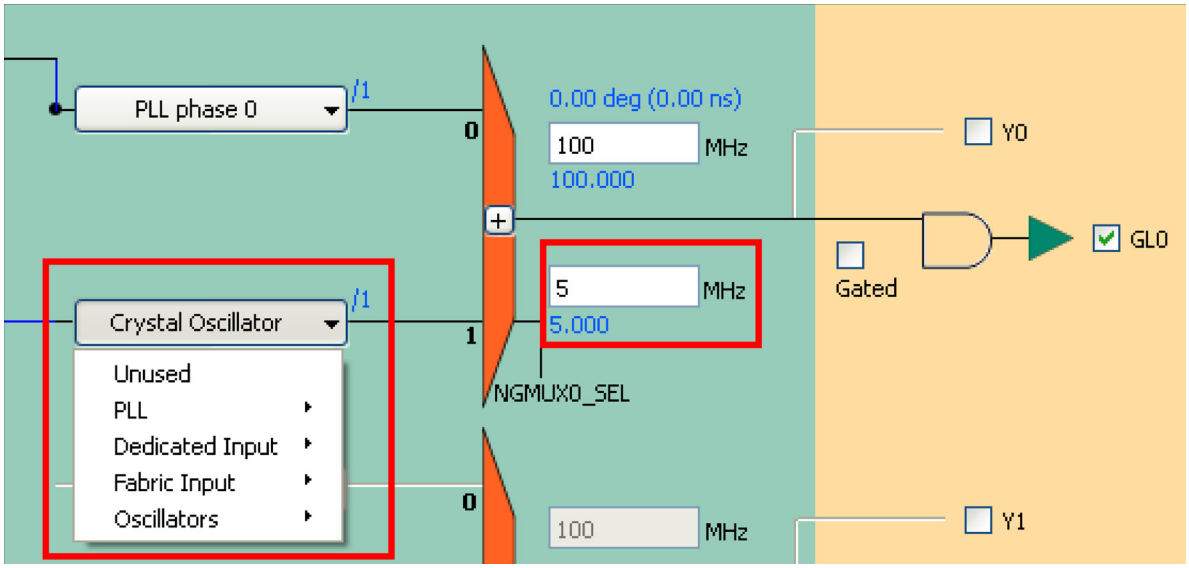


5.7.2.3 Secondary Clock Source Selection

Each enabled clock output can be configured to use a secondary clock source. You must specify the required secondary output clock frequency. Selecting a secondary clock source enables the NGMUX and exposes the input signal NGMUXx_SEL (with x = 0, 1, 2, 3). This signal can be used to dynamically switch between the primary and secondary clock. The transitions from primary to secondary clock and secondary to primary clock are guaranteed to be glitch free.

The following figure shows how to specify the secondary clock source.

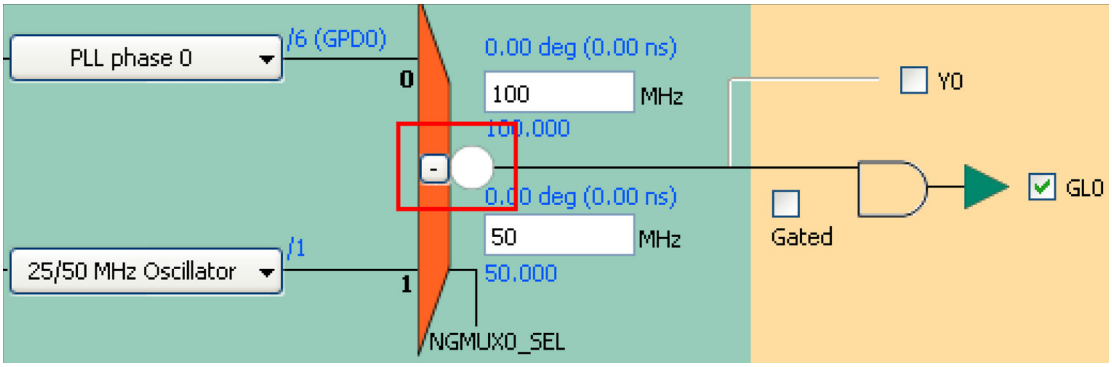
Figure 61 • **Secondary Clock Source and Output Frequency Selection**



5.7.2.4 Inversion Configuration

Each clock output can be inverted if required, as shown in the following figure. The inversion affects both the primary and secondary clock.

Figure 62 • **Clock Inversion Configuration**

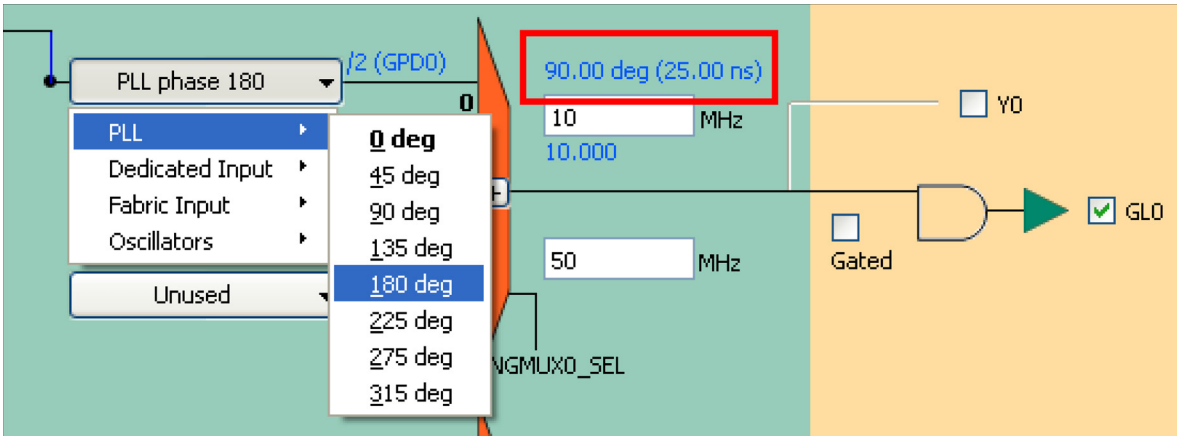


5.7.2.5 Output Clock Phase Adjustment

The output from the fabric CCC can be phase adjusted with respect to the PLL reference clock. The PLL block offers output in eight phases (0° to 315° in 45° steps). You can select a 0°, 45°, 90°, 135°, 180°, 225°, 270°, or 315° phase shift independently for each of the four outputs (GLx/Yx).

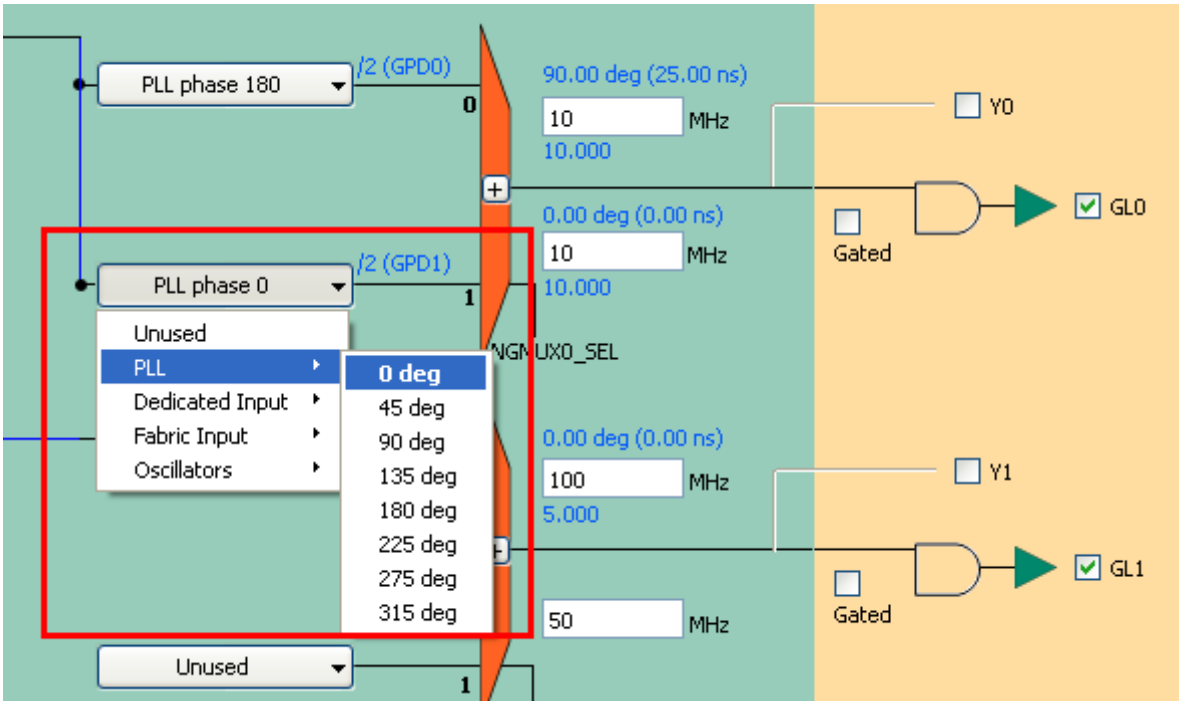
Note: Each one of the phase adjusted signals might also undergo further frequency division through the GPDs located at the outputs of the PLL. The actual phase is not the same as the selected phase, if the GPD division factor is not 1 (actual_phase = selected_phase/GPD division factor). The actual phases are highlighted in blue on the configurator GUI, as shown in the following figure.

Figure 63 • **Primary Clock Phase Adjustment**



If a secondary clock source is enabled for an output, the secondary clock can be phase adjusted with respect to the PLL reference clock by selecting one of the PLL outputs, as shown in the following figure.

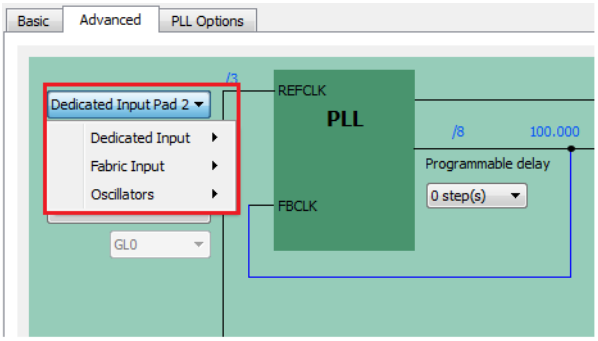
Figure 64 • **Secondary Clock Phase Adjustment**



5.7.2.6 PLL Reference Clock Source Selection

The PLL reference clock source and feedback clock source must be configured if any of the PLL outputs are selected for generating the output clocks. The PLL is bypassed if the PLL outputs are not selected for any of the clock outputs. The PLL reference clock source can be selected from the available CCC input clock sources, as shown in the following figure.

Figure 65 • **PLL Reference Clock Source Selection**



5.7.2.7 Internal PLL Feedback Clock Source

As shown in Figure 66, page 80, if you choose to use the PLL with internal feedback, the feedback path can be one of the following:

- CCC internal feedback from PLL phase 0° output – Default value (Figure 67, page 80)
- PLL internal feedback from PLL phase 0° output – This is a shorter feedback path integrated within the PLL (Figure 68, page 81).

You must use PLL internal feedback when using spread spectrum modulation or if the PLL reference clock frequency is 32 KHz. Programmable delay and output synchronization are not available in this mode.

Figure 66 • **Internal PLL Feedback Clock Source Selection**

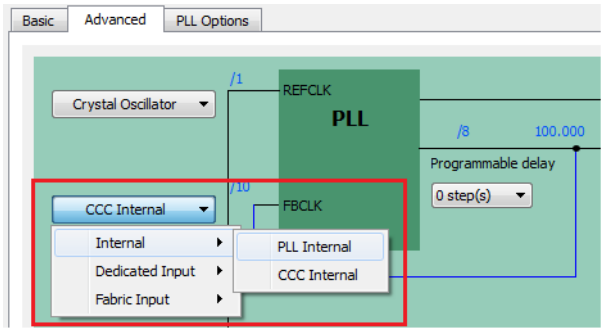


Figure 67 • **CCC Internal Feedback**

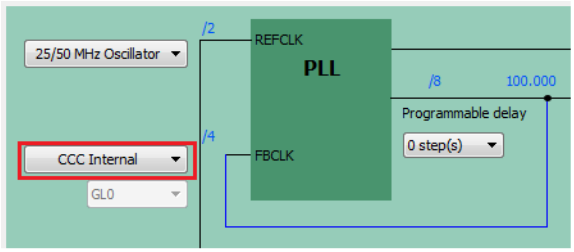
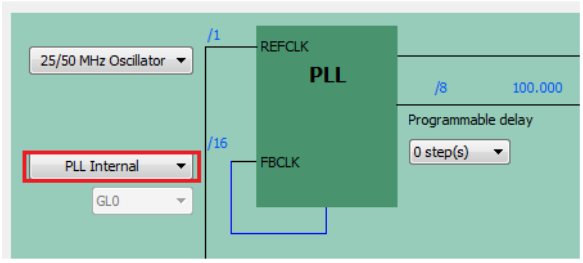


Figure 68 • **PLL Internal Feedback**



5.7.2.8 External PLL Feedback Clock Source

If the PLL is used with an external feedback source, all computations are based on the assumption that the external feedback is driven from the selected GLx/Yx fabric CCC output. This feedback may be driven through the FPGA fabric from one of the four FPGA fabric inputs (Figure 69, page 81) or externally to the chip, using one of the four dedicated global I/Os (Figure 70, page 81).

Figure 69 • **GL0 as External Feedback through FPGA Fabric Input 0**

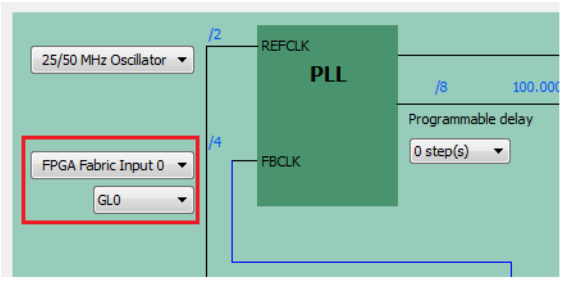
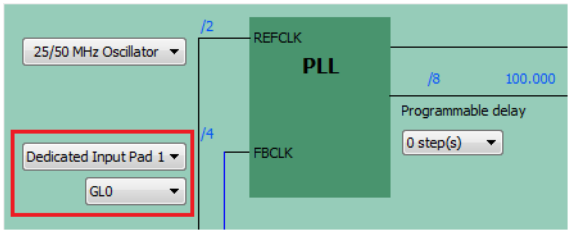


Figure 70 • **GL0 as External Feedback through Dedicated Global I/O Pad 1**

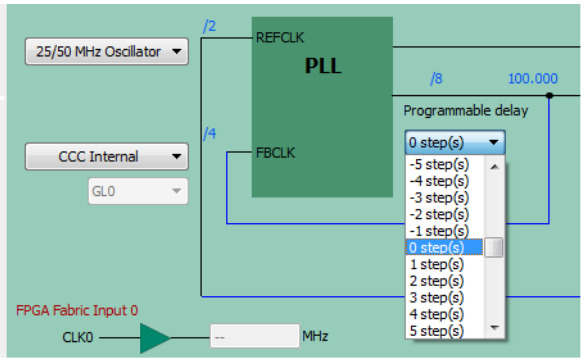


5.7.2.9 Clock Delay Adjustment

As shown in the following figure, the programmable delay elements can be enabled for clock delay adjustments. The programmable delay has 64 positive steps when applied to the PLL reference clock path and 64 negative steps when applied to the PLL feedback clock. See *DS0128: SmartFusion2 and IGLOO2 Datasheet* or perform timing analysis for the actual value of each step. The programmable delay steps are not compensated for process, voltage, or temperature (PVT).

The programmable delay elements are not available when the PLL internal feedback path is selected.

Figure 71 • **Programmable Delay Configuration**



5.7.2.10 CCC Input Clock Sources

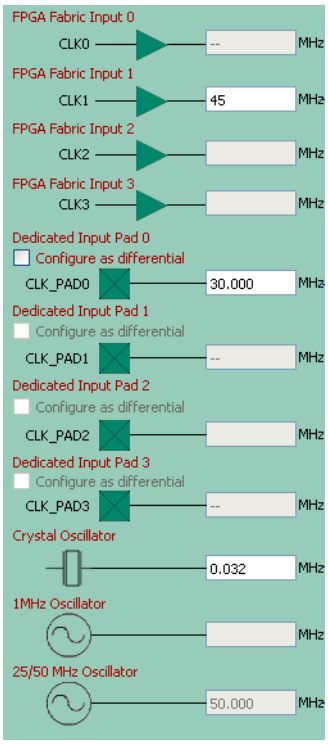
The following sources are available as the CCC inputs for the PLL reference clock, PLL feedback clock, and output direct connection:

- Dedicated input pads: The clock source is one of the four dedicated global I/Os that has a dedicated path to the CCC.
  - Some of the dedicated inputs to the CCC can be configured to use a differential I/O technology.
- FPGA fabric inputs: The clock source can be any signal coming from the FPGA fabric.
- Main crystal oscillator: The source is the main crystal oscillator with an external crystal, ceramic resonator, or RC circuit connected to its external pins.
- 1 MHz RC oscillator: The source is the on-chip 1 MHz oscillator.
- 50 MHz RC oscillator: The source is the on-chip 50 MHz oscillator.

The frequency of each CCC input used in the configuration must be entered in the *CCC* configurator, as shown in the following figure. These frequencies are used to compute the PLL configuration and divider configuration that meet the output frequency requirements. Note the following frequency requirements:

- The on-chip 1 MHz Oscillator clock frequency is fixed to 1 MHz and cannot be changed.
- The on-chip 50 MHz Oscillator clock frequency is fixed to 50 MHz and cannot be changed.
- The main crystal oscillator frequency must be between 0.032 MHz and 20 MHz.
- PLL reference clock frequency must be between 1 MHz and 200 MHz or 32 KHz
  - 32 KHz reference clock frequency is available for one CCC/PLL.
- In PLL Bypass mode, there is no minimum limit on the input frequency sourced from the dedicated global I/Os or FPGA fabric inputs. Maximum output frequency that can be achieved is 400 MHz.

Figure 72 • **CCC Input Clock Sources**



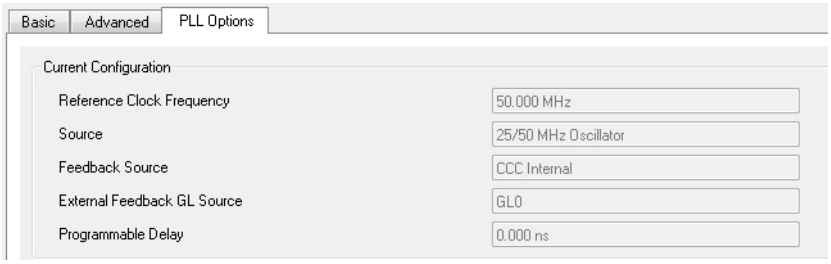
The following subsections explain the advanced configuration options available in the **PLL Options** tab of the CCC configurator. The **PLL Options** tab enables to configure the PLL lock control parameters, spread spectrum modulation parameters, GPD resynchronization options, and to expose various control signals.

5.7.3 PLL Options

5.7.3.1 PLL Current Configuration Summary

As shown in the following figure, the Current configuration section of the **PLL Options** tab summarizes the PLL current configuration according to the selections made in the **Basic** tab and **Advanced** tab of the CCC configurator.

Figure 73 • **PLL Configuration Summary**



5.7.3.2 PLL Lock Control

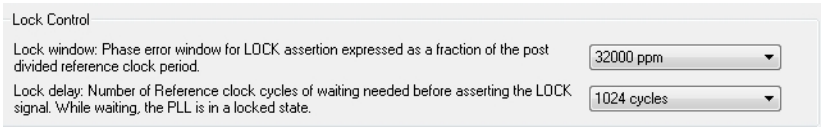
As shown in the following figure, set the PLL lock control parameters, Lock window, and Lock delay, as appropriate for the design.

- Lock Window – Enables configuration of the maximum phase error allowed for the PLL to indicate it has been locked. The lock window is expressed as ppm of the RFCLK frequency.

Note: For SmartFusion 2 and IGLOO 2 devices, if the fabric PLL reference clock is sourced from an internal RC oscillator, the minimum PLL lock window setting should be 64000 ppm and 32000 ppm for 1 MHz RC oscillator and 50 MHz RC oscillator, respectively. If the RFDIV divisor value is 2 or less and a wrong PLL lock window setting is used, then the PLL lock may get toggled.
The combination of high register utilization and toggling rate results high PLL jitter and subsequently the PLL may come out of lock. Use higher PLL lock window for stable PLL operation.

- Lock Delay – Enables configuration of the number of divided RFCLK clock cycles by which the lock is delayed after the PLL has reached the lock condition.

Figure 74 • **PLL Lock Control Settings**

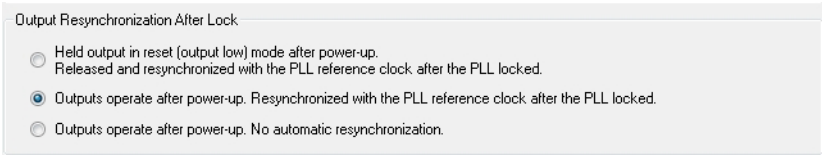


5.7.3.3 Output Resynchronization Configuration

Each fabric CCC contains four GPDs. The GPDs source and division settings are automatically configured based on the frequency requirement specified in the CCC configurator. Microchip recommends to resynchronize the GPDs after the PLL lock is achieved.

The following figure shows the output resynchronization options available in the **PLL Options** tab. See the GPD Operating Modes, page 56 for more information on the GPD synchronization.

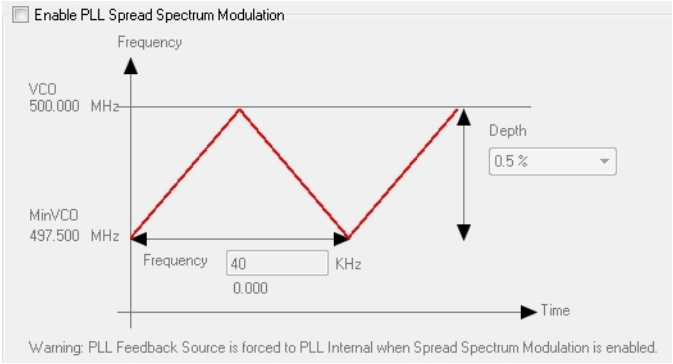
Figure 75 • **GPD Synchronization Configuration**



5.7.3.4 Spread Spectrum Clock Generation

The PLL SSCG functionality can be enabled by selecting **Enable PLL Spread Spectrum Modulation** in the PLL Options tab. When the PLL SSCG functionality is enabled, the output frequency of the PLL is down-spread over time. The SSCG programmable options allow control of the modulation frequency and depth based on the design requirements. The *CCC* configurator forces the PLL feedback source to PLL internal when SSCG is enabled. See the SSCG, page 53 for more information on SSCG.

Figure 76 • **PLL SSCG Configuration**



5.7.3.5 Miscellaneous Options

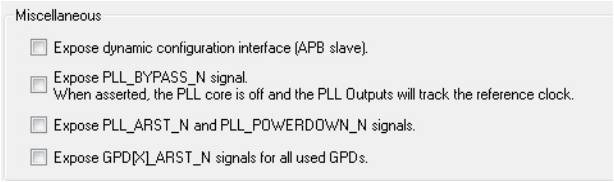
The following control signals can be exposed to the FPGA fabric:

- Dynamic configuration interface (APB3 slave)
- PLL control signals (PLL_BYPASS_N, PLL_ARST_N, and PLL_POWERDOWN_N)
- GPD asynchronous reset signals (GPDx_ARST_N with x= 0 to 3)

These control signals are deasserted by default by the configurator when they are not exposed to the FPGA fabric. APB slave interfaces can be exposed for fabric CCC dynamic configuration. See the Functional Description, page 45 for more information on these control signals.

The following figure shows the miscellaneous options available for exposing fabric CCC control signals.

Figure 77 • **Miscellaneous Options - Fabric CCC Control Signals**



5.7.4 Simulation Support

Microchip Libero SoC design software provides pre-compiled simulation models for the *CCC* macro to show functional behavior of the fabric CCC. The simulation steps include generating the top-level component, which instantiates *CCC* macro, performing simulation for verification with the ModelSim tool, and performing static timing analysis with SmartTime in the Libero SoC software.

If the reference clock input of CCC is an oscillator, then clock constraint is automatically generated for the reference clock input of CCC and generated clock constraints are added in SmartTime for the CCC outputs. If the reference clock input of CCC is from FPGA Fabric Input or Dedicated Input Pad, the clock constraint is not generated for reference clock input of CCC but generated clock constraints are added in SmartTime for the CCC outputs. You need to add constraint on reference clock source output.

Note: The simulation models does not support PLL LOCK DELAY.

5.7.4.1 CCC Simulation Model Limitations

The following table lists the limitations for simulating CCC behavior in IGLOO 2 and SmartFusion 2 devices:

Table 54 • **Simulation Support for IGLOO 2 and SmartFusion 2 Devices**

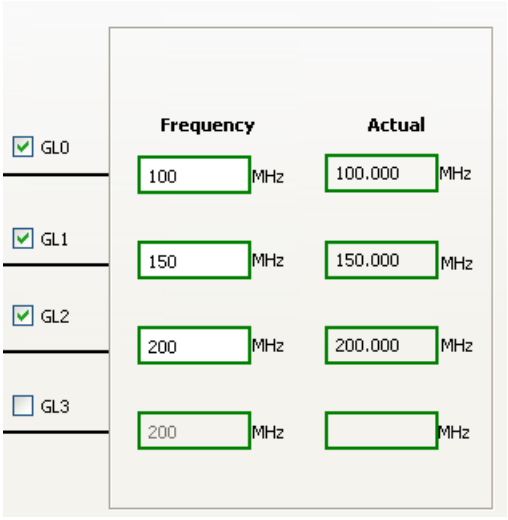
Feature	Simulation Support
Programmable Delay (negative step delays)	Negative step delays are not modelled in PLL/CCC internal feedback mode.
Clock Delay shown in Configurator	“GLx Total Delay” shown in configurator may not have same values in simulation. Refer to Static Timing Analysis path reports for clock generation delay.
Lock generation	This does not necessarily reflect the real lock time of silicon. • In PLL/CCC internal feedback mode, it will come after 5 PFD cycles. • In external feedback mode, it will come after stabilization which depends upon configuration GLx outputs. • Lock Delay & Lock Window fields of configurator will not affect the simulation.
PLL Spread Spectrum	Not Modelled
External feedback Phase alignment	Not Modelled
Phase alignment	Modelled

5.7.5 Clock Frequency Synthesis

Deriving clocks of various frequencies from a single reference clock is known as frequency synthesis. This use model configures one of the on-chip oscillators as a clock source to the fabric CCC(s) and requires the instantiation of both the Chip Oscillators macro and CCC macro in the design. The Chip Oscillators macro must be configured to drive the CCC macro reference clock input. This use model uses the **Basic** tab of the CCC configurator for CCC configuration. See the How to Use On-Chip Oscillators, page 35 for on-chip oscillator configuration. It is also possible to use external clock sources through dedicated global I/Os instead of on-chip oscillators.

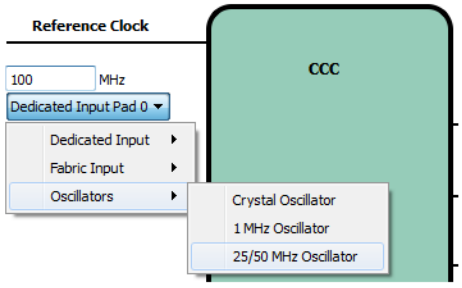
1. In the *CCC* configurator, select the output clocks and enter the desired clock frequencies. These frequencies are used by the *CCC* configurator to compute the configuration of the CCC dividers and PLL to meet the requirements. The **Actual** column displays the actual value that the configurator was able to achieve.
   - GL0 - 100 MHz
   - GL1 - 150 MHz
   - GL2 - 200 MHz

Figure 78 • **Output Clocks Settings**



2. Select the fabric CCC reference clock input from the drop-down menu. For this use model, select the 50 MHz RC Oscillator as the clock source.

Figure 79 • **Fabric CCC Reference Clock Selection**



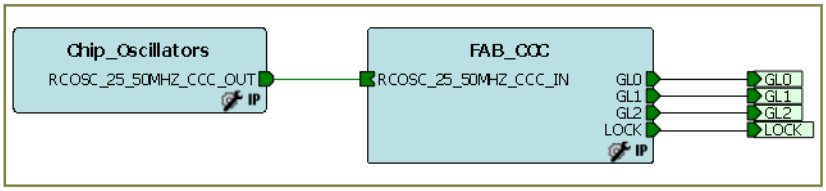
This configuration is sufficient for generating the desired clock frequencies.

3. Click **OK** to close the *CCC* configurator.

4. Connect the Chip Oscillators macro output to the CCC macro reference clock input. The following figure shows the final SmartDesign component with CCC outputs promoted to top-level for simulation.

The generated clocks can then be connected to the fabric logic, external components through I/Os, or as a base clock to the SmartFusion 2 and IGLOO 2 hard IP blocks (MSS/HPMS, FDDR, and SERDESIF).

Figure 80 • **FAB CCC and On-chip Oscillators Connectivity**



Next, perform simulation in ModelSim to verify the generated output frequencies.

5.7.6 Clock Delay Adjustment

For PLLs, delays in the reference clock path enables clock delay and delay in the feedback clock path enables clock advancements with respect to the reference clock. A typical application for clock advancement is to advance the on-chip clock in order to cancel the global network delay and align the internal clock to an external clock on board.

Perform the fabric CCC basic configuration using the **Basic** tab of the CCC configurator as outlined in the Clock Frequency Synthesis, page 86. For clock delay adjustments, configure the fabric CCC programmable delay elements using the **Advanced** tab of the CCC configurator. Selecting a positive delay value enables the delay elements in the reference clock path to delay the output clock with respect to the PLL reference clock (Figure 81, page 88).

Selecting a negative delay value in the CCC configurator enables the delay element in the feedback clock path for the output clock advancement with respect to the PLL reference clock (Figure 82, page 88). Perform the timing analysis using SmartTime for the actual delay between the reference clock and output clocks.

Figure 81 • **Clock Delay Simulation Waveform**

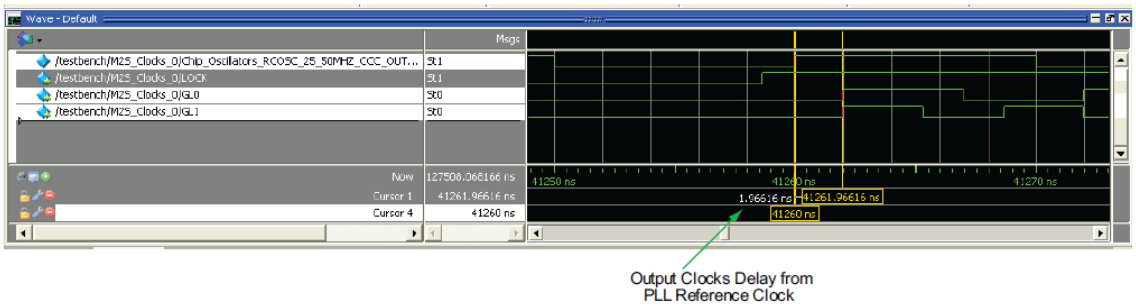
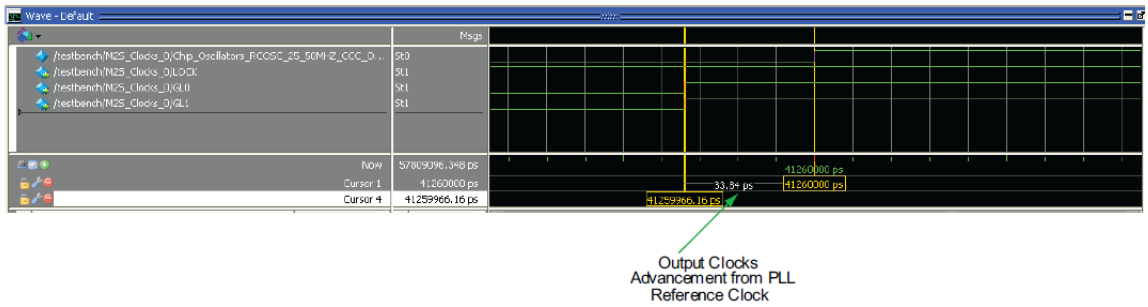


Figure 82 • **Clock Advancement Simulation Waveform**



5.7.7 External Clock Source Through Dedicated Global I/O

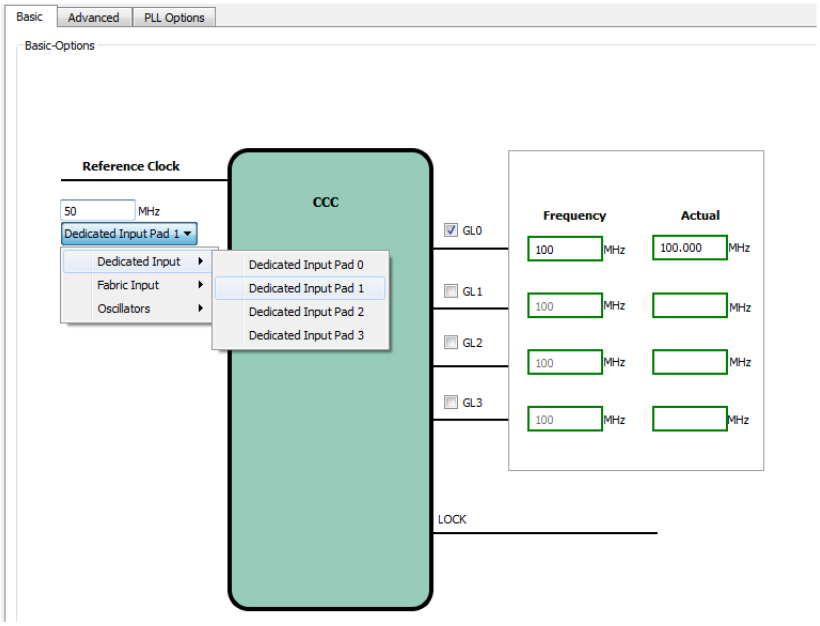
This use model shows how to source the fabric CCC reference clock from an external clock source. Each fabric CCC is associated with four dedicated global I/Os for providing external reference clock.

The mapping of fabric CCCs is done by the Libero SoC place-and-route software based on the dedicated global I/O selected and the reference clock frequency. If the reference clock frequency is set to 32 KHz then the Libero SoC place-and-route software uses the CCC-NE1 and the external clock can come through the dedicated global I/Os associated with the CCC-NE1. The selection of the dedicated global I/Os is performed through I/O Attribute Editor of the Libero SoC design software.

Use the following steps to configure the fabric CCC with external reference clock:

1. Instantiate the CCC macro onto the SmartDesign canvas.
2. In the CCC macro configurator,
   - Select the output clocks and enter the desired clock frequencies.
   - Select one of the four dedicated global input pads as CCC reference clock source and set its frequency, as shown in the following figure.

Figure 83 • **Fabric CCC Configuration**

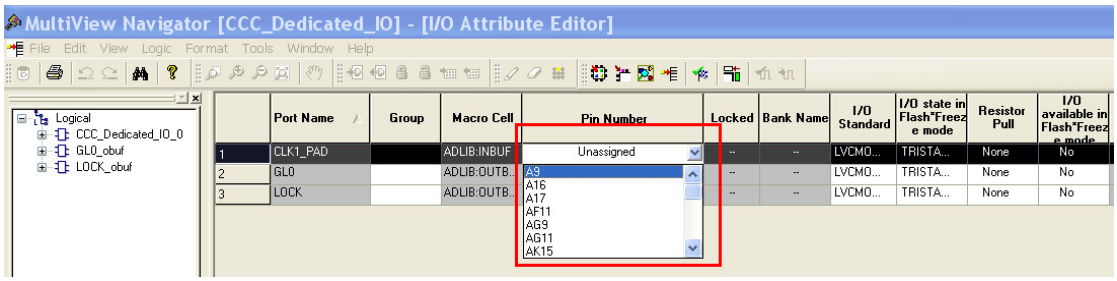


3. Connect the generated fabric CCC output clocks to the desired modules in the design.
4. Generate the top-level SmartDesign component and make sure that the top-level component is generated successfully.
5. Perform the Synthesis of the design.
6. Compile the design to perform legality check and basic netlist optimization.
7. Open **I/O Attribute Editor** (in the Libero SoC Design Flow window, **Implement Design > Constrain Place and Route > Edit I/O Attributes**) to edit I/O attributes.
8. Perform the pin assignment for the fabric CCC dedicated global input pad, as shown in the following figure.

A dedicated global I/O can be selected, for which the dedicated global input pad number (CLKI0, CLKI1, CLKI2, or CLKI3) is the same as the fabric CCC reference clock source selected in Step 2. For example, **Dedicated Input Pad 1** is selected as the reference clock source. The pin assignment can be done to a pin, whose pin name is CCC_xyz_CLKI1, where xy represents the CCC location and z represent the CCC number.

Note: When the reference clock input of CCC is Dedicated Input Pad 2 and is configured as differential, then CCC has two differential inputs—CLK2_PADP, CLK2_PADN. First, the pin assignment for positive polarity pin (CLK2_PADP) must be done to CCC_xyz_CLKI2 pin, where xy represents CCC location and z represents CCC number. The pin assignment for negative polarity pin (CLK2_PADN) is done automatically by the Libero SoC software in the IO Attribute Editor after positive polarity pin assignment. Wrong pin assignment for positive polarity pin leads to DRC error during Place-and-Route.

Figure 84 • **Dedicated Global I/O Selection**



9. Commit the changes in I/O Attribute Editor.
   Perform the place-and-route operation. The place-and-route software instantiates the associated fabric CCC based on the pin assignment.

6 MSS/HPMS Clock Conditioning Circuitry

SmartFusion 2 or IGLOO 2 devices have a dedicated CCC (MSS CCC or HPMS CCC) for generating aligned clocks to all the MSS or HPMS sub-blocks for correct operation and synchronous communication with the user logic in the FPGA fabric. The base clock for the MSS CCC or HPMS CCC (MCCC_CLK_BASE) comes either from a fabric CCC or an external source through the FPGA fabric. The MSS CCC or HPMS CCC is associated with a dedicated PLL (MPLL) for clock synthesis and for deskewing the internal MSS or HPMS clock from MCCC_CLK_BASE. This chapter describes the MSS CCC and HPMS CCC operation, configuration, and use models.

6.1 Features

The MSS CCC and HPMS CCC supports the following configurable features:

- MCCC_CLK_BASE frequency range from 1 MHz to 200 MHz
- MSS (M3_CLK) or HPMS clock (HPMS_CLK) frequency up to 166 MHz
- MDDR clock (MDDR_CLK) frequency up to 333 MHz
  - Multiple—1, 2, 3, 4, 6, or 8—of M3_CLK or HPMS_CLK
- MDDR fabric interface clock (DDR_FIC_CLK) frequency
  - Ratio—1, 2, 3, 4, 6, 8, 12, or 16—of MDDR_CLK
- MSS or HPMS APB0/1 peripherals clock frequency (APB_0_CLK and APB_1_CLK)
  - Ratio—1, 2, 4, or 8—of M3_CLK or HPMS_CLK
- FPGA fabric interface clocks (FIC_0_CLK and FIC_1_CLK)
  - Ratio—1, 2, 4, 8, 16, or 32—of M3_CLK or HPMS_CLK
- Configurable phase error window for MPLL lock assertion
- Configurable MPLL lock delay
- MPLL lock assertion and deassertion interrupts to the Cortex-M3 processor (SmartFusion 2 only)
- MPLL lock generation for the FPGA fabric
- Fabric PLL lock monitor if MCCC_CLK_BASE is generated by a fabric CCC
- MSS or HPMS clock source selection for Flash*Freeze mode
  - One of the on-chip oscillators

The following figures show the system-level block diagram of the MSS CCC and HPMS CCC. The figures show the inputs and outputs for one fabric CCC; each fabric CCC has a similar set of inputs and outputs.

Figure 85 • **MSS CCC System-Level Block Diagram**

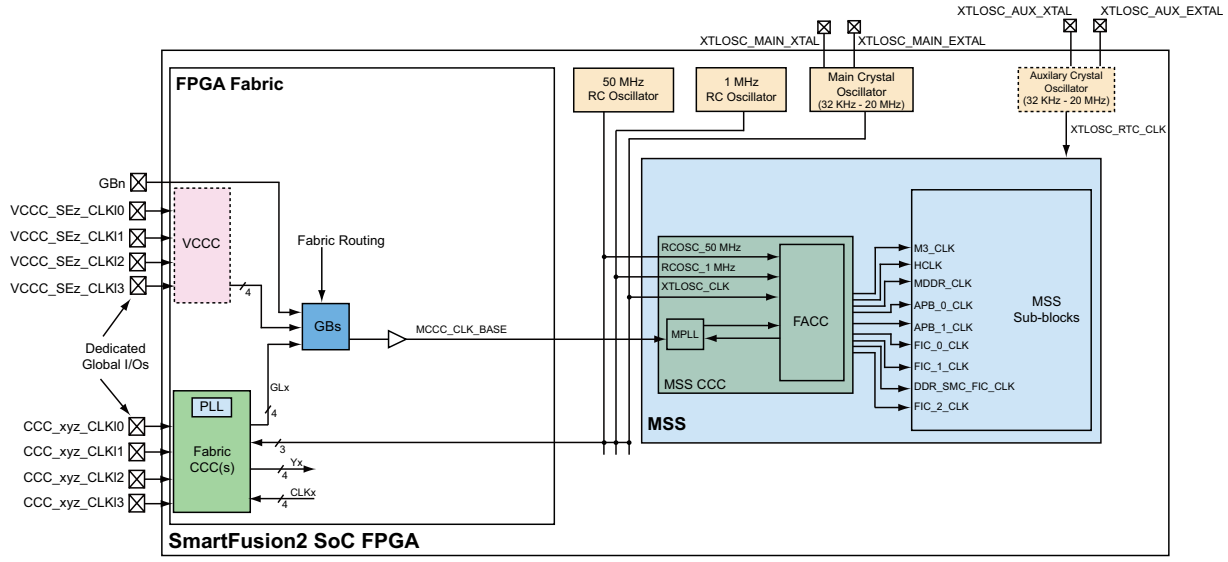
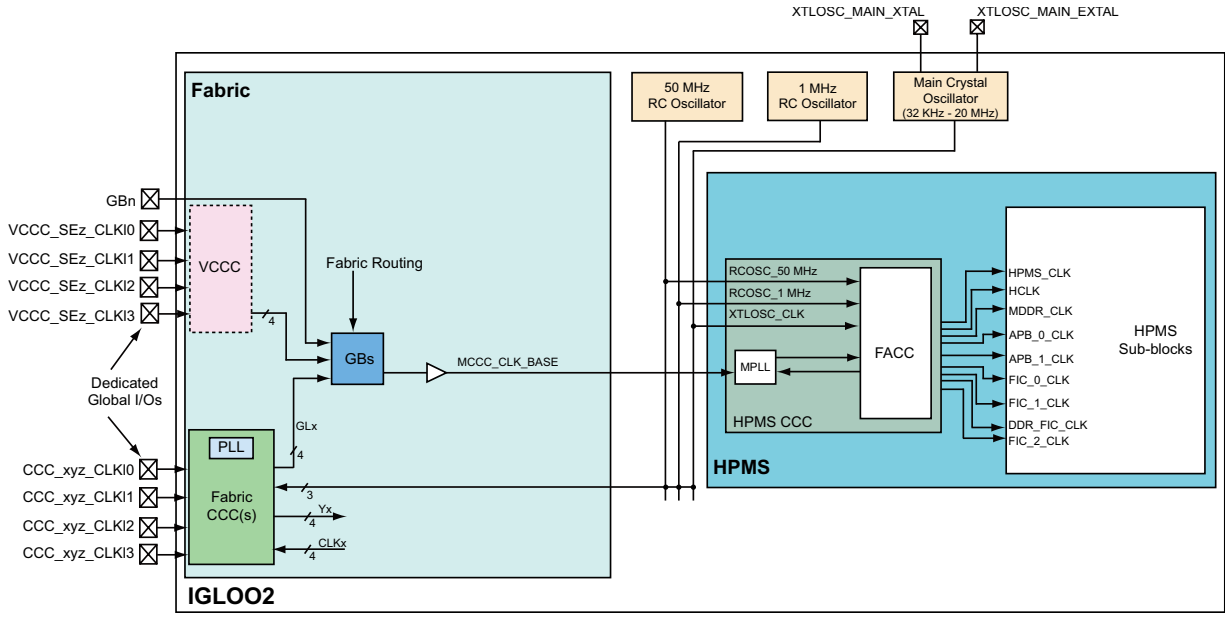


Figure 86 • **HPMS CCC System-Level Block Diagram**



6.2 Functional Description

The MSS CCC or HPMS CCC consists of an MPLL and fabric alignment clock controller (FACC) that generates aligned clocks for MSS or HPMS operation and synchronous communication with the user logic. The following figures show aligned clocks (indicated in blue) generated by the MSS CCC or HPMS CCC and MSS or HPMS clocking scheme.

The following topics are covered in this section:

- MPLL
- FACC
- MSS CCC or HPMS CCC Operational Modes
- PLLs Lock Monitoring

See the MSS CCC or HPMS CCC Configuration, page 100 for information on configuring the MSS CCC and HPMS CCC features.

Figure 87 • **SmartFusion 2 MSS Clocking Scheme**

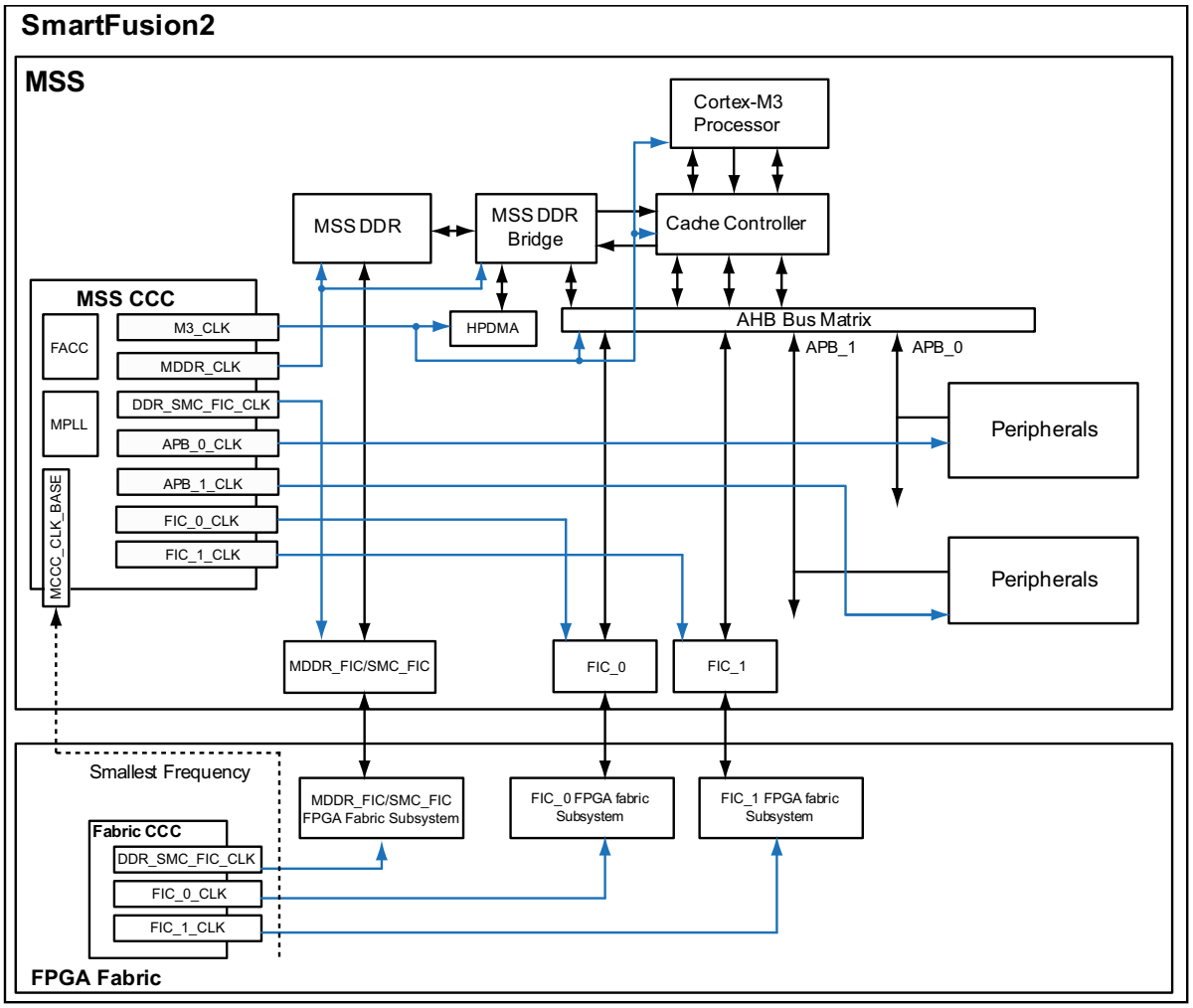
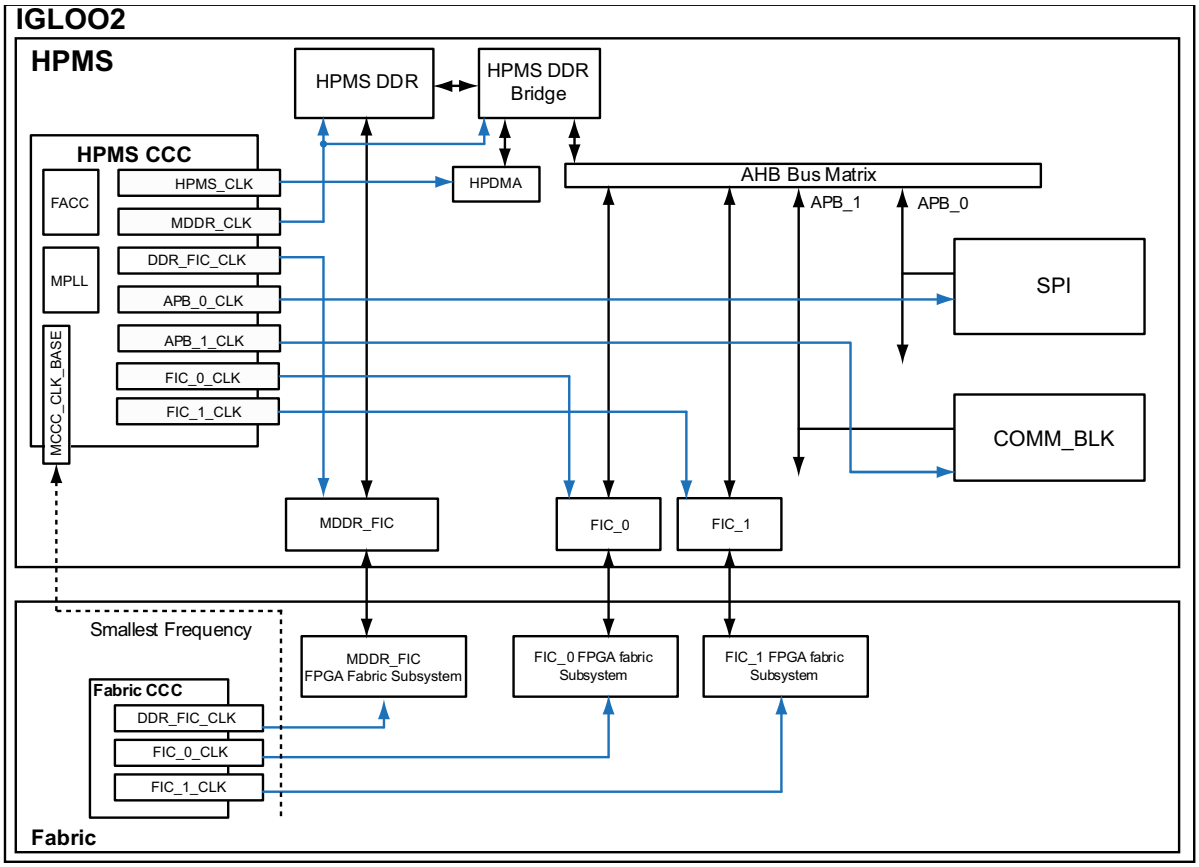


Figure 88 • **IGLOO 2 FPGA HPMS Clocking Scheme**



6.2.1 MPLL

The MCCC_CLK_BASE from the FPGA fabric is used as a reference clock to the MPLL, and is multiplied to generate a clock frequency of up to 333 MHz. The MCCC_CLK_BASE can be generated from a fabric CCC/PLL, one of the on-chip oscillators, or an external source through a dedicated global I/O.

The supplies required to power the MPLL are the VDD for the digital section and analog supply (MDDR_PLL_VDDA) for the analog section. The MDDR_PLL_VDDA can be either 2.5 V or 3.3 V, based on the power supply availability on board. The analog power supply voltage (2.5 V or 3.3 V) does not impact the MPLL frequency range. The PLL analog supply voltage is internally regulated, which provides independence from the external supply. See *SmartFusion2 and IGLOO2 Datasheet* for the MPLL operational range and characteristics.

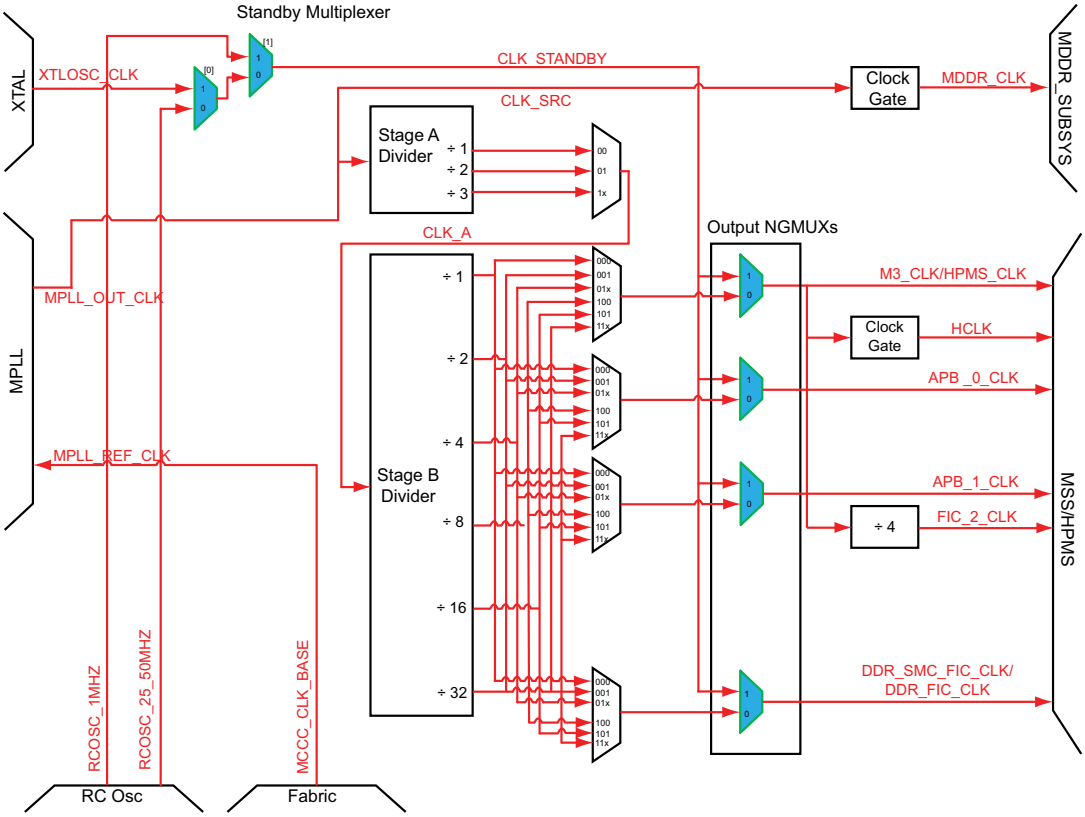
The MPLL generates a lock signal (MCCC_MPLL_LOCK) to indicate that the MPLL is locked on to the MCCC_CLK_BASE signal. The precision of the MCCC_MPLL_LOCK discrimination can be adjusted using the lock window controls. The lock window represents the phase error window for lock assertion. The lock window can be adjusted between 500 ppm and 32,000 ppm in powers of 2. The integration of the lock period can be adjusted using a built-in lock counter. The lock counter or lock delay indicates the number of divided reference clock cycles to wait after the MPLL is locked before asserting the MCCC_MPLL_LOCK signal. The lock delay is useful for avoiding false toggling of the MPLL lock signal. The lock counter can be configured between 32 and 32,768 cycles in multiples of 2.

There are two interrupts connected to the NVIC of the Cortex-M3 processor to indicate the MPLL lock assertion and deassertion. The MCCC_MPLL_LOCK signal can be monitored by the user logic in the FPGA fabric.

6.2.2 FACC

Within the MSS or HPMS CCC, the FACC is responsible for interfacing with the MPLL, generating the various aligned clocks required by the MSS or HPMS sub-blocks, and controlling the alignment of the FPGA fabric interface clocks. The following figure shows the functional block diagram of the FACC.

Figure 89 • **FACC Functional Block Diagram of SmartFusion 2 and IGLOO 2**



The following table lists clocks generated by the FACC.

Table 55 • **Clocks Generated by FACC**

Clock Name	Description
MDDR_CLK	Clocks the MDDR subsystem and the MSS or HPMS DDR bridge. It can be operated up to 333 MHz depending on DDR memory type that is present in the system.
M3_CLK or HPMS_CLK and HCLK	M3_CLK is used within the MSS to clock the Cortex-M3 processor, HPDMA, and the AHB bus matrix. HPMS_CLK is used within the HPMS to clock the AHB bus matrix and HPDMA. HCLK always at the same frequency as the M3_CLK or HPMS_CLK but is a gated version of it. M3_CLK/HPMS_CLK maximum frequency is dependent on device speed grade. For -1 speed grade, the maximum frequency is 166 MHz and for standard speed grade, it is 142 MHz.
APB_0_CLK/PCLK0	Clocks all the peripherals connected on the APB_0 bus.
APB_1_CLK/PCLK1	Clocks all the peripherals connected on the APB_1 bus.
FIC_0_CLK and FIC_1_CLK	Clock the MSS or HPMS FIC_0 and FIC_1 interfaces respectively. These clocks define the frequency at which the connected FPGA fabric subsystem is intended to operate.
FIC_2_CLK	Clocks the APB configuration interface (FIC_2), FIC_2 clock is M3_CLK or HPMS_CLK divided by 4.

Table 55 • Clocks Generated by FACC (continued)

Clock Name	Description
DDR_SMC_FIC_CLK or DDR_FIC_CLK	Clocks the MDDR_FIC or SMC_FIC, and defines the frequency at which the connected FPGA fabric subsystem is intended to operate.

6.2.2.1 FACC Dividers

The FACC consists of two stages of clock dividers: Stage A and Stage B. These stages divide the high-speed clock coming from the MPLL to generate the aligned clocks according to the configured division ratios. These aligned clocks can be at different frequencies to each other, but the rising edges of slower clocks occur coincidentally with the rising edges of faster clocks.

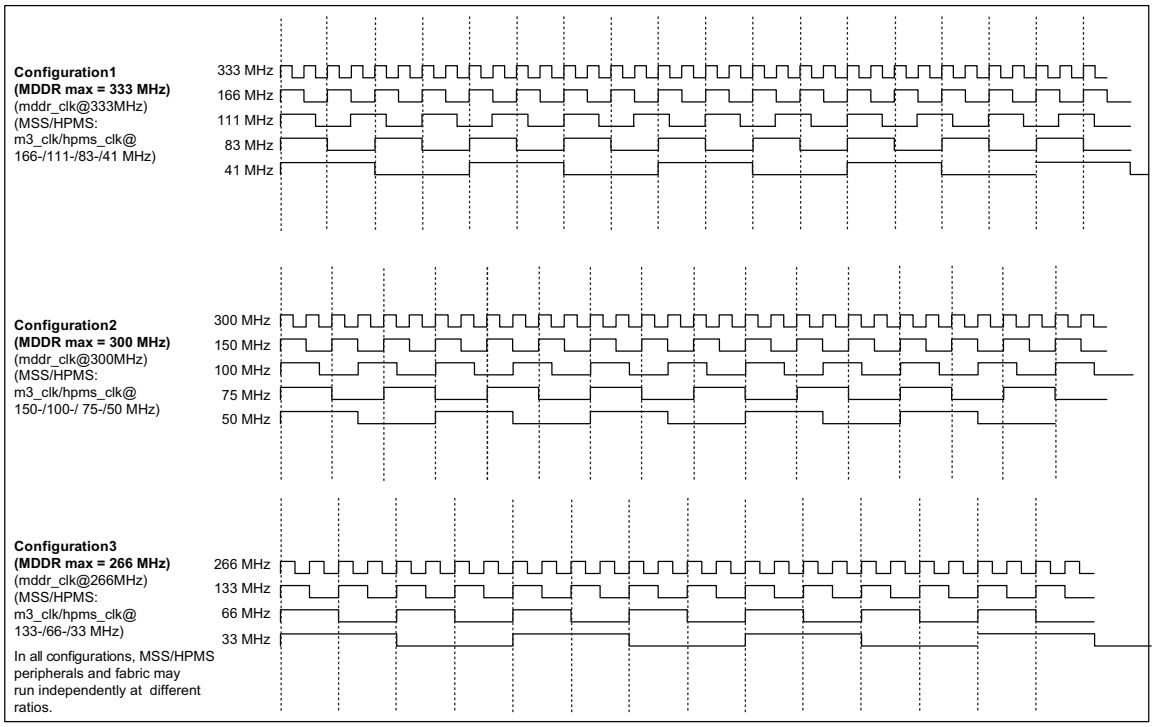
As shown in Figure 89, page 94, the Stage A divider divides the MDDR_CLK by 1, 2, or 3, to feed CLK_A into the Stage B divider. The Stage B divider has a number of outputs, which select the division by 1, 2, 4, 8, 16, or 32. Each of the MSS and HPMS clocks can be configured to select one of these divider taps.

Following are the MSS and HPMS clocks configuration constraints. These constraints are automatically checked by the MSS_CCC and IGLOO 2 System Builder in the Libero SoC software.

1. Maximum frequency of MDDR_CLK is 333 MHz.
2. Maximum frequency of M3_CLK and HPMS_CLK is 166 MHz for -1 speed grade and 142 MHz for standard speed grade.
3. Maximum frequency of DDR_SMC_FIC_CLK and DDR_FIC_CLK is 200 MHz
4. M3_CLK must be greater than 30 MHz when the MSS USB interface is enabled.
5. The MCCC_CLK_BASE must be the lowest of any aligned clocks interfacing between the MSS or HPMS and the FPGA fabric.
6. The possible MDDR_CLK:M3_CLK or HPMS_CLK ratios are 1:1, 2:1, 3:1, 4:1, 6:1, or 8:1.
7. The APB_0_CLK and APB_1_CLK may have different frequencies from each other but both must be equal to, or slower than M3_CLK or HPMS_CLK. The possible M3_CLK or HPMS_CLK:APB_0_CLK/APB_1_CLK ratios are 1:1, 2:1, 4:1, or 8:1.
8. Only in the SmartFusion 2 device, in soft memory controller (SMC) mode (that is, when the SMC_FIC block is operational), the MDDR_CLK and DDR_SMC_FIC_CLK must run at the same speed as to the M3_CLK.
9. When MDDR_FIC is operational, DDR_FIC_CLK must run at the same speed or slower than the MDDR_CLK. Possible configurations for DDR_FIC_CLK:
   - If MDDR_CLK is 3, 6 multiples of M3_CLK or HPMS_CLK then DDR_FIC_CLK must be MDDR_CLK/n where n={3,6,12}
   - If MDDR_CLK is 1, 2, 4, 8 multiples of M3_CLK or HPMS_CLK then DDR_FIC_CLK must be MDDR_CLK/n where n={1,2,4,8,16}
10. Each FPGA fabric FIC subsystem must be driven by a clock with a frequency matching the respective FIC interface clock frequency defined in the MSS CCC or HPMS CCC.
11. Only in the SmartFusion 2 device, if the CAN peripheral within the MSS is operational, then APB_1_CLK must be a multiple of 8 MHz, which implies that M3_CLK must be a multiple of 8.

The following figure shows several example aligned clock configurations for MDDR_CLK and M3_CLK or HPMS_CLK. In all the configurations, the MSS or HPMS peripherals and fabric subsystems can run at different clock ratios independently as per the constraints listed above.

Figure 90 • **Aligned Clock Configuration Examples**



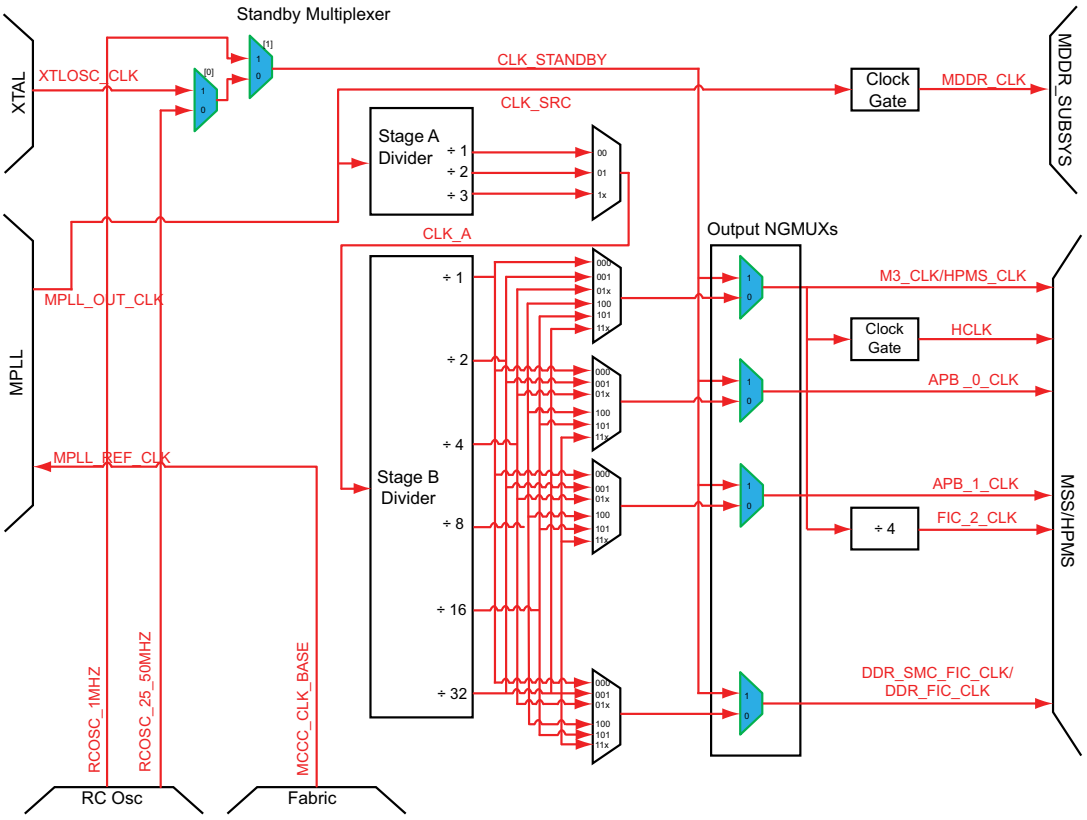
6.2.2.2 No-Glitch Multiplexers

The FACC includes no-glitch multiplexers (NGMUXs) to feed the MSS or HPMS clocks with a standby clock (CLK_STANDBY) during the MPLL initialization, Flash*Freeze mode, or in the second stage of in-application programming (IAP), when the FPGA fabric is not valid (MCCC_CLK_BASE is not usable) as shown in the following figure.

The standby multiplexer selects the output of one of the three on-chip oscillators as the standby clock. During MPLL initialization, the standby multiplexer is automatically configured to select the 50 MHz RC oscillator as the standby clock. The standby clock during Flash*Freeze mode can be configured to come from the 1 MHz RC oscillator, or 50 MHz RC oscillator. For IAP, the standby multiplexer must be configured to select the 50 MHz RC oscillator as the clock source, before requesting IAP from the System Controller.

There are four other NGMUXs available at outputs to switch the MSS or HPMS clocks cleanly between the CLK_STANDBY and the MPLL derived clock, as shown in the following figure. All four of these NGMUXs switch at the same time.

Figure 91 • **FACC Functional Block Diagram—No-Glitch Multiplexers**



When CLK_STANDBY is used for MSS or HPMS clocking:

- M3_CLK or HPMS_CLK and APB_0_CLK/APB_1_CLK are the same as CLK_STANDBY
- MDDR_CLK is gated off

For Flash*Freeze mode, the switching of the clock source to use the selected standby clock is done automatically by the System Controller upon receiving a Flash*Freeze entry request message. When a Flash*Freeze exit condition is detected, the System Controller switches the clock source of the MSS or HPMS back to the MPLL output without any glitches. The System Controller also switches the clock source to the standby clock during the second stage of IAP.

6.2.2.2.1 MSS Clock Switching from User Clock to Standby Clock

In SmartFusion 2 devices, when the clock source for the M3_CLK switches from user clock (CLK_SRC) to standby clock (CLK_STANDBY), there is a possibility for both M3_CLK and FIC_2_CLK to halt if a wrong Stage B divisor is used within the FACC, see Figure 91, page 97. The Stage B divider must be set with correct divisor value before initiating MSS clock switching from user clock to standby clock. The following table lists required Stage B divider setting for proper MSS clock switching from user clock to standby clock.

Table 56 • **Stage B Divider setting for MSS Clock Switching from User Clock to Standby Clock**

SmartFusion® 2 Device	Stage B Clock Divisor to use
5	Divide by 32
10	Divide by 32
25	Divide by 32
50	Divide by 32
60	Any divider except divide by 32

Table 56 • **Stage B Divider setting for MSS Clock Switching from User Clock to Standby Clock**

SmartFusion® 2 Device	Stage B Clock Divisor to use
90	Divide by 32
150	Divide by 32

If you have requested for fabric digest check, IAP, and Flash*Freeze system service using SmartFusion 2 MSS System Services software driver (v2.9.100), the firmware driver changes the Stage B clock divisor value according to the preceding table and no action is required. Use the correct clock divisor setting before switching from CLK_SRC to CLK_STANDBY when the firmware driver is not used.

6.2.3 MSS CCC or HPMS CCC Operational Modes

In order to have clock alignment between the FPGA fabric and the MSS or HPMS, it is necessary to use the MPLL to perform deskewing of the MSS or HPMS clocks. After the MPLL is initialized, it typically takes over 500 divided reference clock cycles to achieve lock. The MPLL lock assertion time is also dependent on the MPLL lock parameters (lock window and lock delay). Furthermore, if the MPLL reference clock is derived from a fabric PLL, the fabric PLL must first come into lock. It is desirable for the MSS or HPMS to be usable in advance to the MPLL coming into lock.

The MSS CCC or HPMS CCC is designed to facilitate two operational modes:

- PLL Initialization Mode
- Run-Time Mode

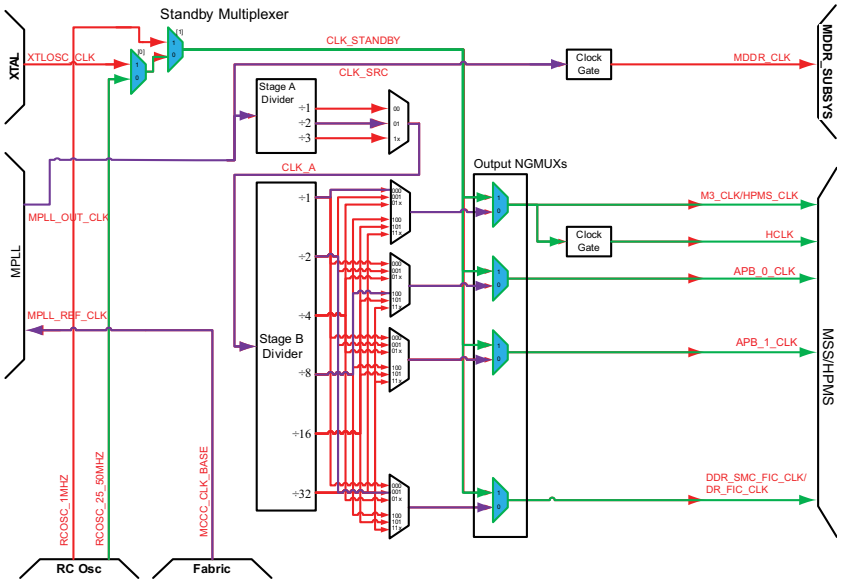
The System Controller performs the clock sequencing that is waiting for the MPLL to come into lock and switches the FACC configuration from PLL Initialization mode to run-time mode. See the Startup Clock Configuration Sequence, page 100 for more information on clock sequencing after power-up.

6.2.3.1 PLL Initialization Mode

The default configuration of the FACC, immediately after power-on reset, selects the 50 MHz RC oscillator as the source of the MSS or HPMS clocks: M3_CLK or HPMS_CLK, APB_0_CLK, and APB_1_CLK. These are all configured as 1:1 ratios for the duration of the initialization time. During this time, the MSS or HPMS cannot communicate with either the FPGA fabric or the MDDR subsystem. However, the Cortex-M3 processor can boot from the embedded nonvolatile memory (eNVM) and perform system initialization, if desired.

The following figure shows clock configuration within the FACC during PLL initialization time.

Figure 92 • **FACC Configuration During PLL Initialization Time**



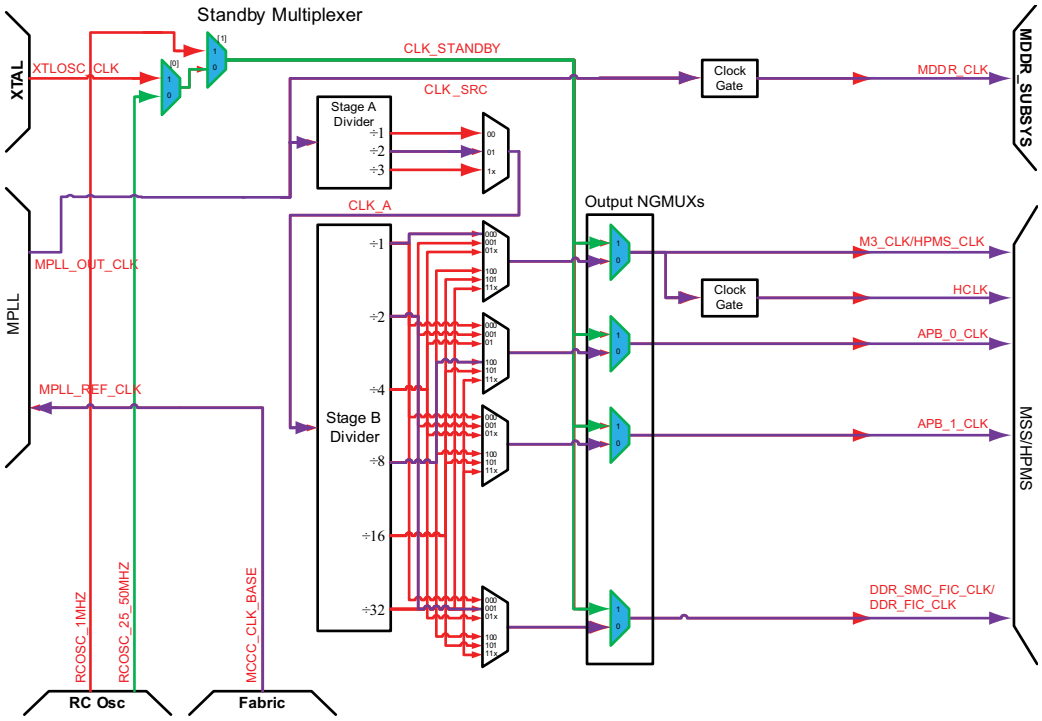
6.2.3.2 Run-Time Mode

When both the fabric PLL (if being used) and MPLL have come into lock, the System Controller switches the MSS or HPMS clocks' source in a glitch free manner, such that they originate from the MPLL and FACC divider combination and are therefore fully aligned with the FPGA fabric clocks. In this mode, the MSS or HPMS is fully operational and it is safe to communicate with the FPGA fabric and the MDDR subsystems.

The FACC clock configuration in Run-time mode is entirely dependent on the configuration set in the **SmartFusion 2 and IGLOO 2 System Builder Clocks** tab. The FACC configuration is initialized by flash configuration bits and must not be overwritten by any MSS or HPMS master during run-time.

The following figure shows clock configuration within the FACC configuration during run-time.

Figure 93 • **FACC Configuration During Run-Time**



6.2.4 PLLs Lock Monitoring

The MSS CCC or HPMS CCC has an input (MCCC_CLK_BASE_PLL_LOCK) to monitor the fabric PLL lock. It must be connected to the lock signal generated by the fabric PLL which is used to generate the base clock to the MSS CCC or HPMS CCC. During system initialization, the System Controller monitors the lock signals of both MPLL and fabric PLL. After system initialization, the MPLL lock (MCCC_MPLL_LOCK) signal must be monitored by the Cortex-M3 processor or user logic in the FPGA fabric for loss of lock. It is sufficient to monitor only the MCCC_MPLL_LOCK signal since the MPLL is able to stay in lock even with variations in its reference clock. There is no hardware support provided within the FACC for automatic switching of clock source or resetting of the FACC or the MSS/HPMS when the MPLL lock is lost. A recovery scheme must be implemented using the Cortex-M3 processor firmware or FPGA fabric logic to switch the MSS/HPMS clocks to a standby clock in the event of loss of the MPLL lock. The recovery scheme must also incorporate the logic to prevent user logic from attempting to communicate with the HPMS until the MPLL lock is achieved again.

In SmartFusion 2 devices, within the MSS, there are four interrupts related to the PLL lock. For each of the lock signals (that is, fabric PLL lock and MPLL lock) there is a lock interrupt, indicating lock achieved, and a lock lost interrupt. Each of these four interrupts has a corresponding interrupt enable bit in the MSS system registers. It is also possible to read the state of the two PLL lock signals through the MSS system registers.

In the event of loss of MPLL lock, even though its output is not exactly in phase lock with the reference, the MPLL still generates a clock. Due to the fact that the clock continues to run (though possibly not fully frequency—or phase—aligned with the reference clock), loss of lock is not considered a catastrophic event for the MSS or HPMS CCC. In SmartFusion 2 devices, since the Cortex-M3 processor is running, firmware can use the loss of lock interrupt to take precautionary action, in a controlled way such as the following:

- Executing a recovery routine from embedded SRAM (eSRAM) by switching the clock source to the 50 MHz RC oscillator
- Disabling communications with the FPGA fabric (as the fabric and MSS clocks may be misaligned, if the PLLs are not in lock).

User logic in the FPGA fabric can use the MCCC_MPLL_LOCK signal to prevent user logic from attempting to communicate with the MSS during this time.

6.3 Startup Clock Configuration Sequence

This section summarizes the SmartFusion 2 MSS or IGLOO 2 HPMS clock configuration sequence from power-on reset.

1. After power-on reset, the MSS or HPMS is clocked by the 50 MHz RC oscillator through the CLK_STANDBY path.
   - MDDR is in reset during clock start-up.
   - User logic in the FPGA fabric can use the MPLL lock signal to prevent logic from attempting to communicate with the MSS or HPMS during this time.
   - The MSS or HPMS can continue to run in a limited mode (no FIC transactions) for executing initialization code.
2. Flash configuration bits configure fabric CCC/PLL, MPLL, and FACC dividers for desired clock configuration.
   - MCCC_CLK_BASE is provided to the MPLL reference input.
   - Waits for the fabric CCC/PLL lock, if it is being used.
   - Waits until MPLL comes into lock.
3. Upon lock, the System Controller switches M3_CLK or HPMS_CLK, HCLK, APB_0_CLK, APB_1_CLK, and DDR_FIC_CLK NGMUXs to source from the FACC clock dividers.
4. Releases reset of the MDDR and fabric interfaces.

6.4 MSS CCC or HPMS CCC Configuration

The MSS or HPMS clocks are statically configured through the MSS clock configurator (MSS_CCC) or **IGLOO 2 System Builder** (Clocks tab) in the Libero SoC software. The MSS_CCC configurator or IGLOO 2 System Builder derives the necessary MPLL internal divider ratios and selects the FACC dividers ratios based on the MCCC_CLK_BASE and the desired output frequencies. After power-on reset, the MSS CCC or HPMS CCC is initialized with these configuration values.

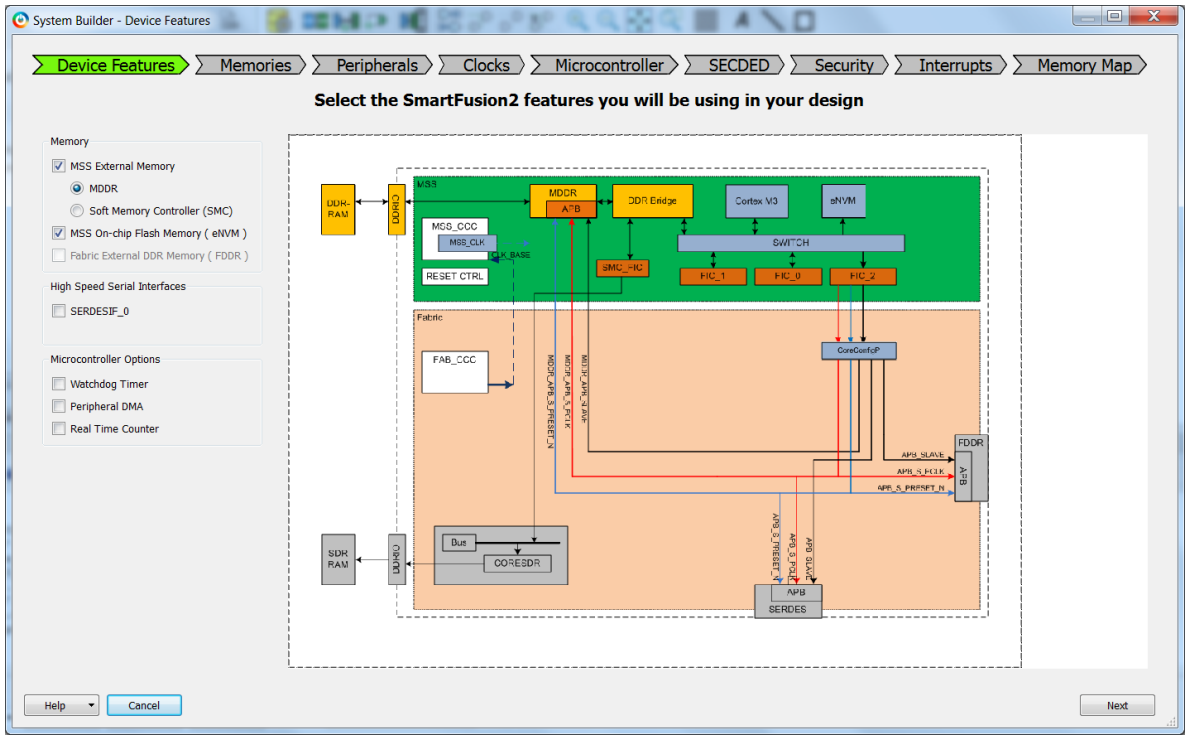
The MSS CCC or HPMS CCC configuration is controlled by the SYSREG control registers. However, Microchip recommends that the MSS CCC or HPMS CCC must be configured at the programming stage through flash bits set by the MSS_CCC configurator or IGLOO 2 System Builder and not to be modified dynamically during run-time. It is possible to read the state of the MPLL lock signal through the MSS system registers. See the SYSREG Control Registers, page 110 for HPMS CCC configuration.

6.5 How to Use MSS CCC

This section describes how to use the clocks in the SmartFusion 2 devices. To configure the SmartFusion 2 device features and then build a complete system, use the **System Builder** graphical design wizard in the Libero SoC software.

The following figure shows the initial **System Builder** window where you can select the device features that you require. For details on how to launch the **System Builder** wizard and a detailed information on how to use it, see *SmartFusion2 System Builder User Guide*.

Figure 94 • **System Builder Window**

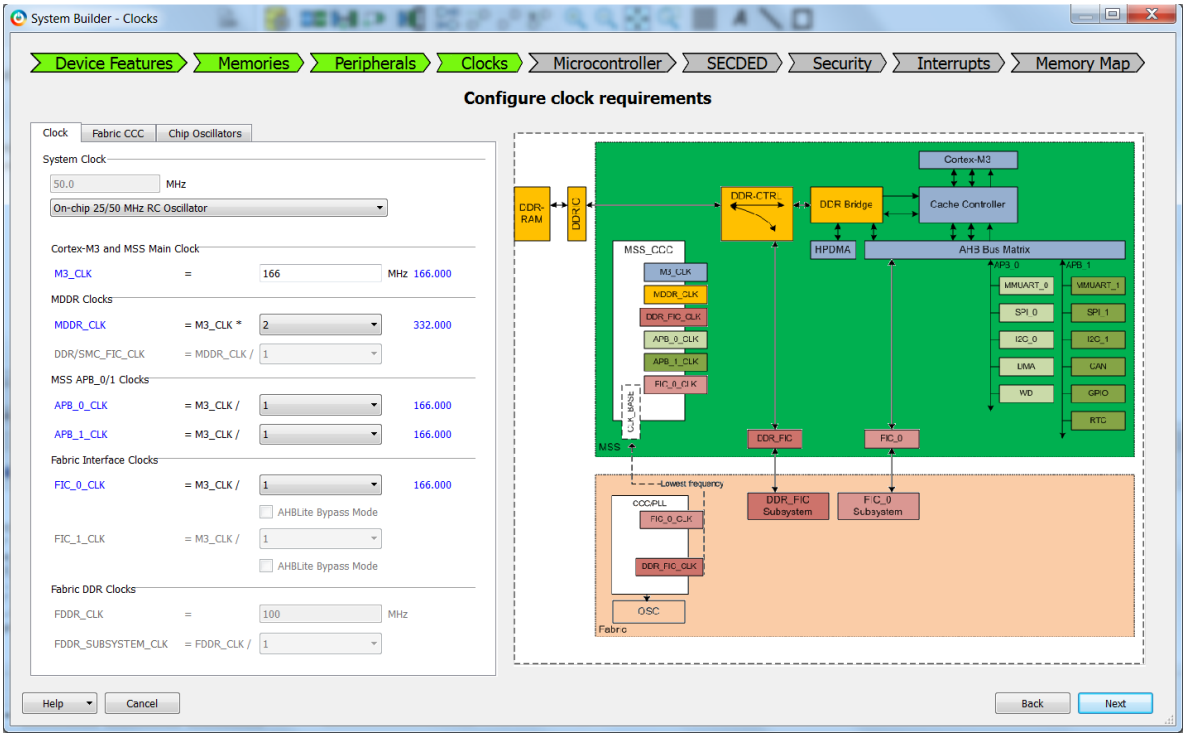


6.5.1 Configuring Clocks

The following steps describe how to use the clocks in the user application.

1. Click **Next** to navigate to the **Clocks** tab in the **System Builder** wizard.

Figure 95 • **System Builder - Clocks Tab**



Configure the following MSS CCC options in the Clocks tab:

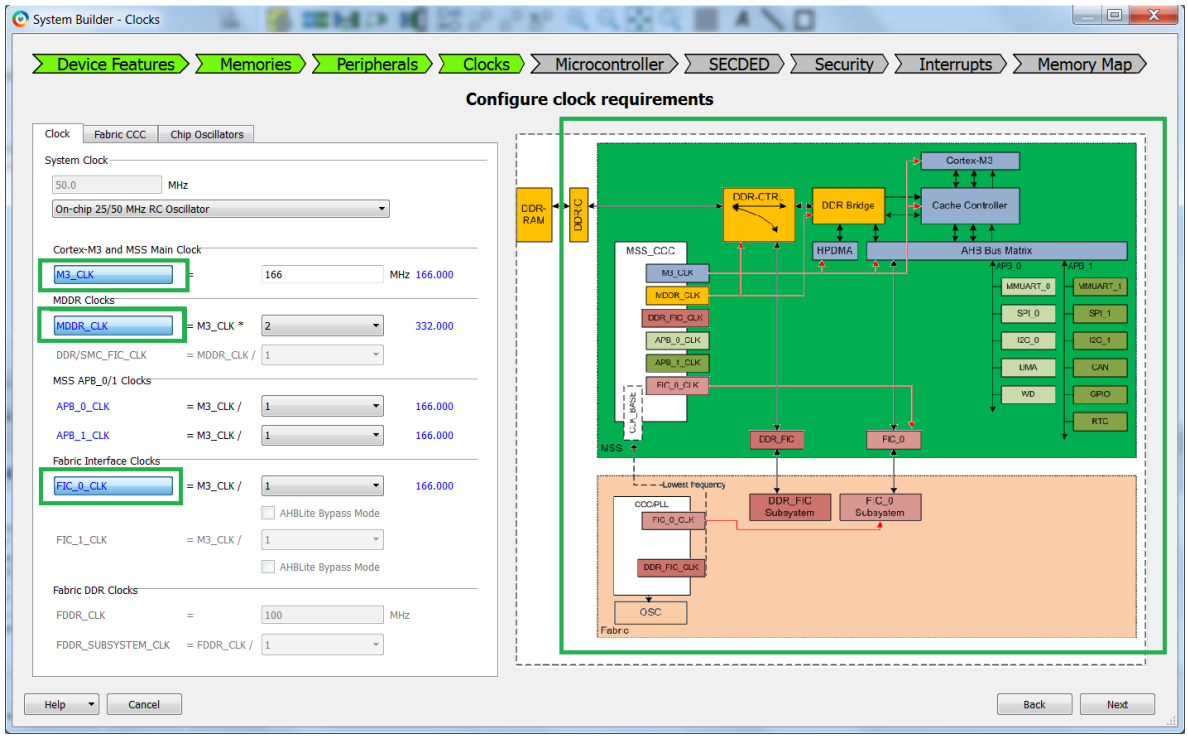
- System clock source selection
- System clock frequency selection
- The MSS and Cortex-M3 clock (M3_CLK)
- The MDDR clocks (MDDR_CLK and DDR_SMC_FIC_CLK)
- The MSS APB_0 peripheral clock (APB_0_CLK)
- The MSS APB_1 peripheral clock (APB_1_CLK)
- The FIC clocks (FIC_0_CLK and FIC_1_CLK)
- FDDR clocks (FDDR_CLK and FDDR_SUBSYSTEM_CLK)

For MSS clocks configuration constraints, see the FACC Dividers, page 95.

Note: Only the clocks used in your design are configurable in the **Clocks** tab. Ensure to enable and correctly configure all the MSS sub-blocks you intend to use in your design before configuring the MSS CCC sub-block using the **Clocks** tab.

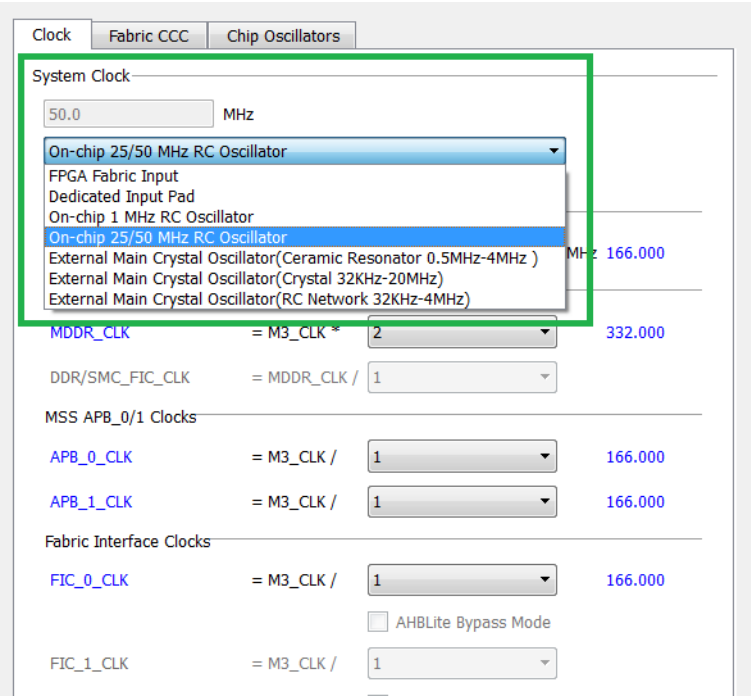
The **Clocks** tab displays a high level block diagram of your design based on the configuration set in the **System Builder** wizard. The following figure shows the clock domains (each clock domain is in a different color) within the HPMS and the clock domains that cross into the FPGA fabric. Clicking any of the clocks highlights (blue labels) that particular clock domain on the **Clocks** configuration dialog.

Figure 96 • **Clocks Tab Showing the Clock Paths Preview**



2. Select the desired **System Clock** source. Based on the clock source selection, system builder instantiate and configure the CCC macro and chip oscillators macro.

Figure 97 • **Clock Source Selection**



3. Configure the clock frequency of the MSS and Cortex-M3 processor (M3_CLK). This specifies the frequency of the Cortex-M3 processor, AHB bus matrix, HPDMA, eNVM, and eSRAM.
4. MDDR clocks consist of two clock domains: MDDR_CLK and DDR_SMC_FIC_CLK
   - Configure **MDDR_CLK** as a multiple—1, 2, 3, 4, 6, or 8—of the M3_CLK.
   - Configure **DDR/SMC_FIC_CLK** as a ratio—1, 2, 3, 4, 6, 8, 12, or 16—of the MDDR_CLK.

5. Configure the **MSS APB_0 Clock** for peripherals connected to APB_0 bus.
6. Configure the **MSS APB_1 Clock** for peripherals connected to APB_1 bus.
7. Configure the **Fabric Interface Clocks** (FIC_0_CLK and FIC_1_CLK) as a ratio—1, 2, 4, 8, 16, or 32—of M3_CLK such that the generated frequencies meet the timing requirements of the logic implemented in the FPGA fabric for each FIC subsystem.
8. Configure the FIC subsystem clocks as a ratio—1, 2, 4, 8, 16, or 32—of M3_CLK. Verify the timing for each FIC subsystem if it meets the selected clock frequency by performing timing analysis of the design using SmartTime.

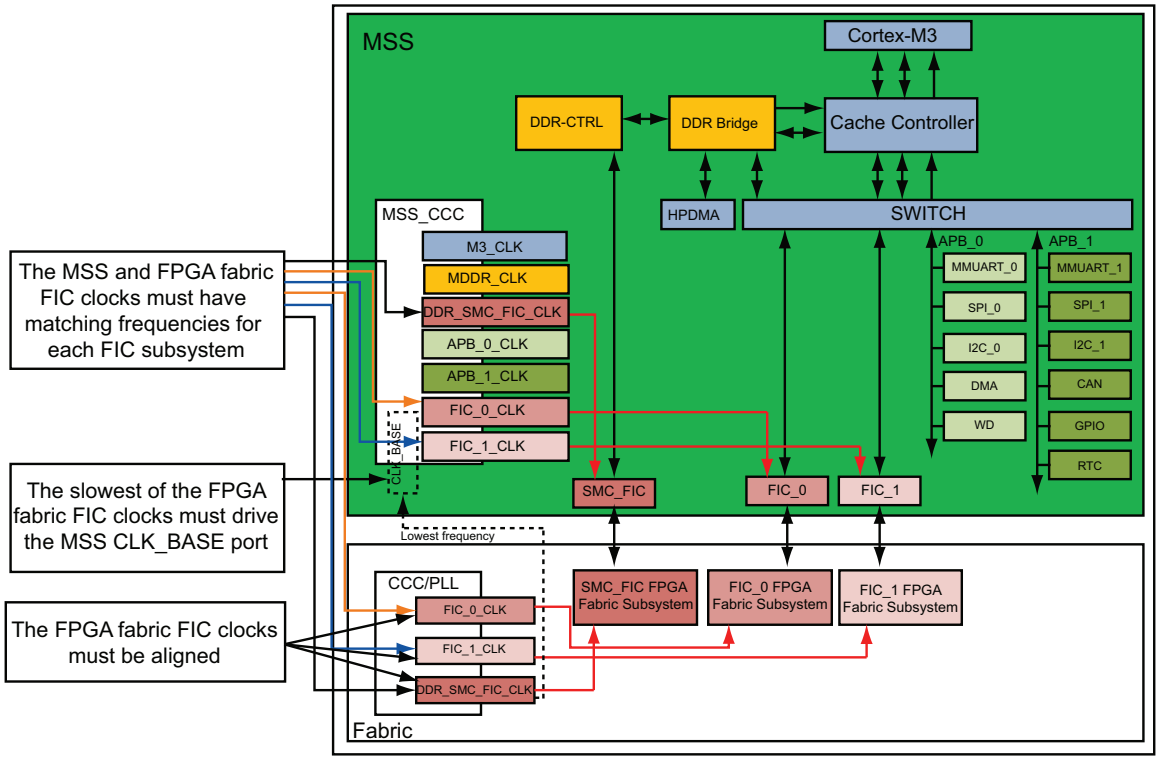
6.5.2 How to Clock MSS and FIC Subsystems for Synchronous communication

The following rules of the SmartFusion 2 architecture must be followed for synchronous communication between the MSS and FPGA fabric FIC subsystems.

- Drive each FPGA fabric FIC subsystem by a clock with a frequency that matches the frequency defined for that particular subsystem in the **Clocks** configuration tab in **System Builder** wizard.
- The FPGA fabric FIC subsystem clock with the smallest frequency must drive the MSS CLK_BASE.
- Align all the FPGA fabric FIC subsystem clocks precisely; the clocks could be of different frequencies, but align the rising-edges of the slower clocks to the rising-edges of the fastest clocks.

The following figure shows the clocking scheme that must be followed for synchronous communication between MSS and FPGA fabric.

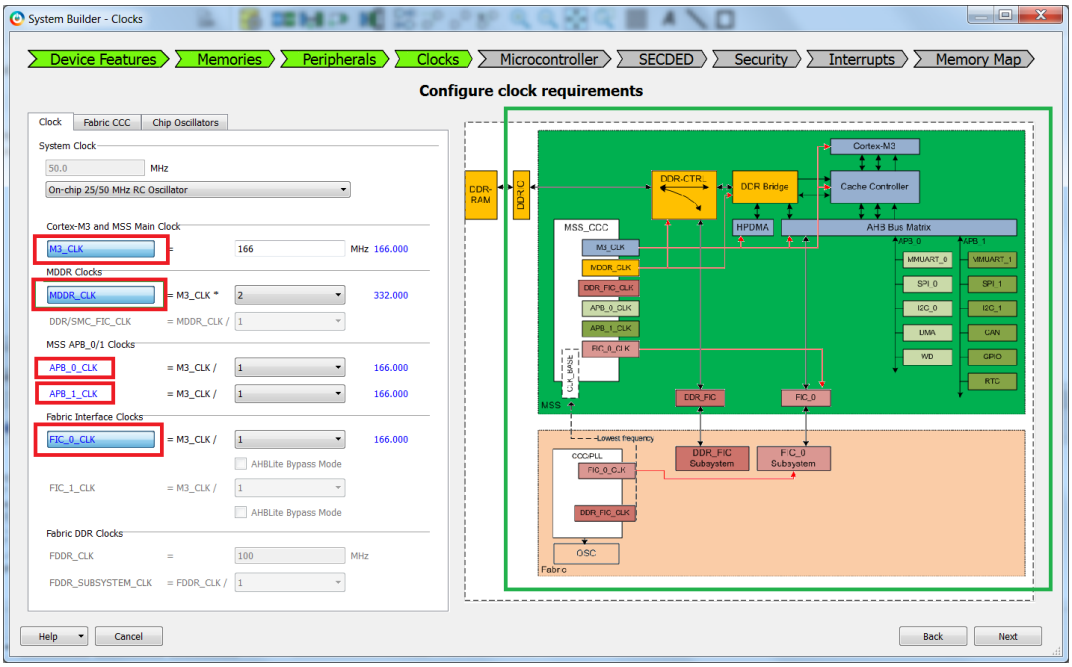
Figure 98 • **Clocking Scheme for Synchronous Communication Between MSS and FPGA Fabric**



Use the following steps to configure the clock networks for all the FIC subsystems.

1. Configure the SmartFusion 2 MSS peripheral components and FIC interfaces (FIC_0, FIC_1, and/or DDR_FIC) according to the design requirements using the System Builder.
   See *SmartFusion2 System Builder User Guide* for more information on how to create FIC subsystem in FPGA fabric.
2. Configure the MSS Clock, MDDR Clocks, MSS APB Clocks, and Fabric Interface Clocks in the **Clocks** tab.

Figure 99 • **MSS CCC Clock Configuration**

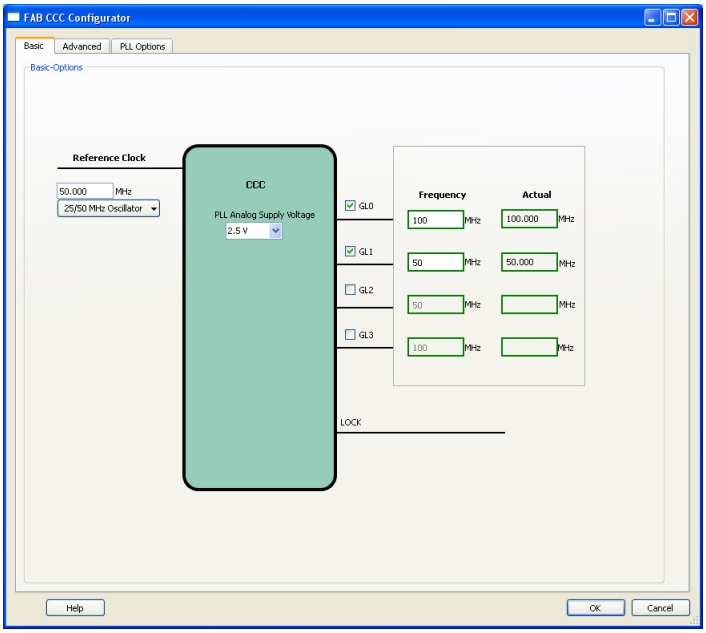


3. Proceed with the rest of the configurations in the **System Builder** wizard and create the subsystem.
4. The System Builder generated subsystem exposes the FIC clocks (FIC_0_CLK and FIC_1_CLK) and corresponding lock signals (FIC_0_LOCK and FIC_1_LOCK) to the FPGA fabric. Connect the FIC clocks to the fabric master and/or slave clock inputs for synchronous operation with the MSS. The FIC_0_LOCK and FIC_1_LOCK signals can be used to monitor the state of the FIC clocks.

6.6 How to Use HPMS Clocks

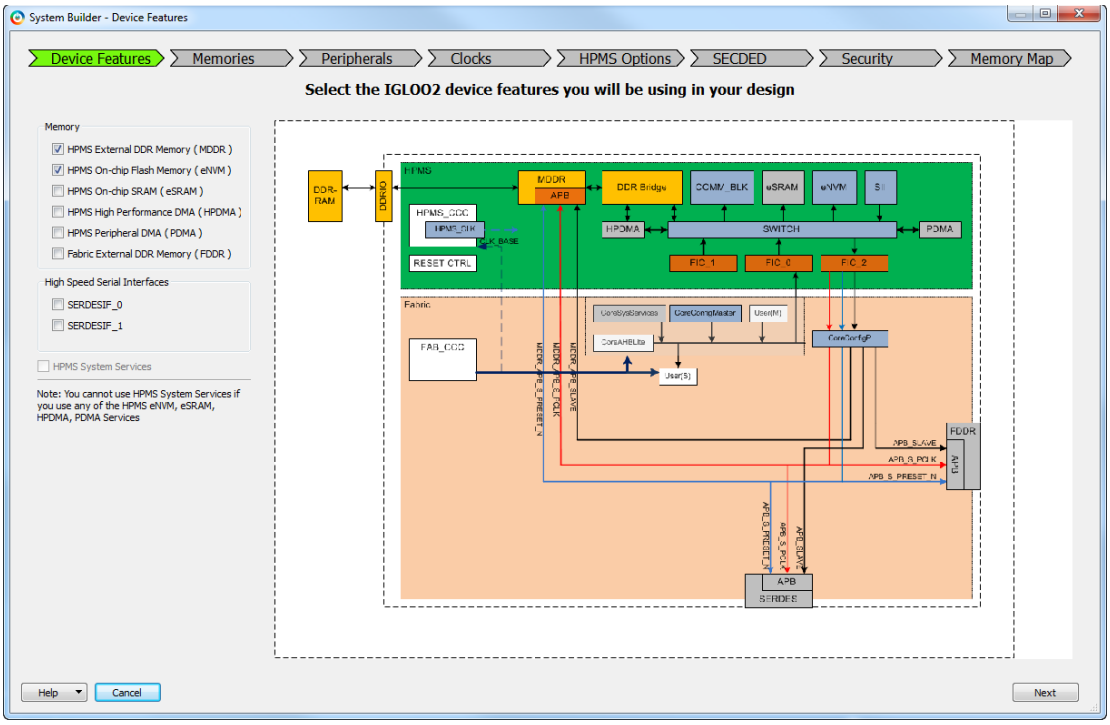
This section describes how to use the clocks in the IGLOO 2 devices. To configure the IGLOO 2 device features and then build a complete system, use the **System Builder** graphical design wizard in the Libero SoC software.

Figure 100 • **Fabric Clocks Configuration**



The following figure shows the initial **System Builder** window where you can select the device features that you require. For details on how to launch the **System Builder** wizard and a detailed information on how to use it, refer the *IGLOO2 System Builder User Guide*.

Figure 101 • **System Builder Window**

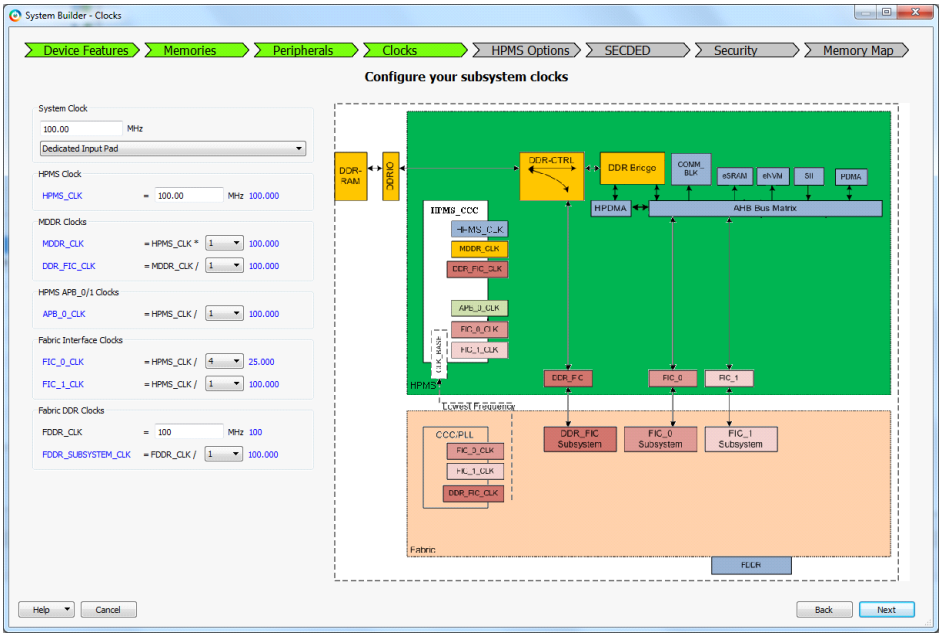


6.6.1 Configuring Clocks

The following steps describe how to use the clocks in the user application.

1. Click **Next** to navigate to the **Clocks** tab in the **System Builder** wizard.

Figure 102 • **System Builder - Clocks Tab**



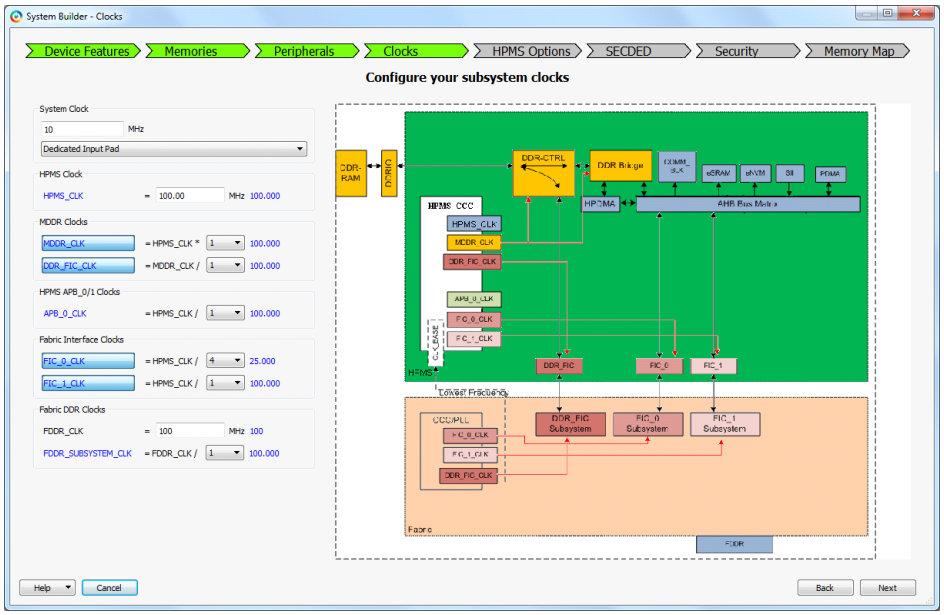
Configure the following HPMS CCC options in the Clocks tab:

- System clock frequency selection
- System clock source selection
- The HPMS clock (HPMS_CLK)
- The MDDR clocks (MDDR_CLK and DDR_FIC_CLK)
- The HPMS APB_0 peripheral clock (APB_0_CLK)
- The FIC clocks (FIC_0_CLK and FIC_1_CLK)

For HPMS clocks configuration constraints, see the FACC Dividers, page 95.

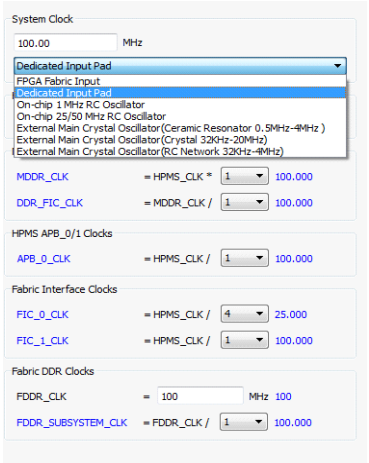
Note: Only the clocks used in your design are configurable in the **Clocks** tab. Ensure to enable and correctly configure all the HPMS sub-blocks you intend to use in your design before configuring the HPMS CCC sub-block using the **Clocks** tab.
The **Clocks** tab displays a high level block diagram of your design based on the configuration set in the **System Builder wizard**. The following figure shows the clock domains (each clock domain is in a different color) within the HPMS and the clock domains that cross into the FPGA fabric. Clicking any of the clocks highlights (blue labels) that particular clock domain on the **Clocks** configuration dialog.

Figure 103 • **Clocks Tab Showing the Clock Paths Preview**



2. Select the desired **System Clock** source. Based on the clock source selection, system builder instantiate and configure the CCC macro and chip oscillators macro.

Figure 104 • **Clock Source Selection**



3. Configure the clock frequency of the HPMS clock (HPMS_CLK). This specifies the frequency of the AHB bus matrix, HPDMA, eNVM, and eSRAM.
4. MDDR clocks consist of two clock domains: MDDR_CLK and DDR_FIC_CLK
   - Configure **MDDR_CLK** as a multiple—1, 2, 3, 4, 6, or 8—of the HPMS_CLK.
   - Configure **DDR_FIC_CLK** as a ratio—1, 2, 3, 4, 6, 8, 12, or 16—of the MDDR_CLK.
5. Configure the **HPMS APB_0 Clock** for peripherals connected to APB_0 bus.
6. Configure the **Fabric Interface Clocks** (FIC_0_CLK and FIC_1_CLK) as a ratio—1, 2, 4, 8, 16, or 32—of HPMS_CLK such that the generated frequencies meet the timing requirements of the logic implemented in the FPGA fabric for each FIC subsystem.
7. Configure the FIC subsystem clocks as a ratio—1, 2, 4, 8, 16, or 32—of HPMS_CLK. Verify the timing for each FIC subsystem if it meets the selected clock frequency by performing timing analysis of the design using SmartTime.

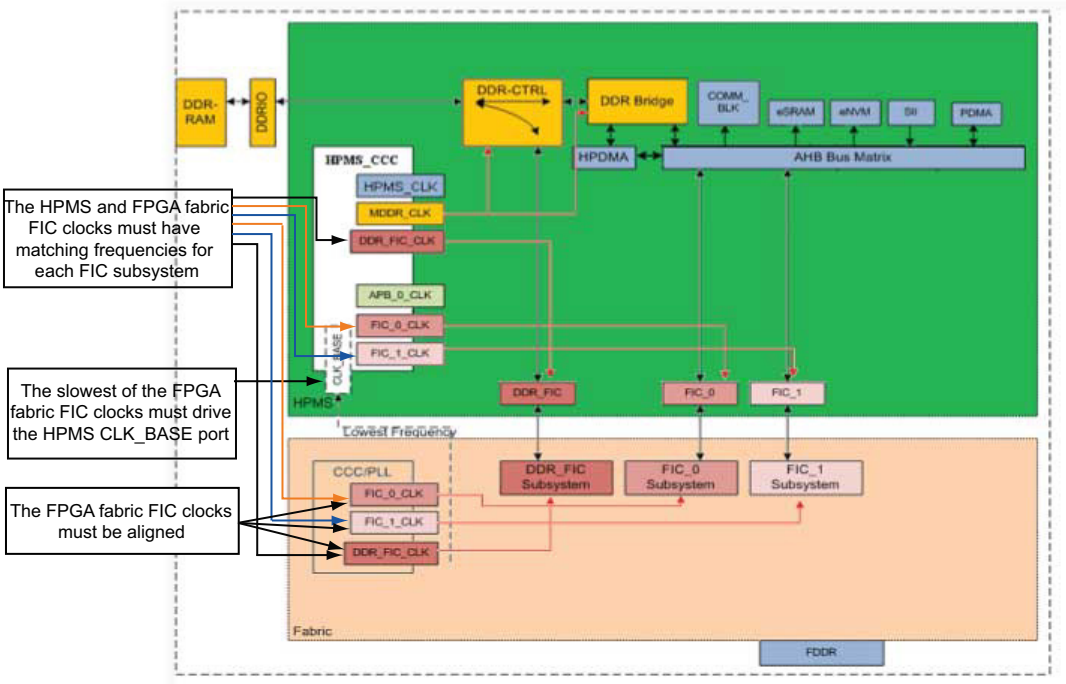
6.6.2 Clocking HPMS and FIC Subsystems

The following rules of the IGLOO 2 architecture must be followed for synchronous communication between the HPMS and FPGA fabric FIC subsystems.

- Drive each FPGA fabric FIC subsystem by a clock with a frequency that matches the frequency defined for that particular subsystem in the **Clocks** configurator tab in **System Builder** wizard.
- The FPGA fabric FIC subsystem clock with the smallest frequency must drive the HPMS CLK_BASE.
- Align all the FPGA fabric FIC subsystem clocks precisely; the clocks could be of different frequencies, but align the rising-edges of the slower clocks to the rising-edges of the fastest clocks.

The following figure shows the clocking scheme that must be followed for synchronous communication between HPMS and the FPGA fabric.

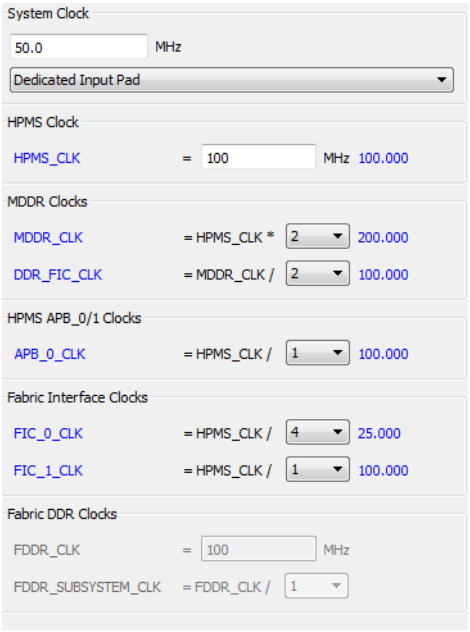
***Figure 105* • Clocking Scheme for Synchronous Communication between HPMS and FPGA Fabric**



Use the following steps to configure the clock networks for all the FIC subsystems.

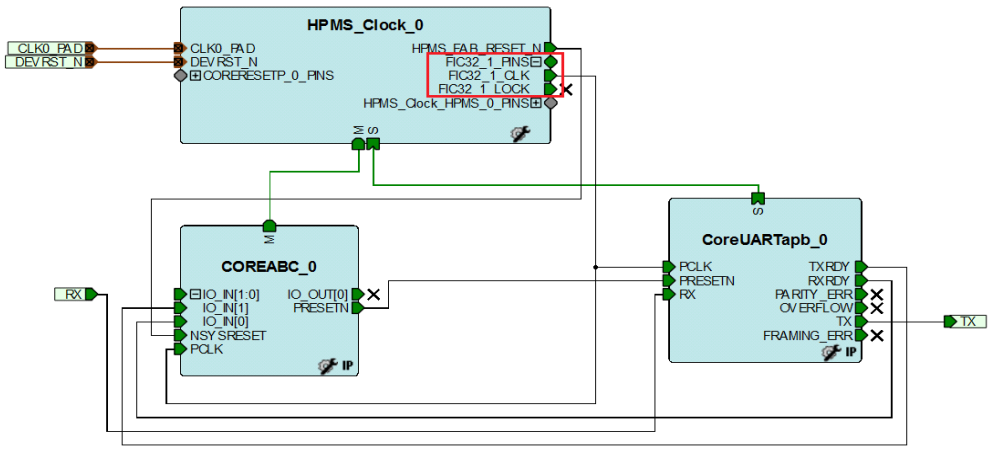
1. Configure the IGLOO 2 HPMS peripheral components and FIC interfaces (FIC_0, FIC_1, and/or DDR_FIC) according to the design requirements using the System Builder.
   See *IGLOO2 System Builder User Guide* for more information on how to create FIC subsystem in FPGA fabric.
2. Configure the HPMS Clock, MDDR Clocks, HPMS APB Clocks, and Fabric Interface Clocks in the **Clocks** tab as shown in the following figure.

Figure 106 • **HPMS CCC FIC Clock Configuration**



3. Proceed with the rest of the configurations in the **System Builder** wizard and create the subsystem.
4. The following figure shows an example system created with FIC_1 interface enabled. The System Builder generated subsystem exposes the FIC_1 clock (FIC32_1_CLK) and corresponding lock signal (FIC32_1_LOCK). Connect FIC32_1_CLK to User Fabric master and/or slave. If the User Fabric Master or Slave has a PLL LOCK input, connect it to FIC32_1_LOCK. FIC32_1_LOCK asserts when FIC32_1_CLK is valid.

Figure 107 • **Example Design with FIC_1 Subsystem in the FPGA Fabric**



6.7 SYSREG Control Registers

The following tables list the SYSREG control registers related to MSS CCC and HPMS CCC configuration The base address of the SYSREG control registers is 0x40038000. See the “System Registers Block” chapter of the *UG0331: SmartFusion2 Microcontroller Subsystem User Guide* or *UG0448: IGLOO2 FPGA High Performance Memory Subsystem User Guide* for detailed bit definitions of these registers. These system registers should not be modified during run-time, but only configured statically at start-up by flash configuration bits set by the MSS_CCC configurator or IGLOO 2 System Builder. A master can read these registers to know the current configuration of the MSS CCC or HPMS CCC or status of the PLLs.

Table 57 • **MSS CCC Configuration Registers in SYSREG Block**

Register Name	Address Offset	Register Type	Flash Write Protect	Reset Source	Description
MSSDDR_FACC1_CR	0x98	RW-P	Field	CC_RESET_N	FACC configuration register 1
MSSDDR_FACC2_CR	0x9C	RW-P	Field	CC_RESET_N	FACC configuration register 2

Table 58 • **HPMS CCC Configuration Registers in SYSREG Block**

Register Name	Address Offset	Register Type	Flash Write Protect	Reset Source	Description
HPMS_FACC1_CR	0x98	RW-P	Field	CC_RESET_N	FACC configuration register 1
HPMS_FACC2_CR	0x9C	RW-P	Field	CC_RESET_N	FACC configuration register 2